



Propeller aerodynamics

AE4135 – Rotor/Wake Aerodynamics

24 March 2025

Tomas Sinnige

Why propellers?

- ...



Why propellers?



- Propellers offer unrivaled propulsive efficiency

- Thrust $T = \frac{dp}{dt} = \dot{m}(V_{\text{jet}} - V_{\infty})$
- $\eta_{\text{propulsive}} = \frac{TV_{\text{in}}}{\Delta \dot{E}_k} = \frac{2}{1 + \frac{V_{\text{jet}}}{V_{\infty}}}$

High efficiency at given thrust?

- Small velocity increment $V_{\text{jet}} - V_{\infty}$
- Large mass flow $\dot{m}$

Large diameter
(high by-pass
ratio)



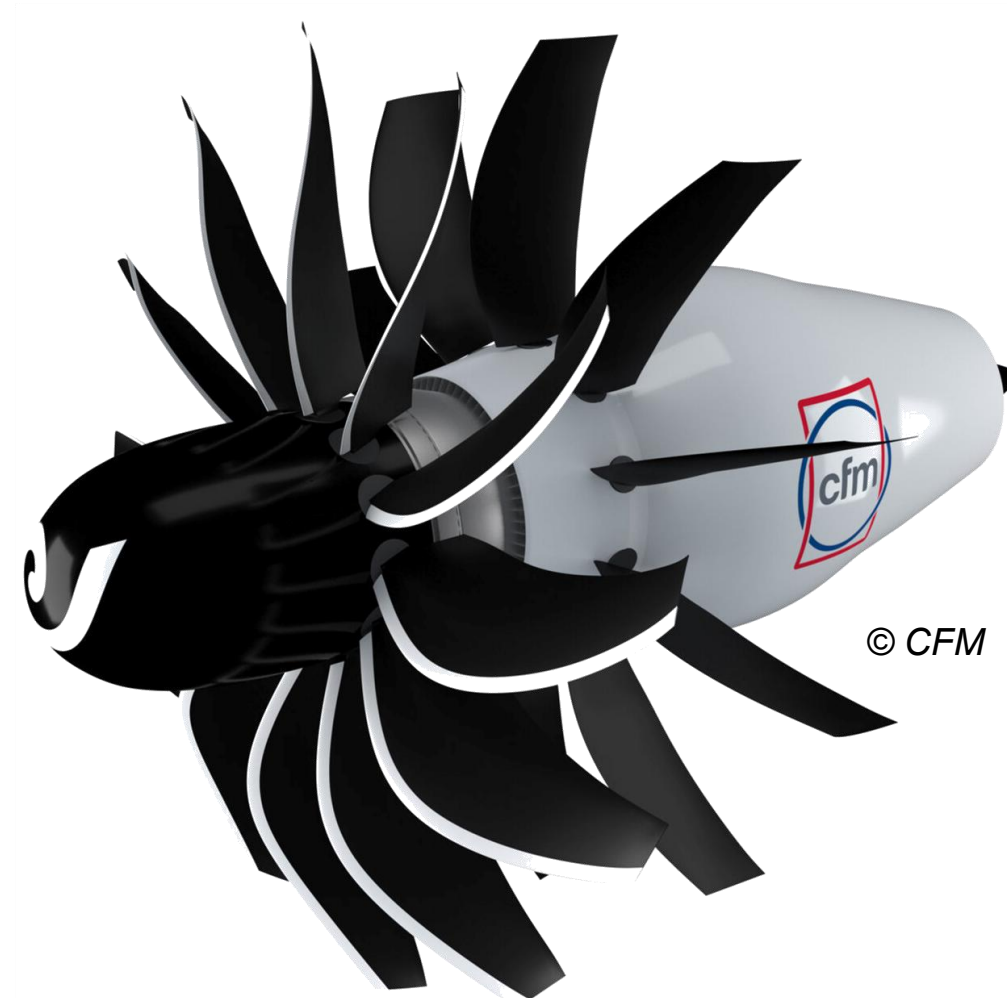
Why propellers?



- Propellers offer unrivaled propulsive efficiency
 - Attractive solution for more sustainable aircraft
 - Integration of propellers with airframe needs to be improved
 - Propeller noise needs to be minimized



Active field of research & development!



Propellers vs wind turbines



Are they that different?

- ...
- ...
- ...
- ...

Propellers vs wind turbines



- | | | |
|-----------------------|-----------------------|------------------------|
| • Purpose | Thrust | Power |
| • Axial force* | Forward | Backward |
| • Power | Consumed | Extracted |
| • Size | ~1-5 m | ~40-200 m |
| • Operation | ‘Low’ tip-speed ratio | ‘High’ tip-speed ratio |

** on the rotor*

Both have rotating blades → aerodynamics are comparable!

Today's lecture

- Fundamentals of propeller aerodynamics:
 1. Propeller blade loading
 2. Propeller performance
- ... and how it compares to wind-turbine aerodynamics



Propeller blade loading

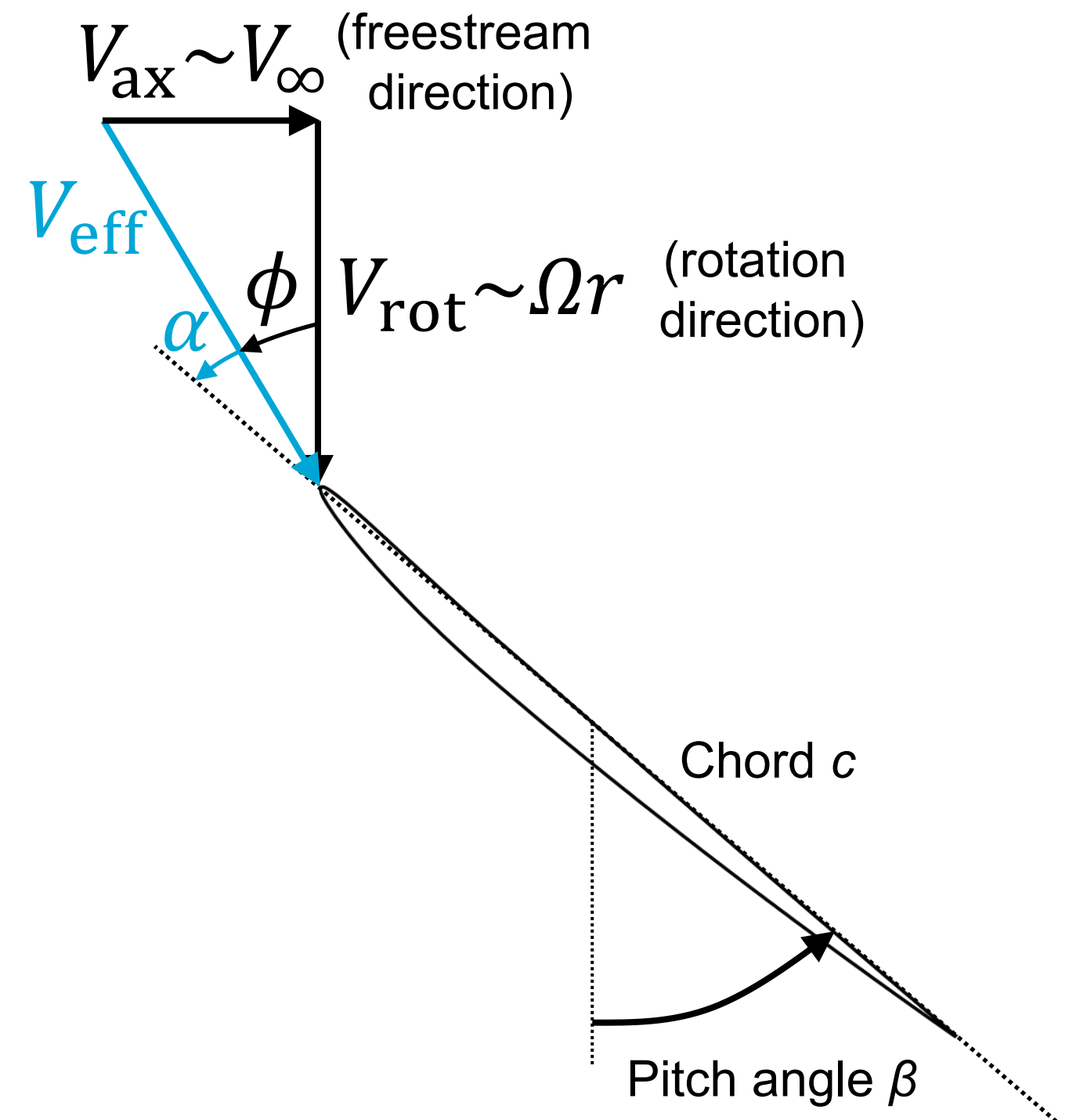
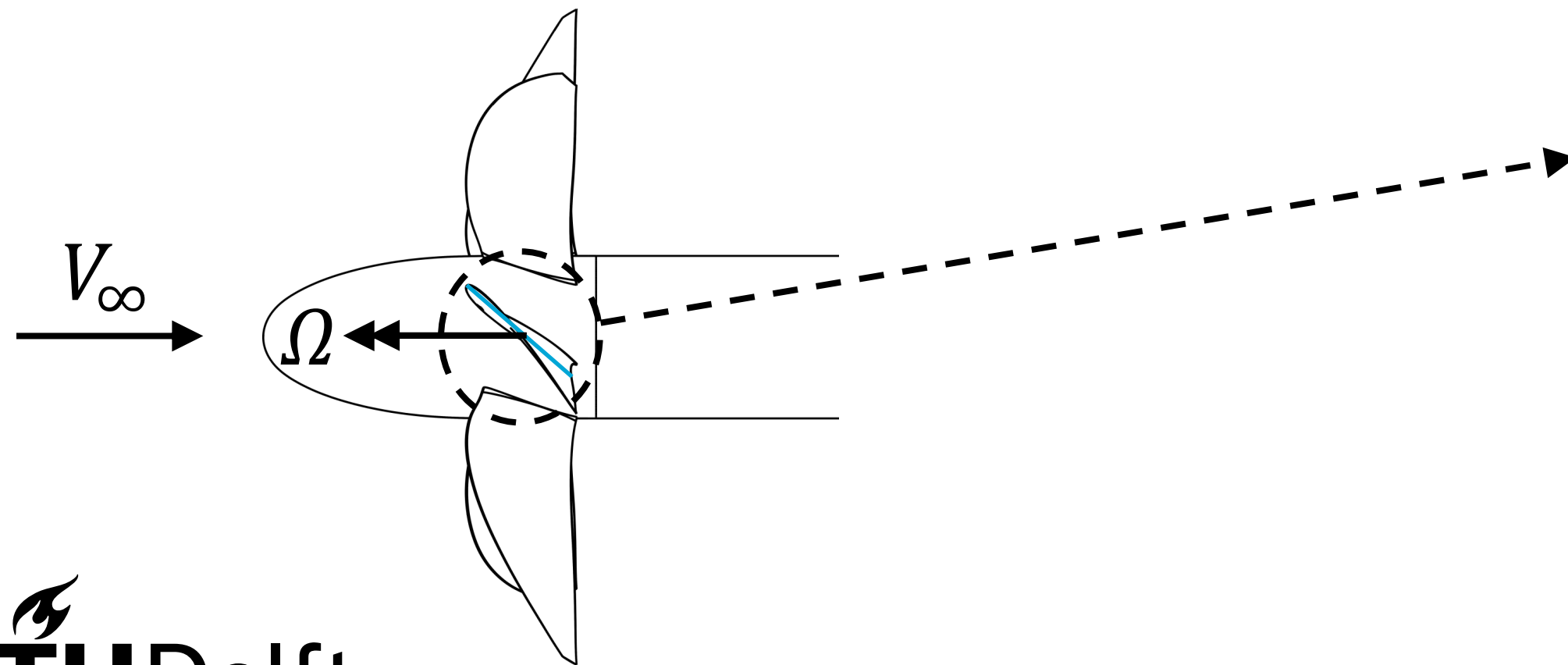
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How do propellers generate thrust?

- Propeller blades are like rotating wings

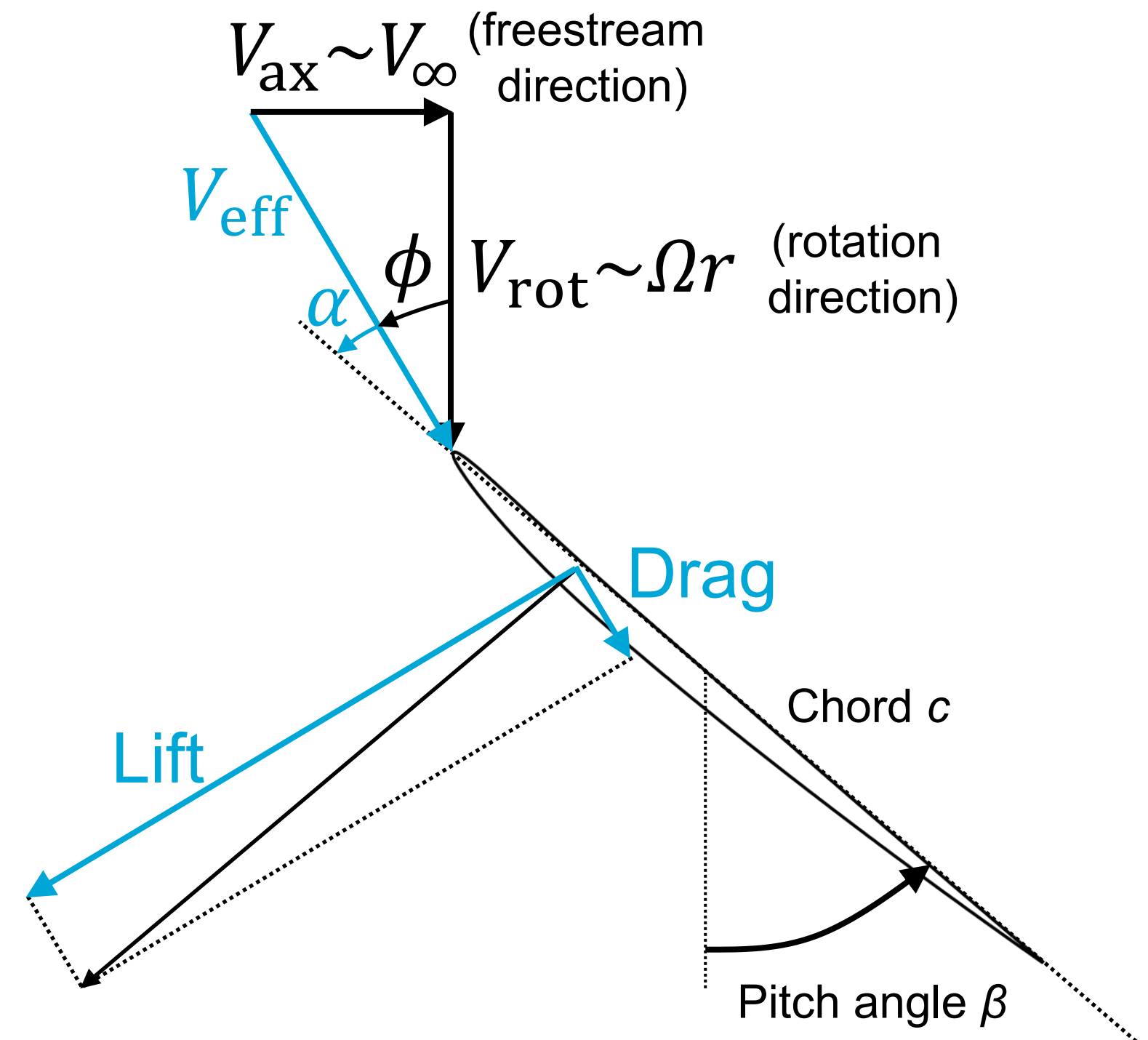
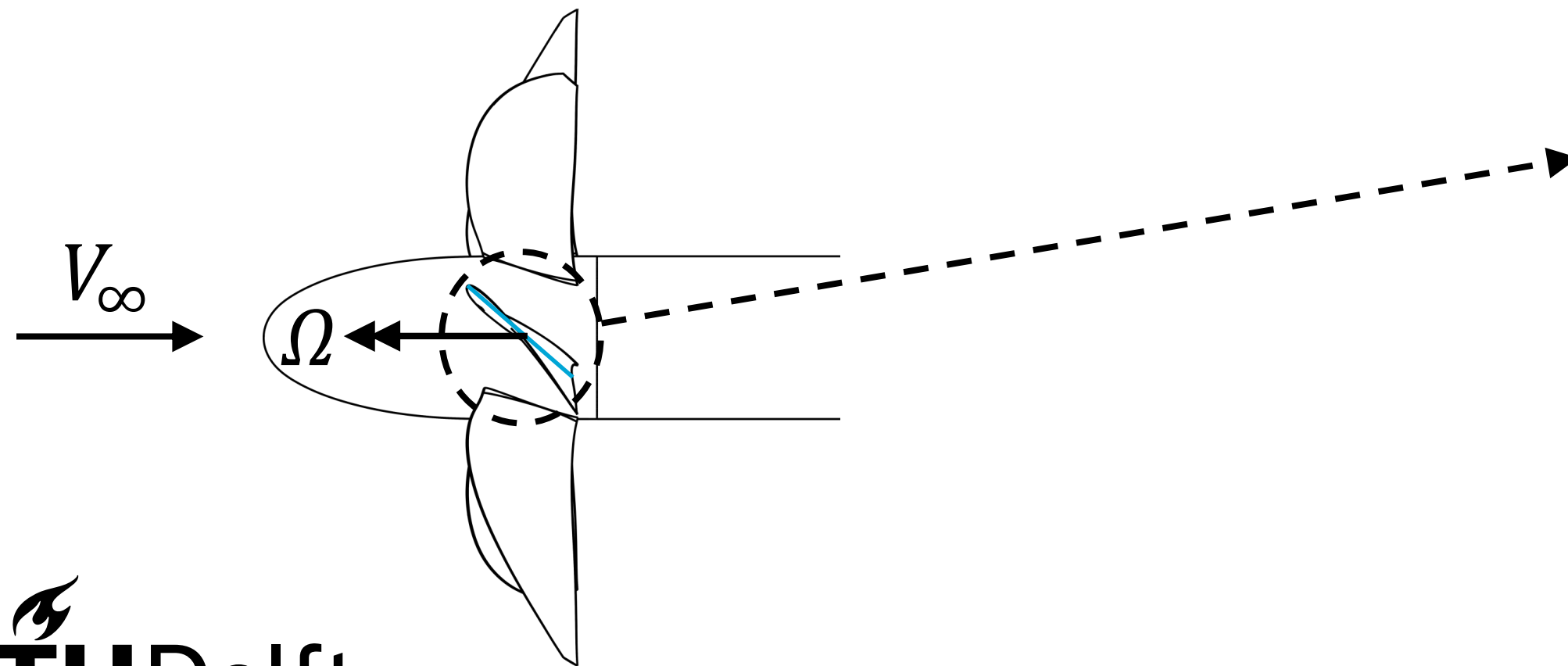
- Effective velocity $V_{\text{eff}} = \sqrt{V_{\text{ax}}^2 + V_{\text{rot}}^2}$
- Inflow angle $\phi = \tan^{-1} \left(\frac{V_{\text{ax}}}{V_{\text{rot}}} \right)$
- Angle of attack $\alpha = \beta - \phi$



How do propellers generate thrust?

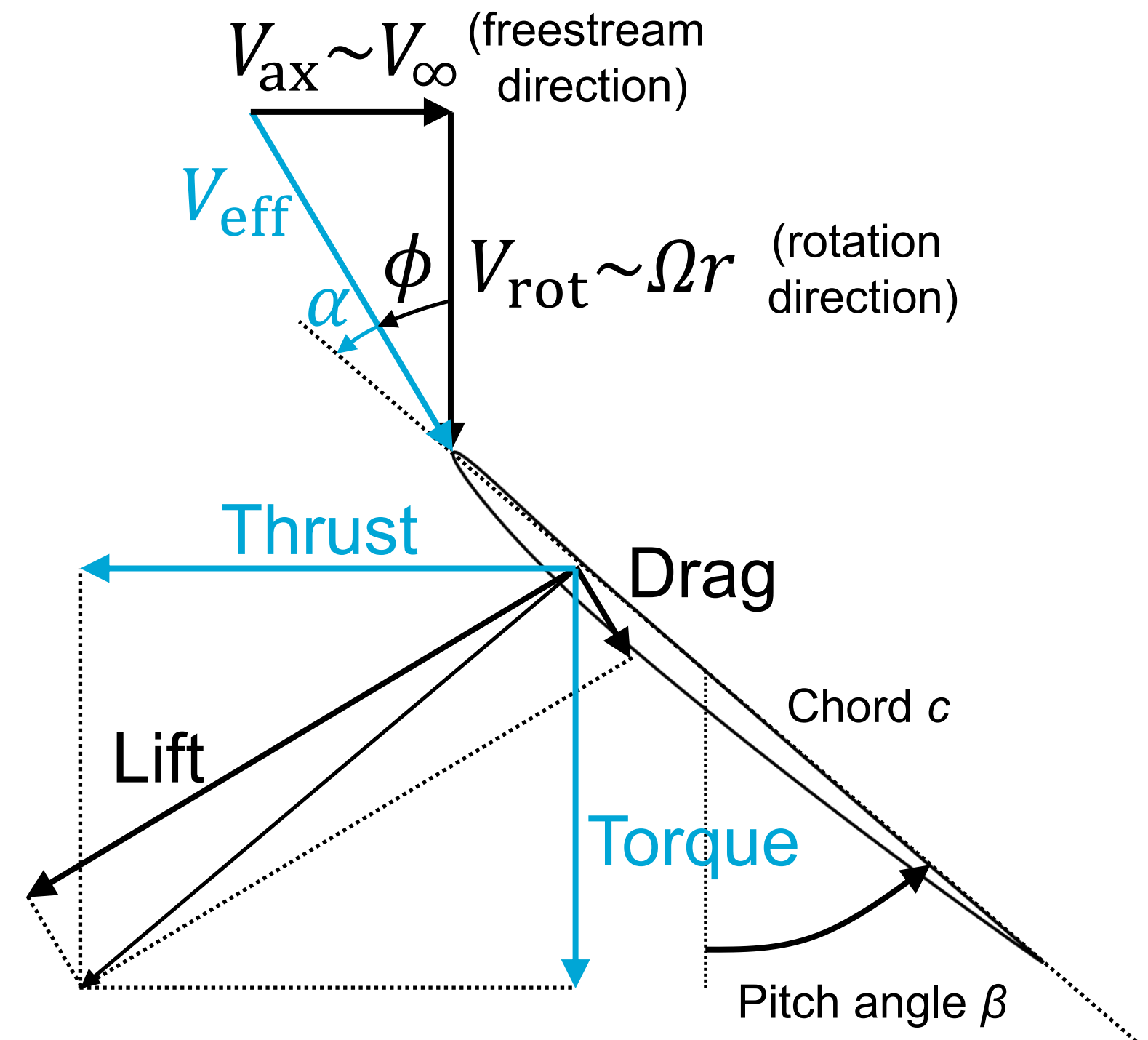
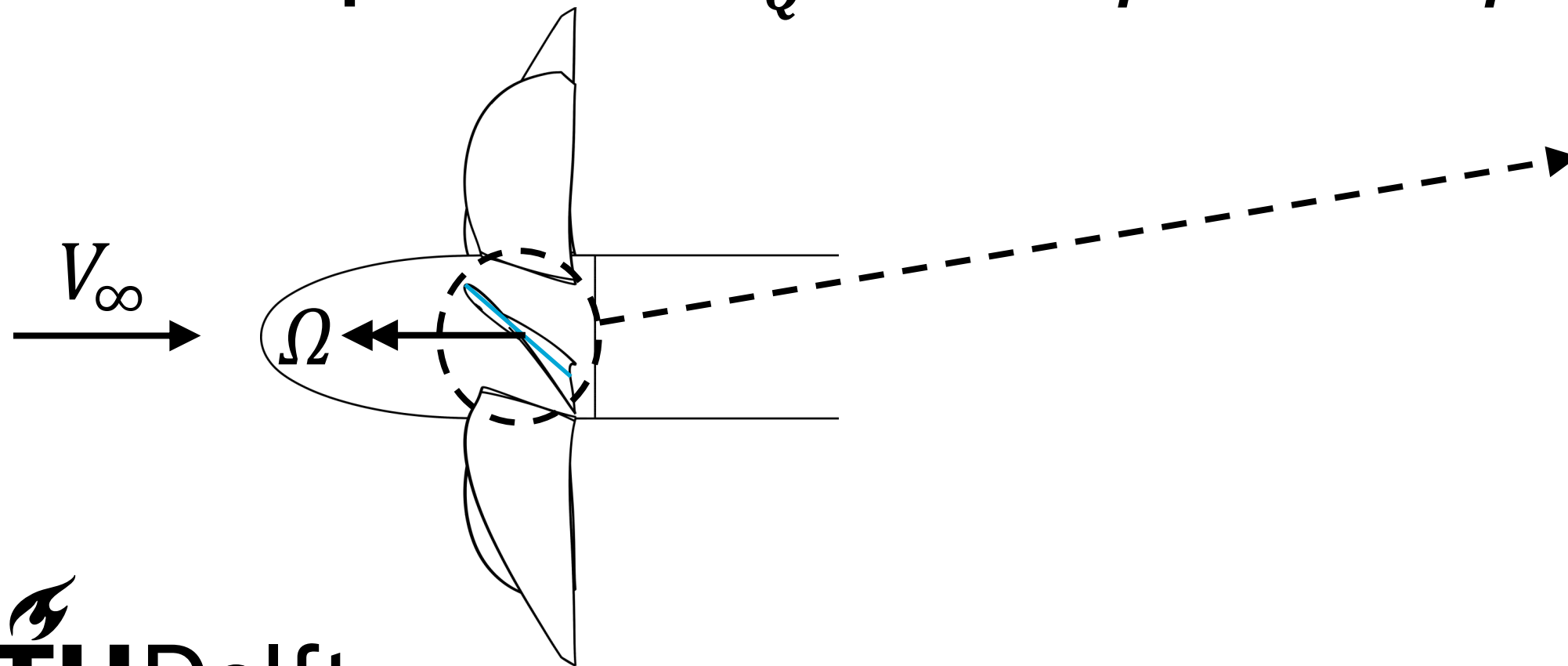
- Propeller blades are like rotating wings

- Lift force $L' = c_l(\alpha, Re, M) \frac{1}{2} \rho V_{\text{eff}}^2 c$
 - Drag force $D' = c_d(\alpha, Re, M) \frac{1}{2} \rho V_{\text{eff}}^2 c$
- (Per unit blade span)



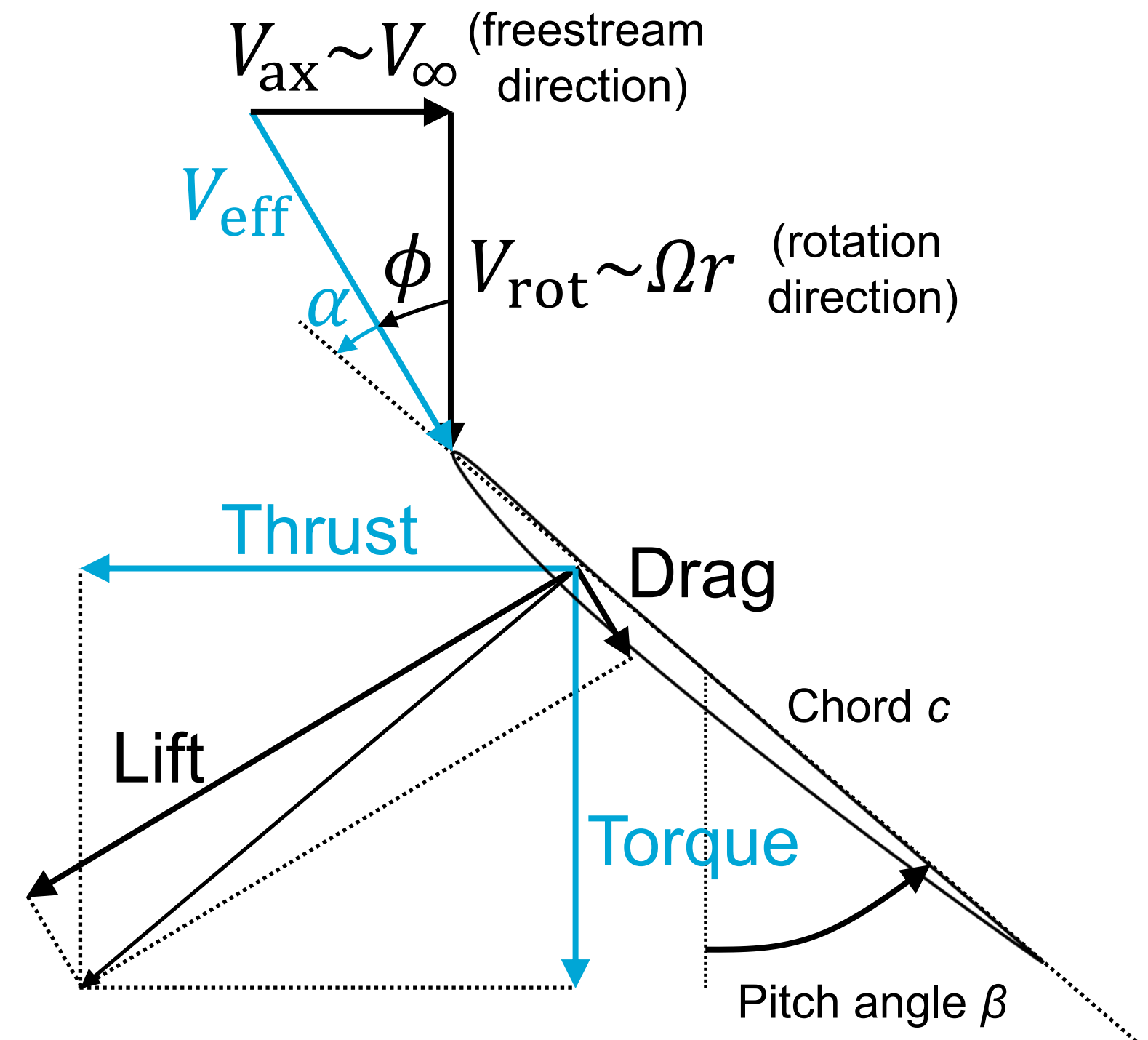
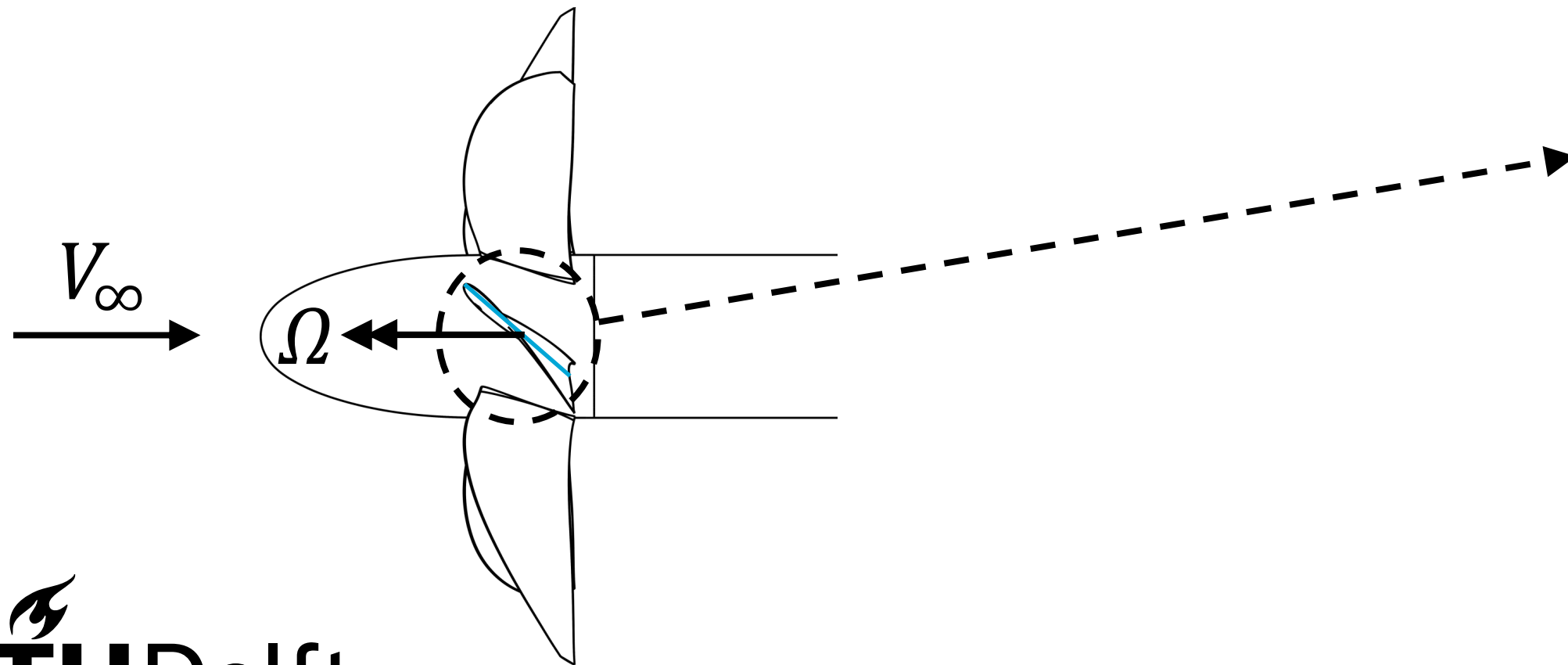
How do propellers generate thrust?

- Propeller blades are like rotating wings
 - Lift force $L' = c_l(\alpha, Re, M) \frac{1}{2} \rho V_{\text{eff}}^2 c$
 - Drag force $D' = c_d(\alpha, Re, M) \frac{1}{2} \rho V_{\text{eff}}^2 c$
 - Thrust force** $T' = L' \cos \phi - D' \sin \phi$
 - Torque force** $F'_Q = L' \sin \phi + D' \cos \phi$



Torque: the price of producing thrust

- Lift and drag forces cause torque about propeller shaft
 - Torque force $F'_Q = L' \sin \phi + D' \cos \phi$
 - **Torque** $Q' = F'_Q \cdot r$
 - Needs to be overcome by engine



Thrust vs Torque

	Thrust	Torque
<i>Contribution blade lift</i>	+	+
<i>Contribution blade drag</i>	-	+
<i>Dominant contribution at typical operational condition</i>	Lift	Lift (!)*

* Except in hover-like condition

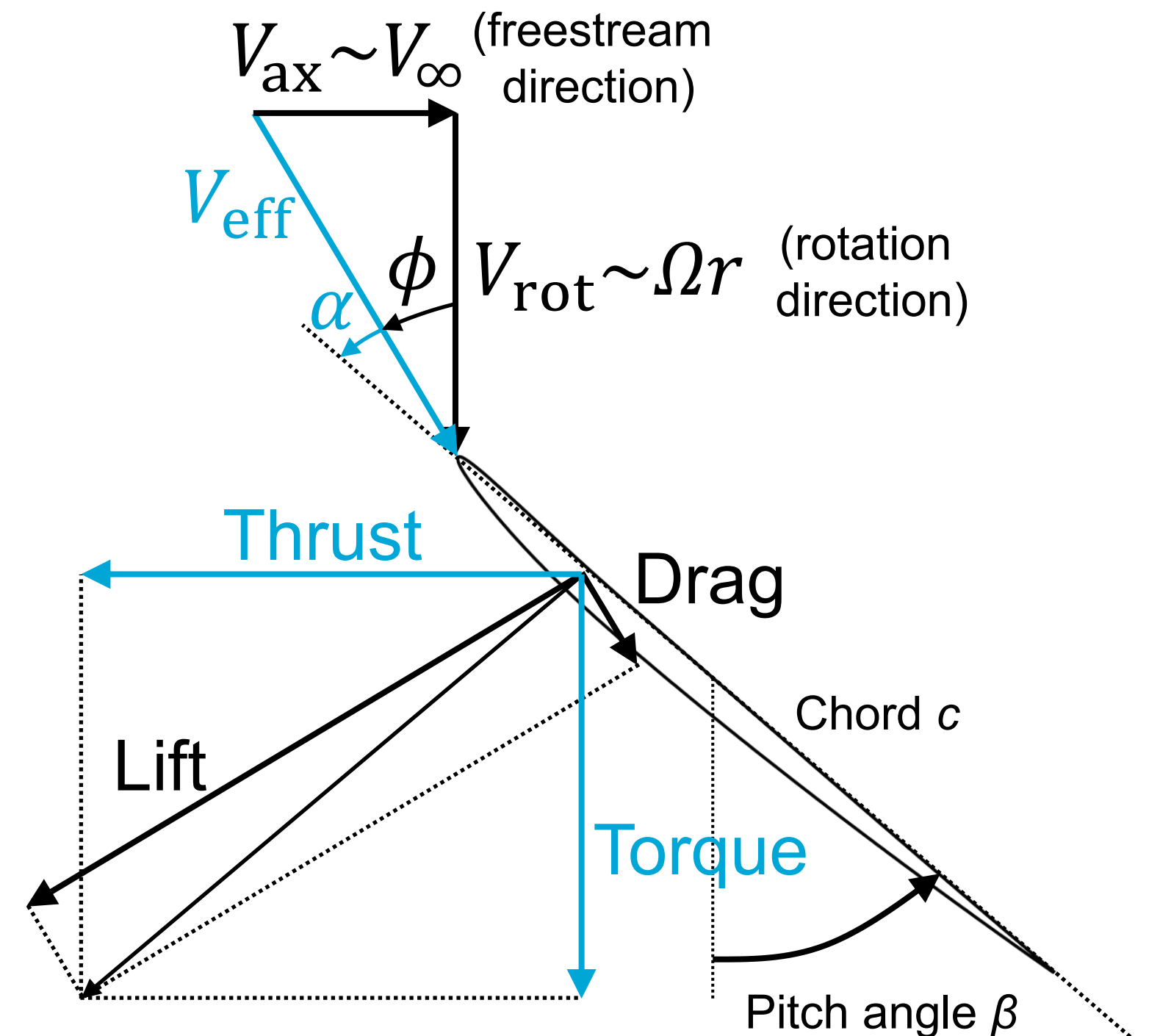
- $\frac{c_l}{c_d} \approx 100^{**}$
 - $\frac{\sin \phi}{\cos \phi} \approx 0.2 - 3^{***}$
- $\left. \begin{array}{l} \frac{c_l}{c_d} \approx 100^{**} \\ \frac{\sin \phi}{\cos \phi} \approx 0.2 - 3^{***} \end{array} \right\} \frac{L' \sin \phi}{D' \cos \phi} \gg 1 \rightarrow \underline{\text{torque is lift dominated}}$

** typical operational point without separation and/or wave drag

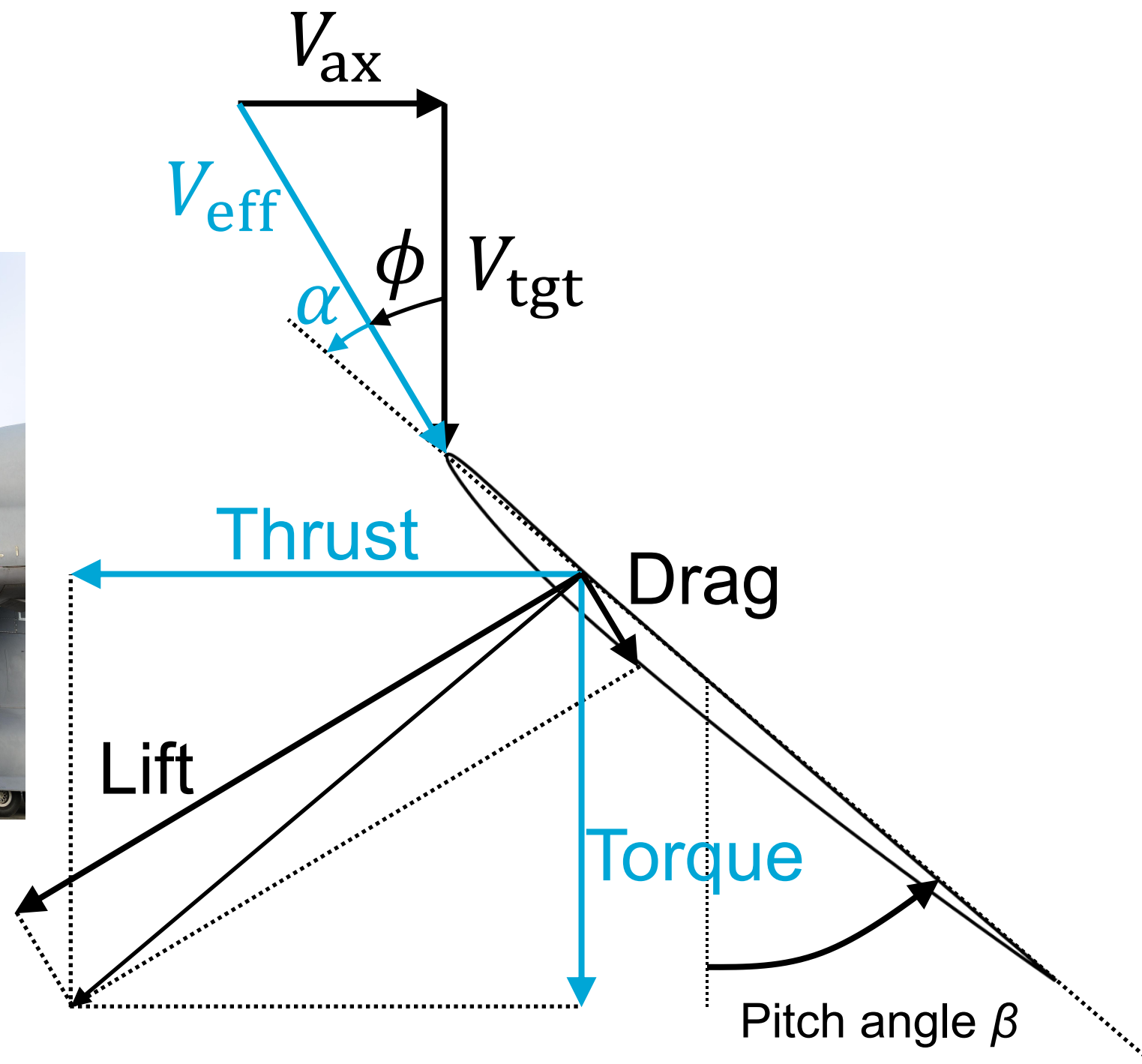
*** value depends on advance ratio and radial coordinate

Thrust $T' = L' \cos \phi - D' \sin \phi$

Torque $F'_Q = L' \sin \phi + D' \cos \phi$



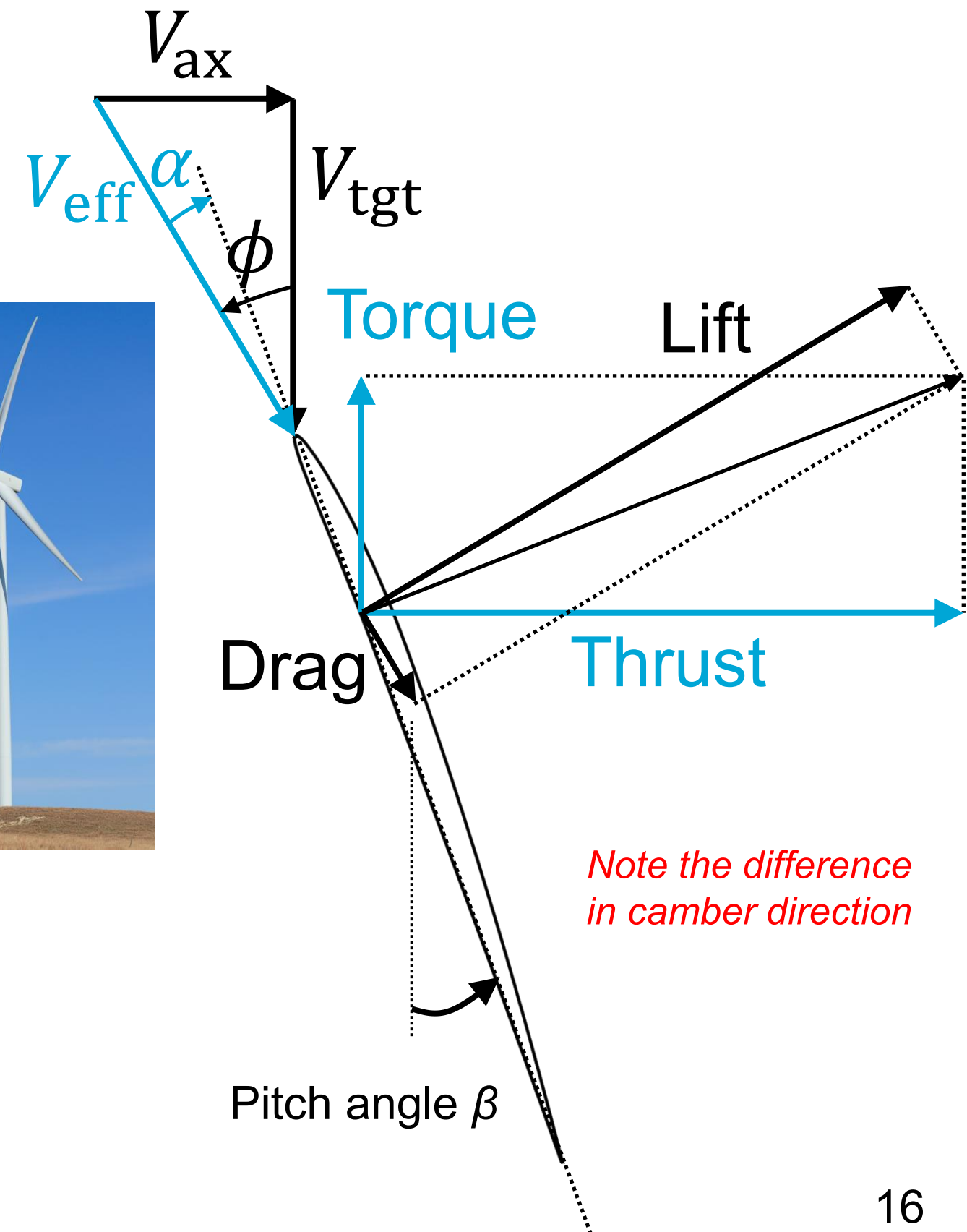
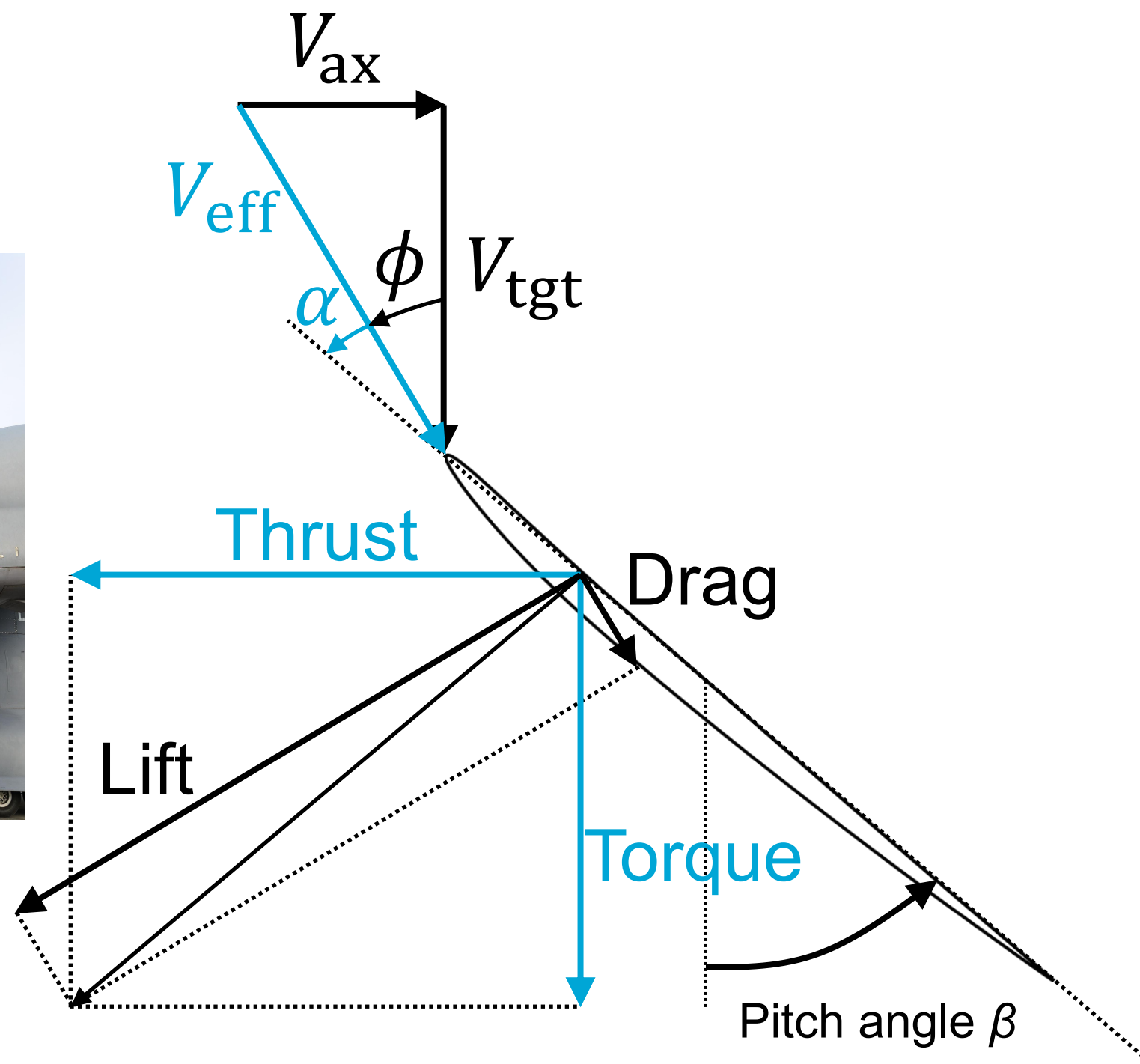
Propellers vs wind turbines



Differences?

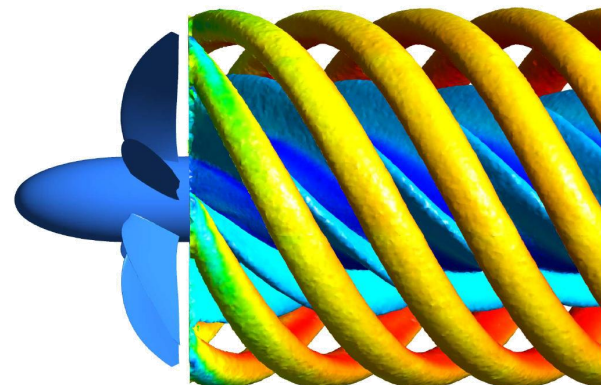
- *Blade camber*
- *Tip-speed ratio*
 $TSR = \lambda = \Omega R / V_{\infty}$
- *Mach number*

Propellers vs wind turbines

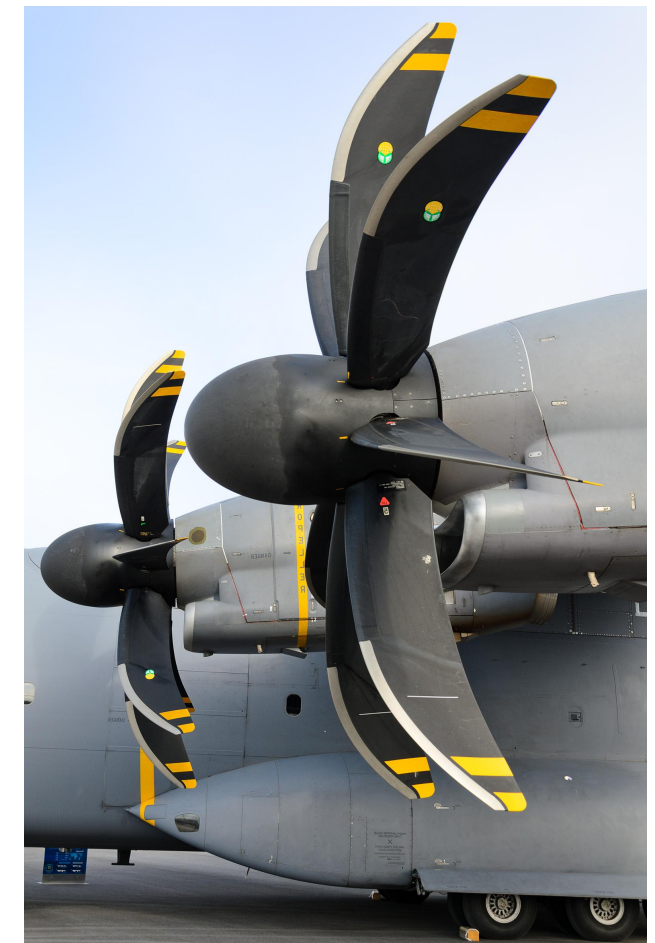
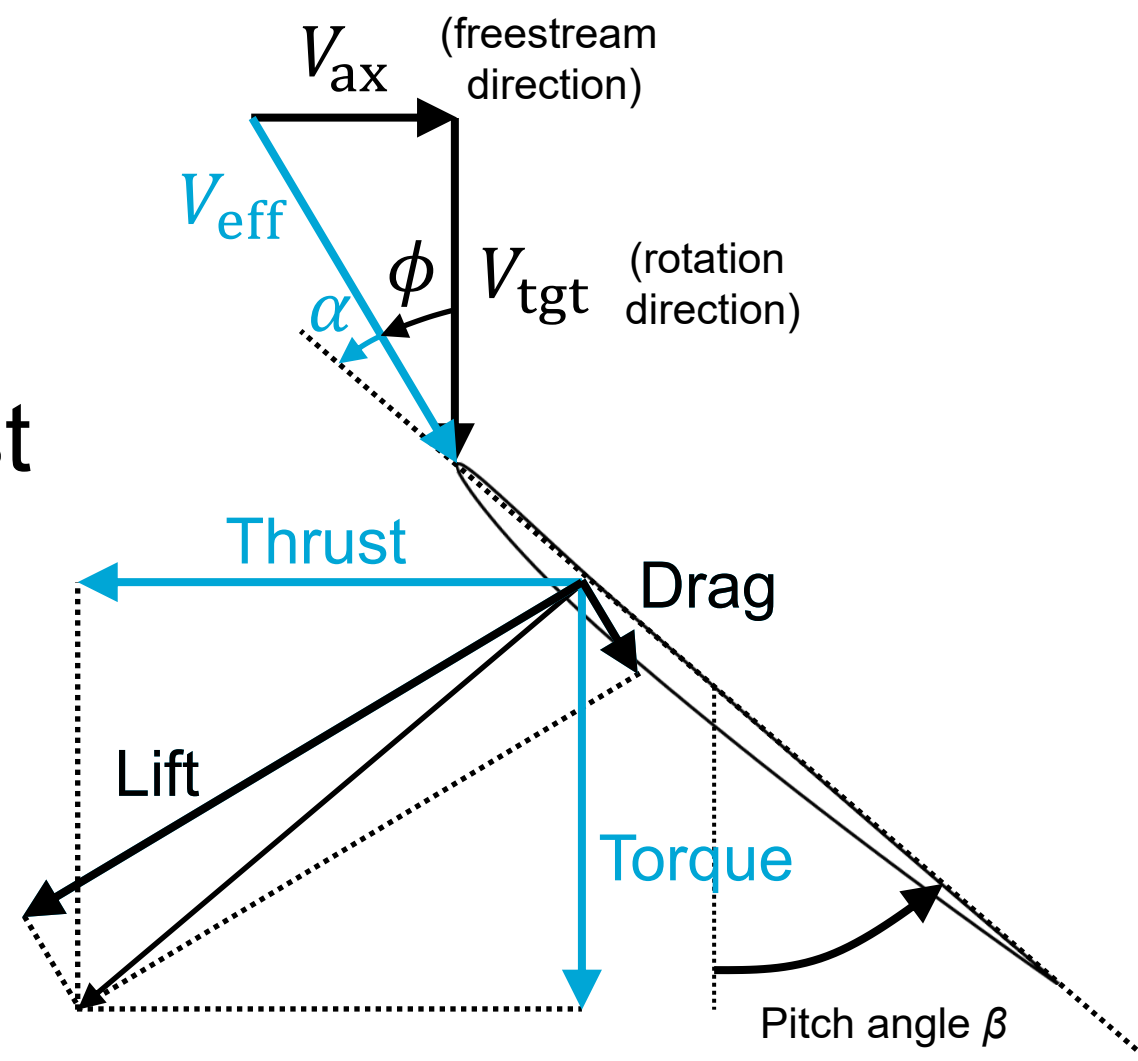


Propellers vs wind turbines

- Propellers typically operated at low tip-speed ratio vs wind turbines
 - $TSR = \lambda = \Omega R / V_\infty = \pi / J \rightarrow$ “Advance ratio”
- Lower tip-speed ratio means
 - Higher inflow angle $\phi \rightarrow$ higher pitch angle β
 - High input shaft torque w.r.t. thrust
 - More ‘stretched’ wake

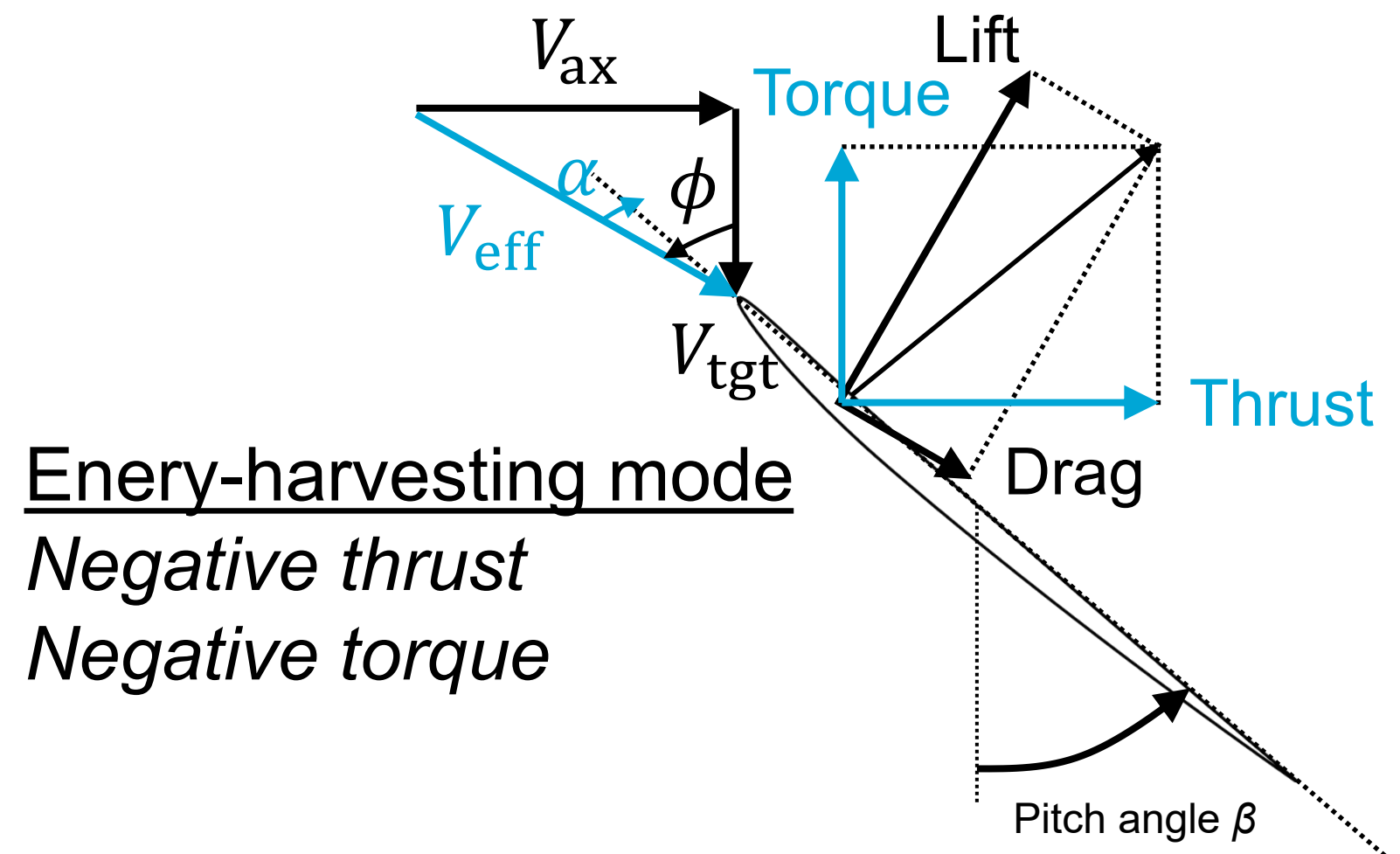
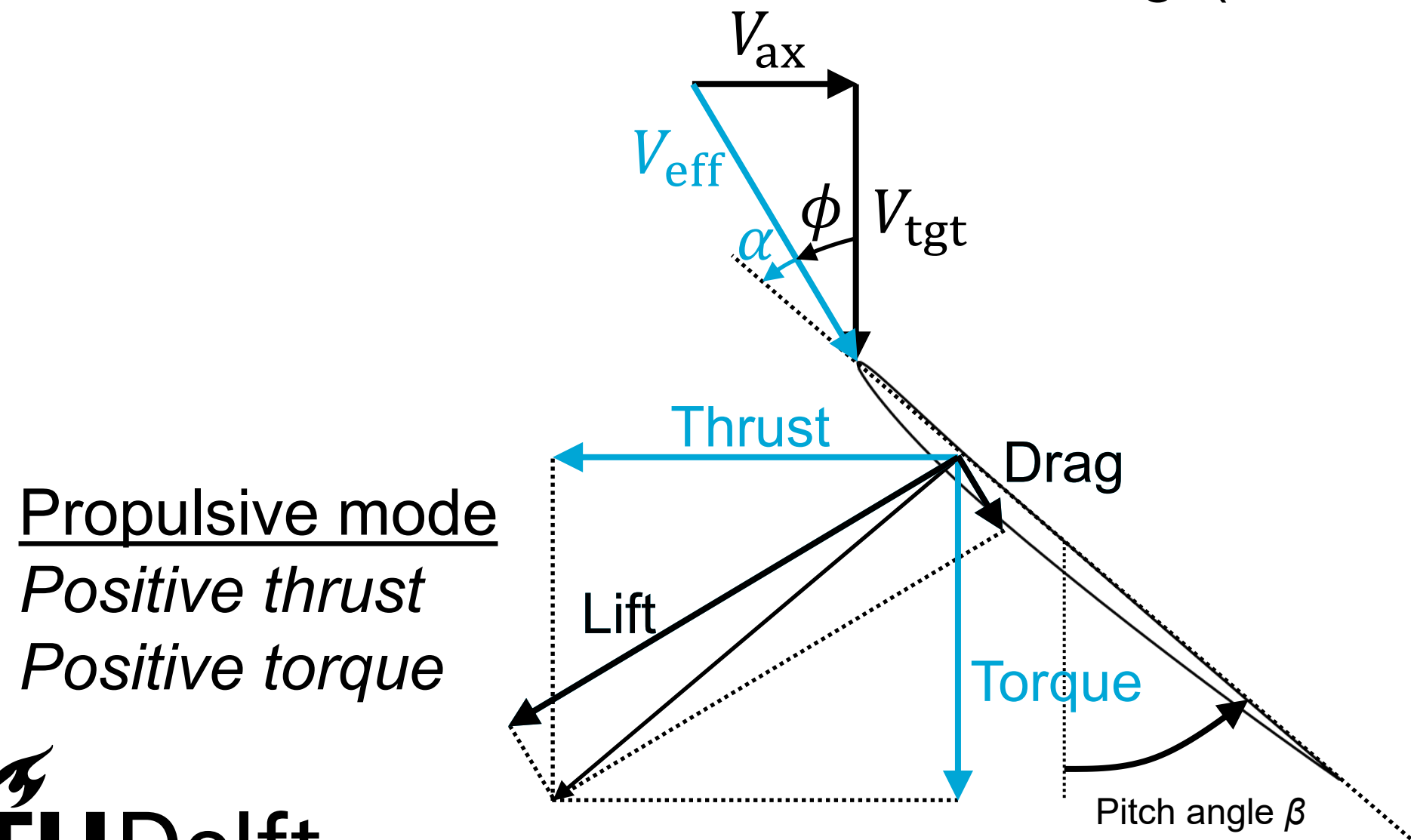


Credit: T.C.A. Stokkermans



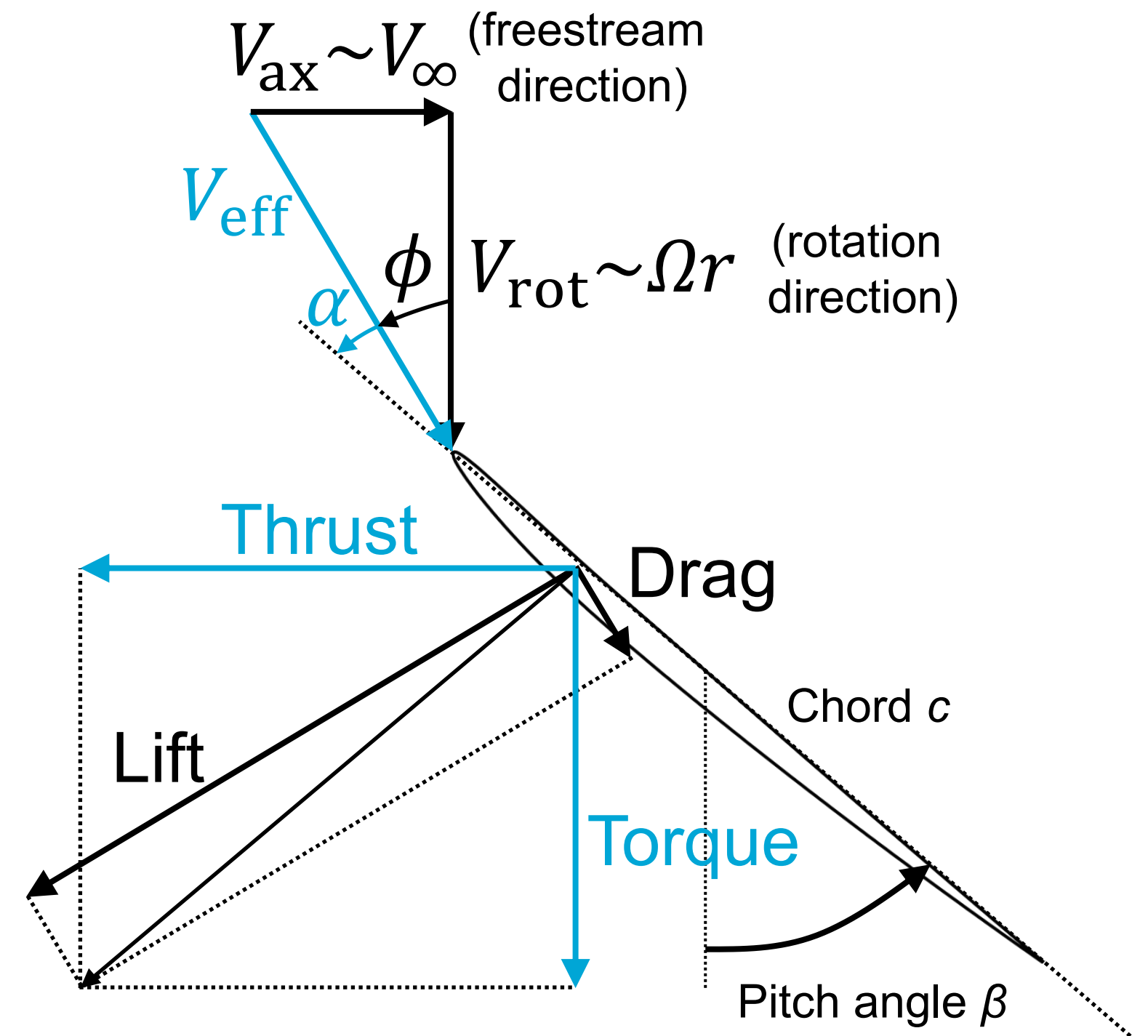
Propellers = wind turbines ?!

- Propeller can be used as energy-harvesting airbrake (electric motor required)
 - Same rotation direction as propulsive mode, but lower tip-speed ratio or pitch
 - Low efficiency vs wind turbines due to negative AoA with positive camber
 - **Not** the same as 'free windmilling' (active braking required)



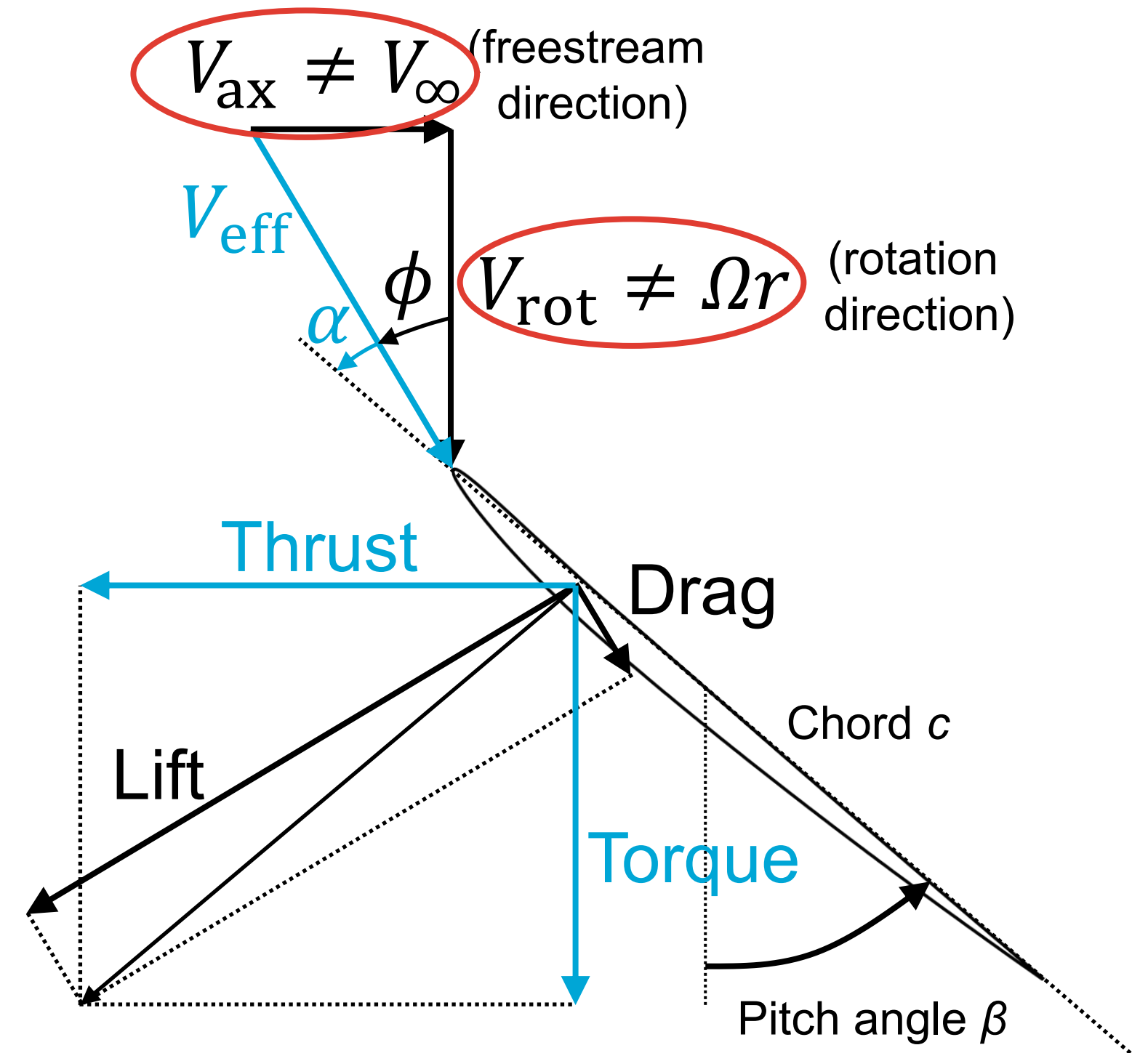
From sectional (2D) to blade (3D) loading

- Thrust $T = N_B \int_{\text{hub}}^{\text{tip}} T' dr$
- Torque $Q = N_B \int_{\text{hub}}^{\text{tip}} F'_Q r dr$
- **BUT: the 2D analysis is incomplete**
 - Loading goes to zero at the blade tip
 - Radial flow
 - ...



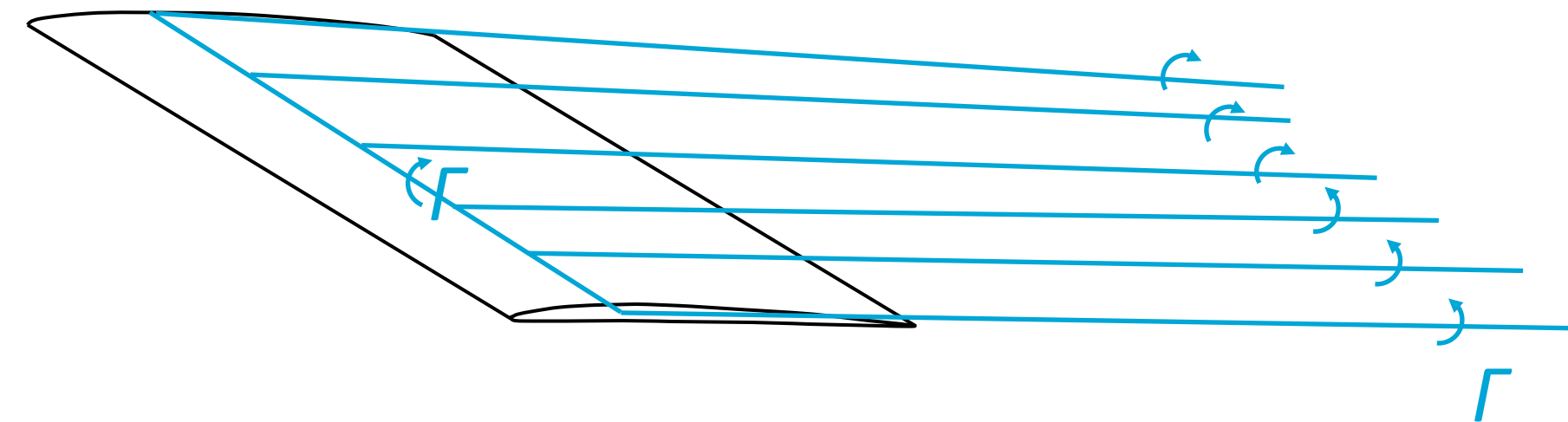
From sectional (2D) to blade (3D) loading

- Thrust $T = N_B \int_{\text{hub}}^{\text{tip}} T' dr$
- Torque $Q = N_B \int_{\text{hub}}^{\text{tip}} F'_Q r dr$
- **BUT: the 2D analysis is incomplete**
 - Loading goes to zero at the blade tip
 - Radial flow
 - **Self-induction changes effective inflow velocity components**

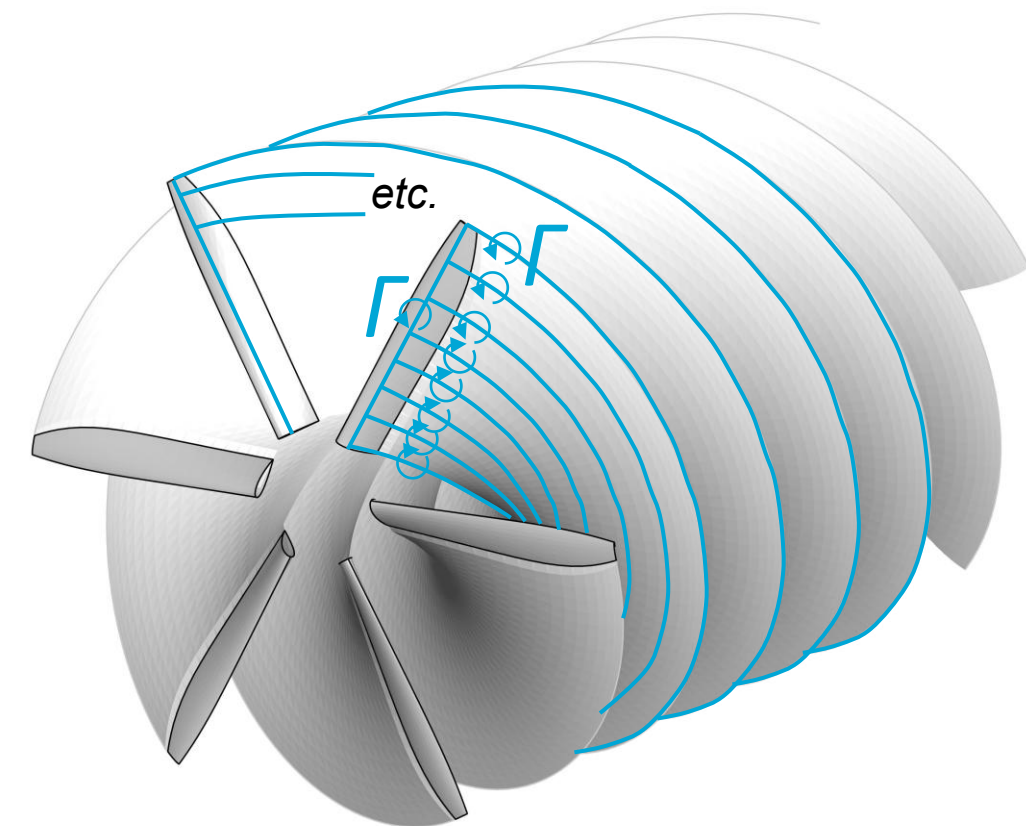


Induced velocities

- Wing
 - Downwash at wing due to trailing vortex sheet (tip vortices)
 - Reduction in 2D angle of attack
 - Reduction in lift, generation of induced drag



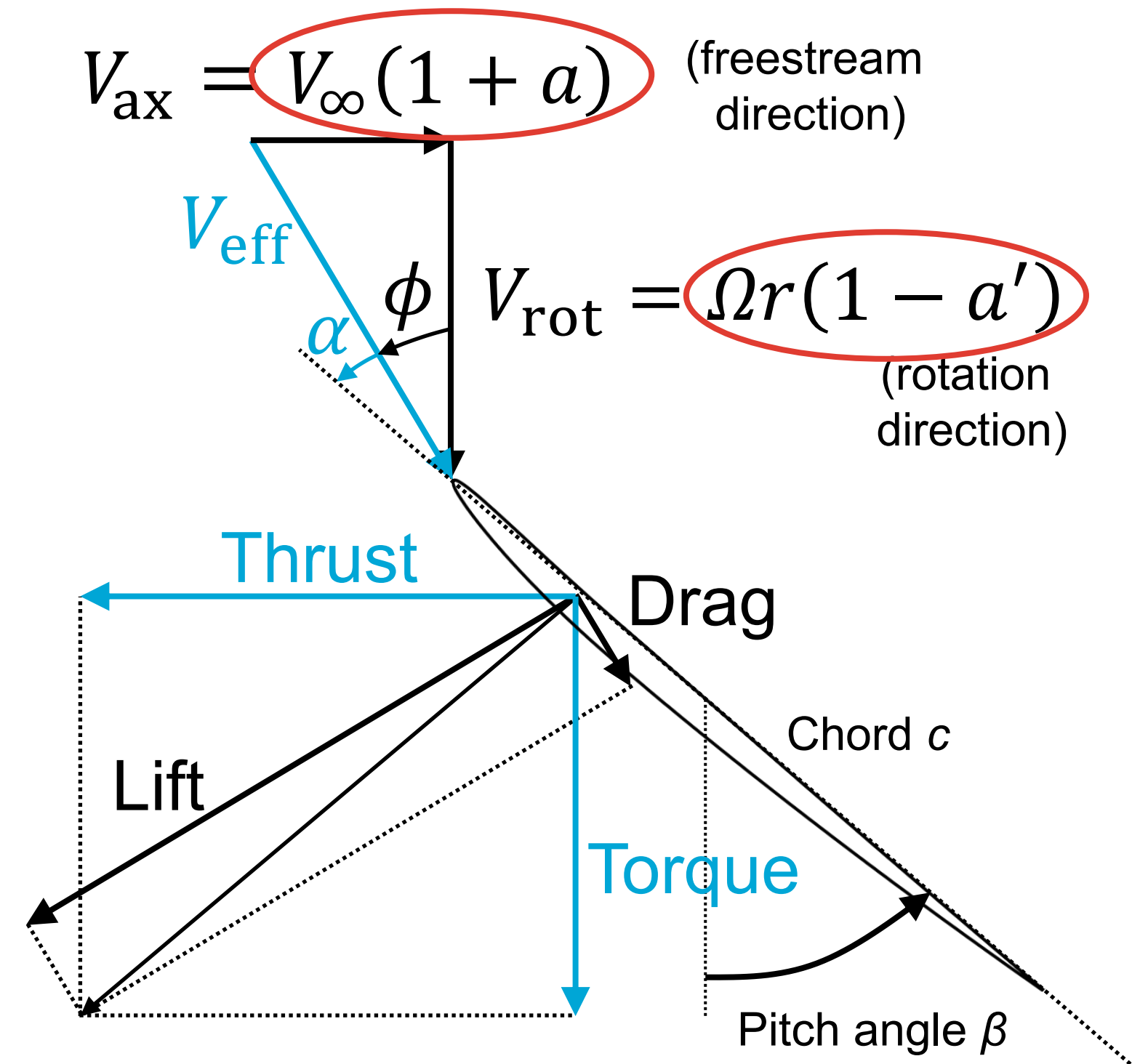
- Propeller
 - Induced velocities at blade by vortex sheet behind propeller
 - Accelerated inflow due to thrust
 - Swirl due to torque



Credit: Gabriel Margalida

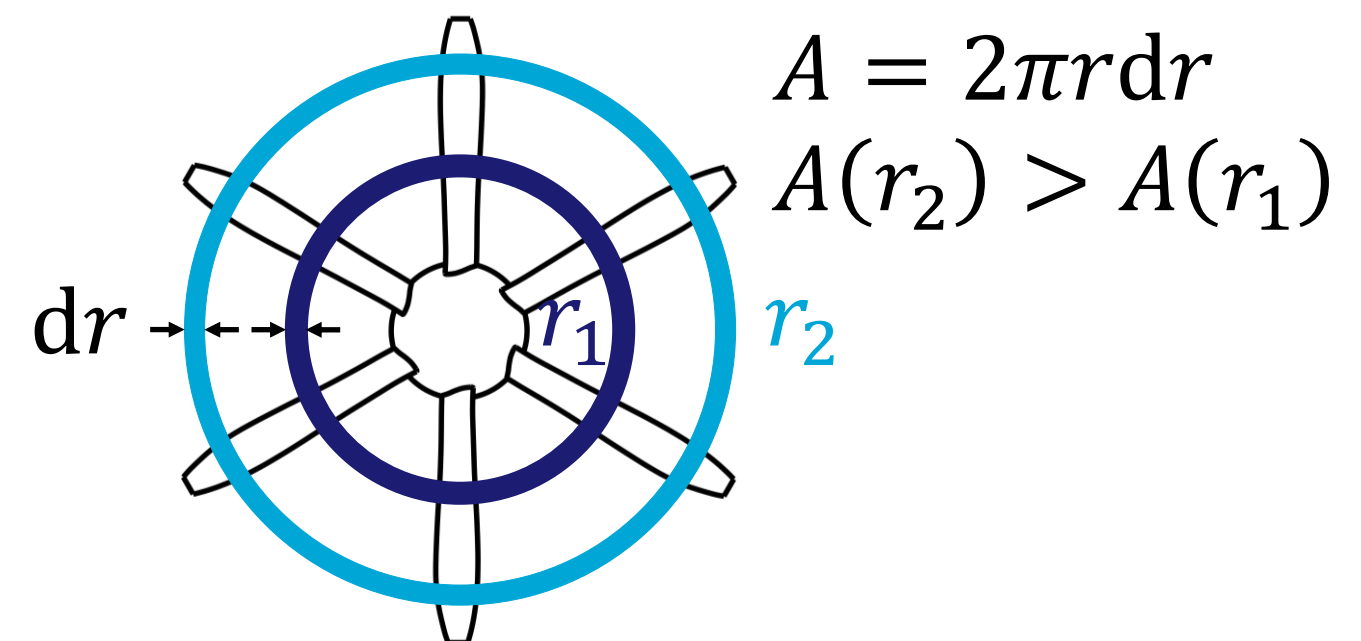
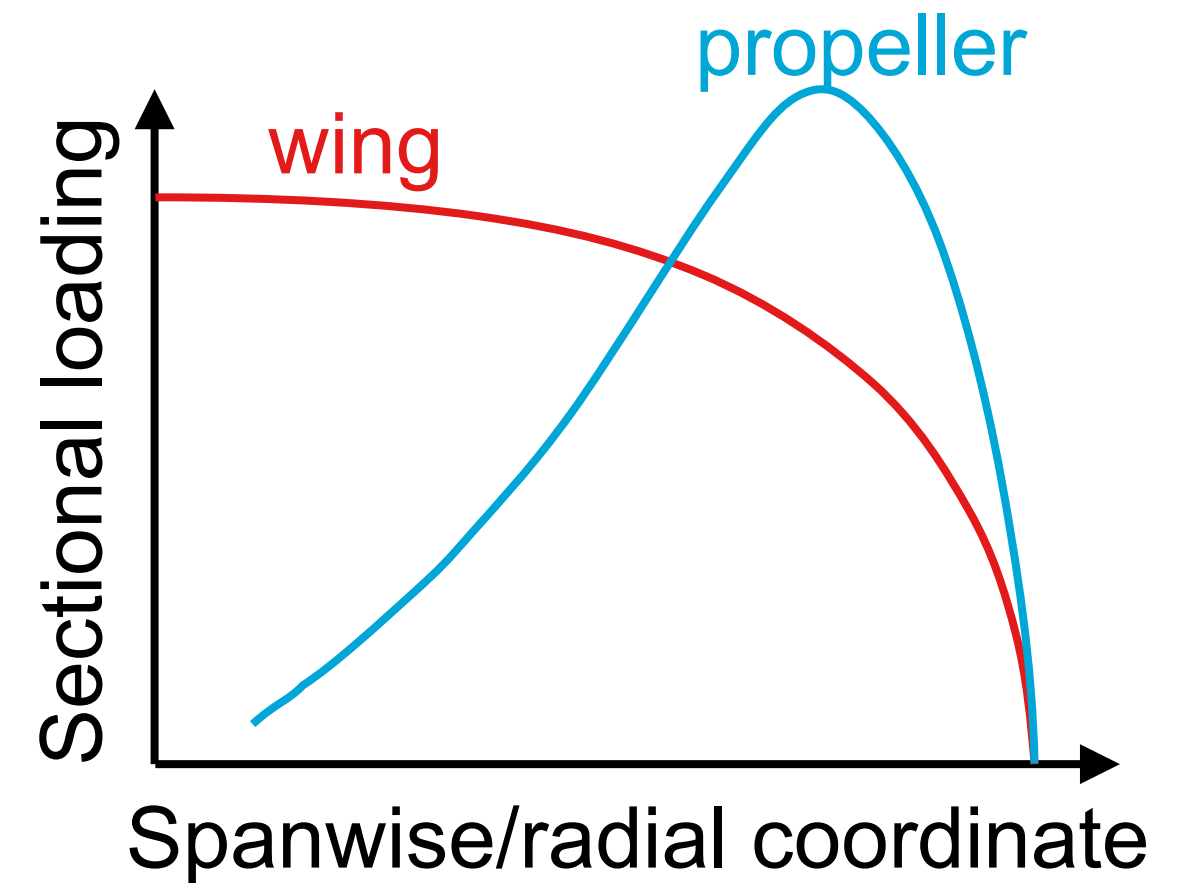
Induced velocities

- Inflow velocities due to wake induction
 - $V_{ax} = V_{\infty} + v_{ax} = V_{\infty}(1 + a)$ *Opposite sign convention vs wind turbines*
 - $V_{rot} = \Omega r - v_{rot} = \Omega r(1 - a')$ *Opposite sign convention vs wind turbines*
- Induction factors
 - $a = \frac{v_{ax}}{V_{\infty}} \rightarrow$ scales with thrust
 - $a' = \frac{v_{rot}}{\Omega r} \rightarrow$ scales with torque
- Propeller blades experience ...
 - Increased axial velocity component
 - Reduced rotational velocity component



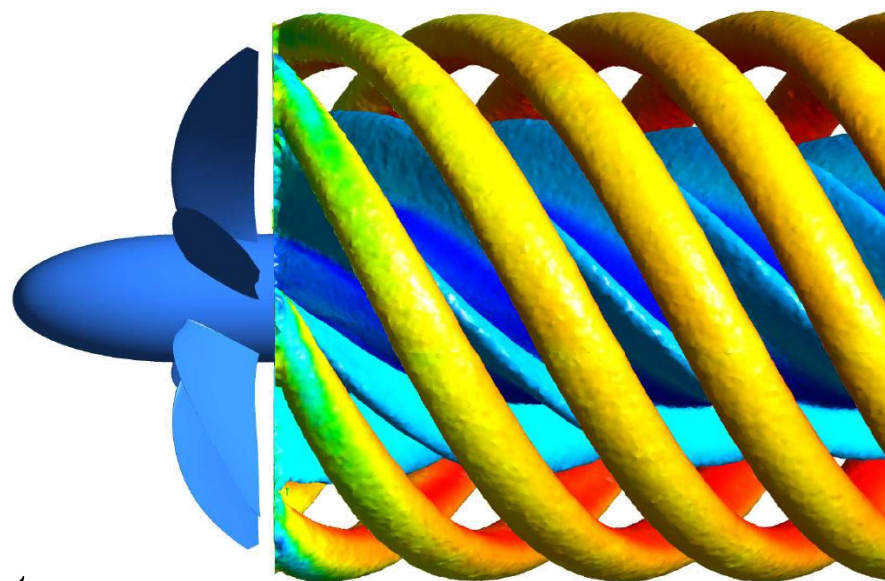
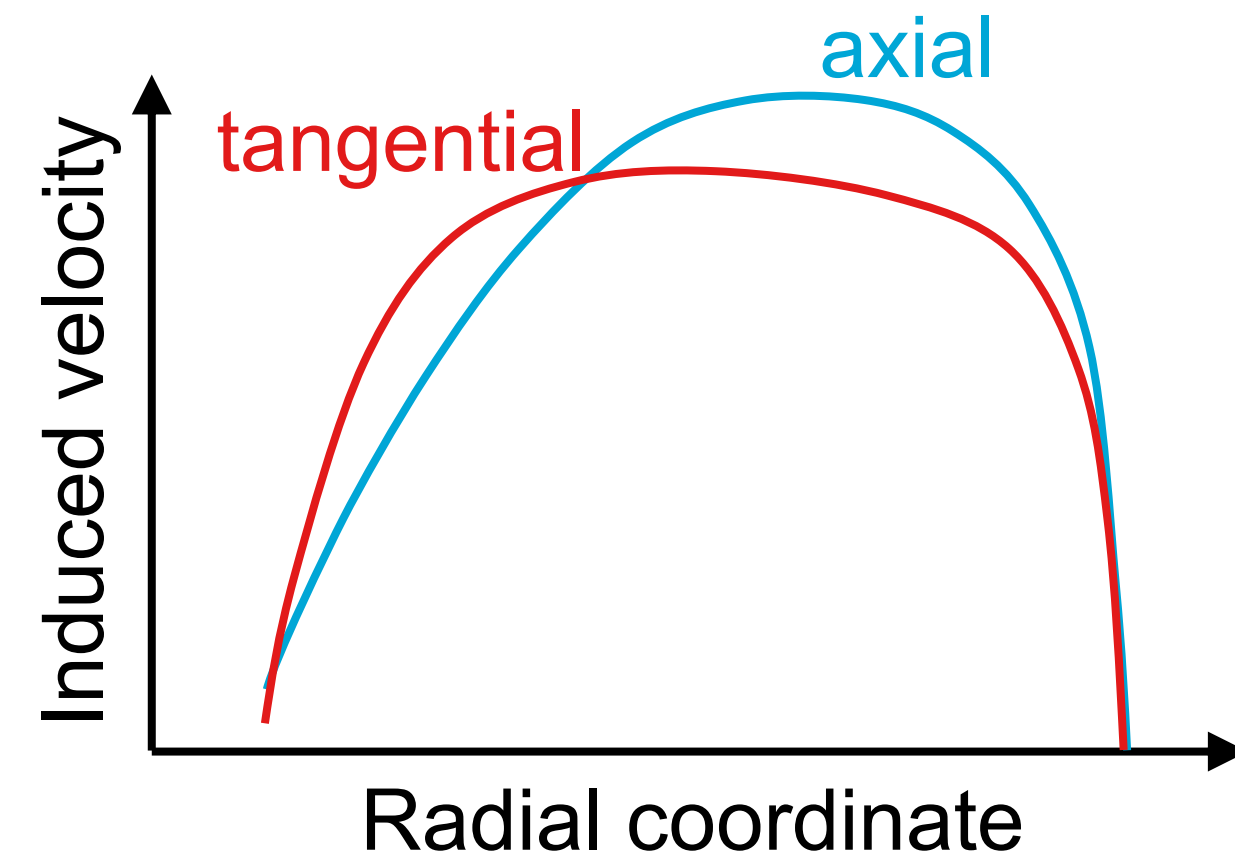
From sectional (2D) to blade (3D) loading

- Radial variation in inflow velocity due to
 - Geometry (rotational velocity component)
 - Radial variation in induction
- Combination with tip effects leads to outboard maximum in loading distribution
- Disk loading distribution differs from blade loading distribution due to radial variation of swept areas

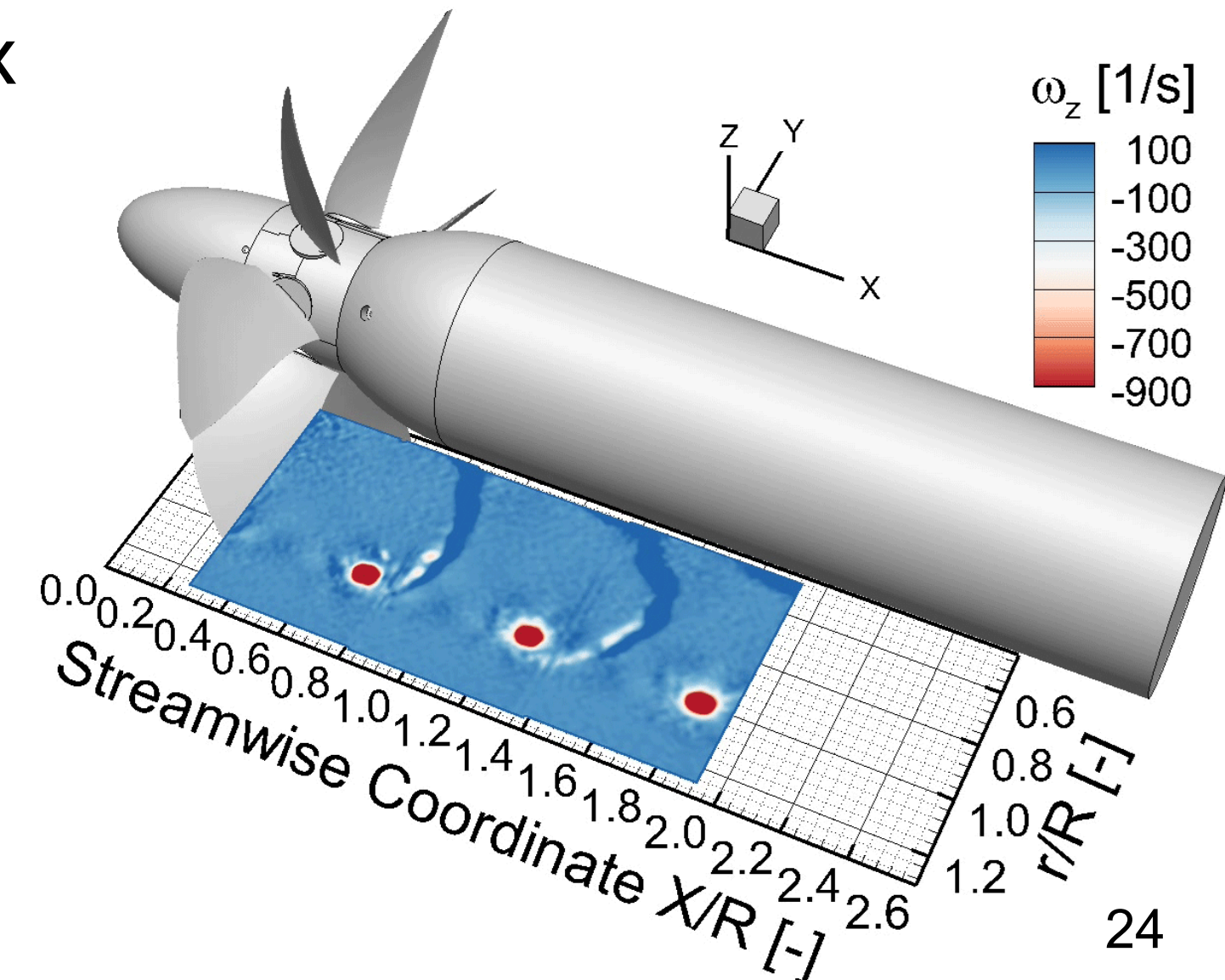
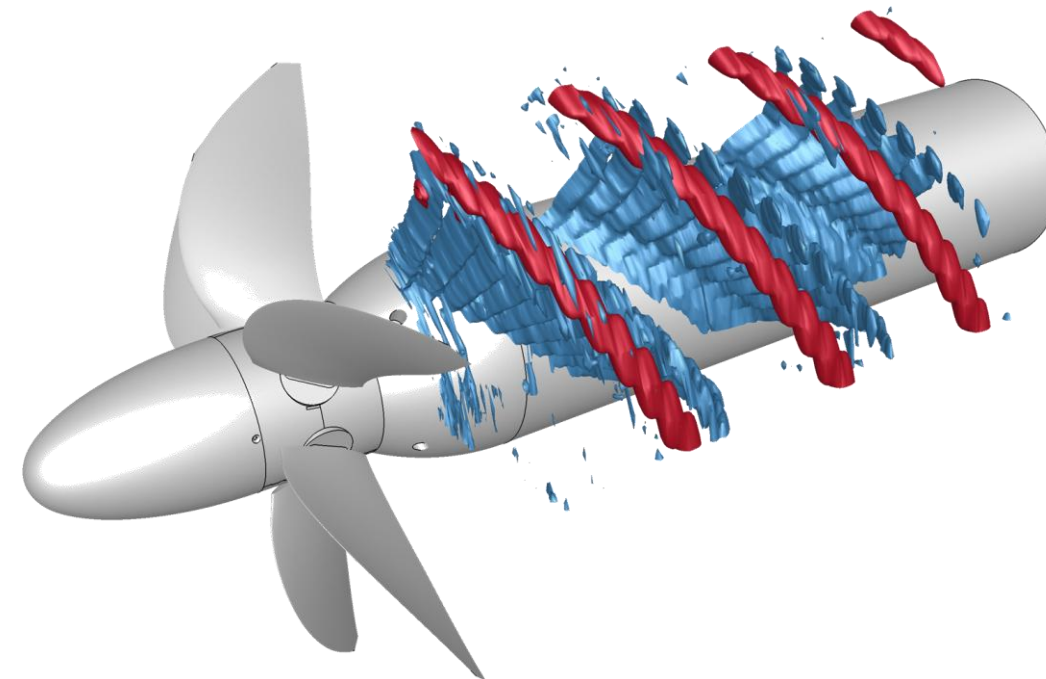


The propeller slipstream

- Thrust causes axial induced velocities
- Torque causes tangential induced velocities (swirl)
- Induced velocities vary in radial direction
 - Radial distribution like loading distribution
- Each blade generates a wake, tip vortex, root vortex
 - Slipstream flowfield is time-dependent!



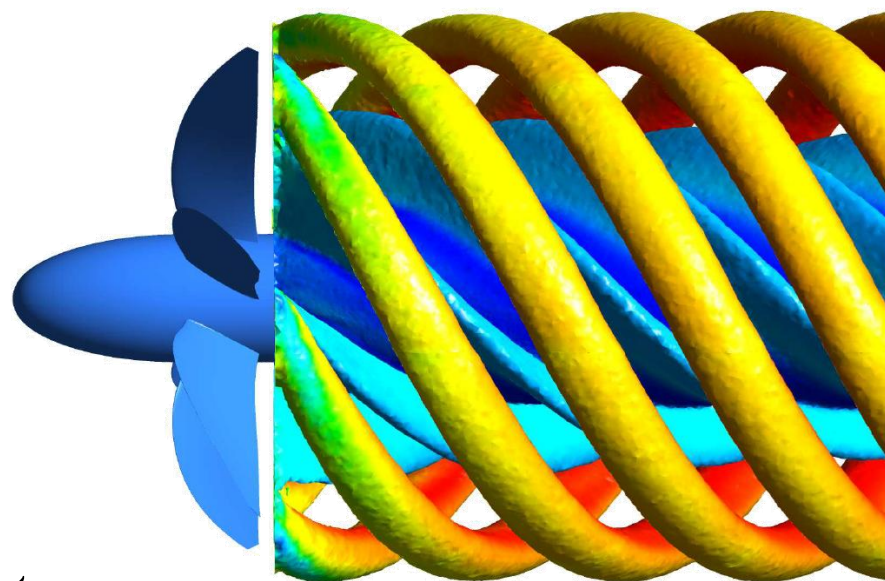
Credit: T.C.A. Stokkermans



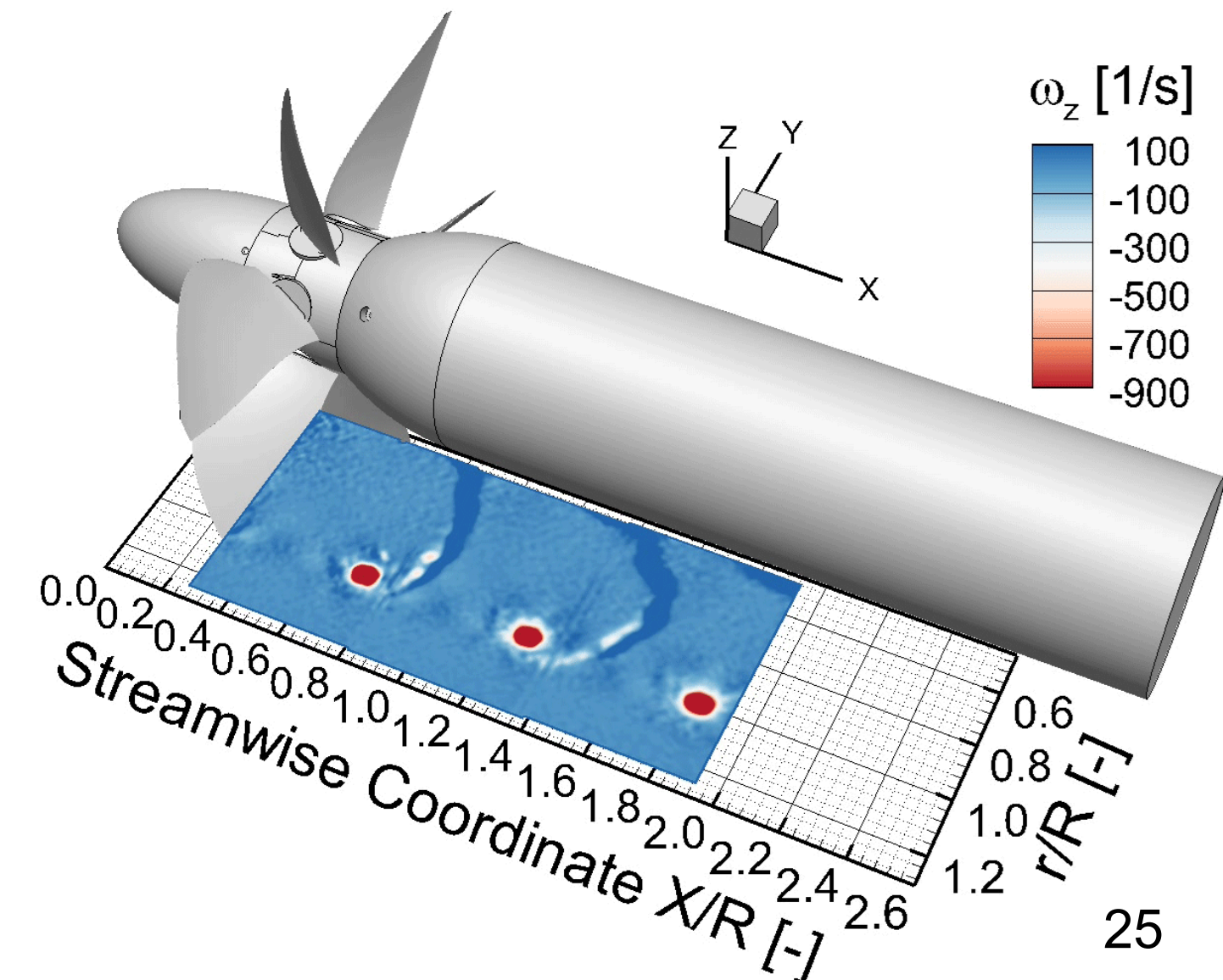
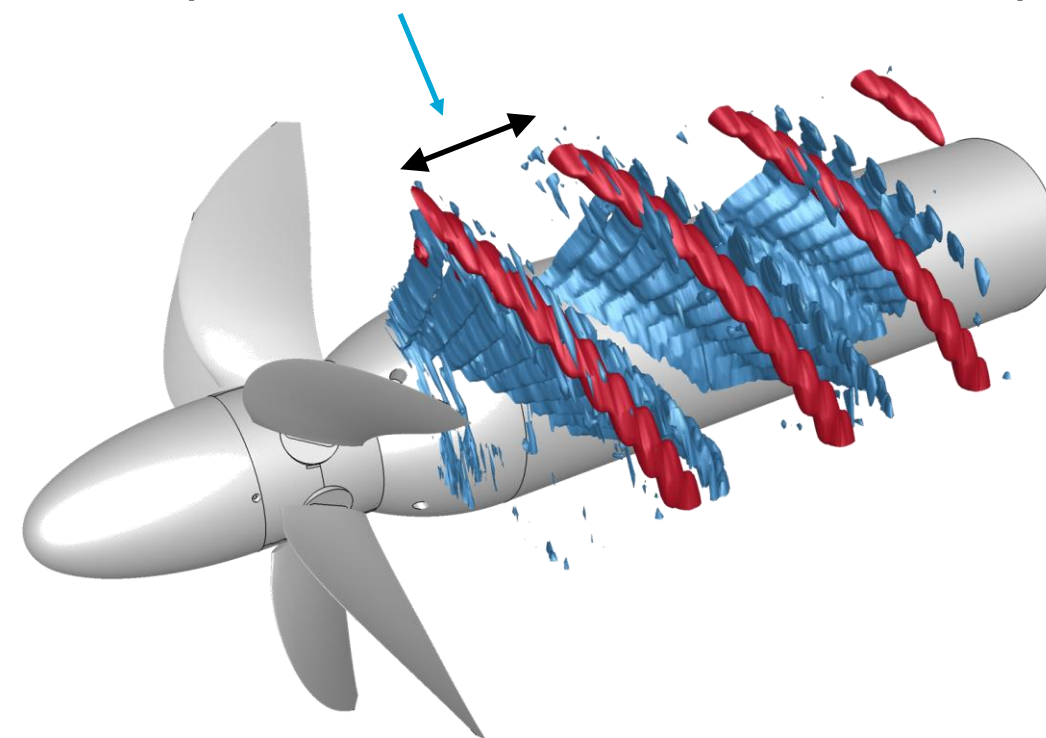
The propeller slipstream

- Spacing between tip vortices directly related to advance ratio ($\sim 1/\text{tip-speed ratio}$)
 - Propeller slipstream much more stretched out than wind-turbine wake

Distance between subsequent tip vortices scales with advance ratio (inverse of tip-speed ratio)

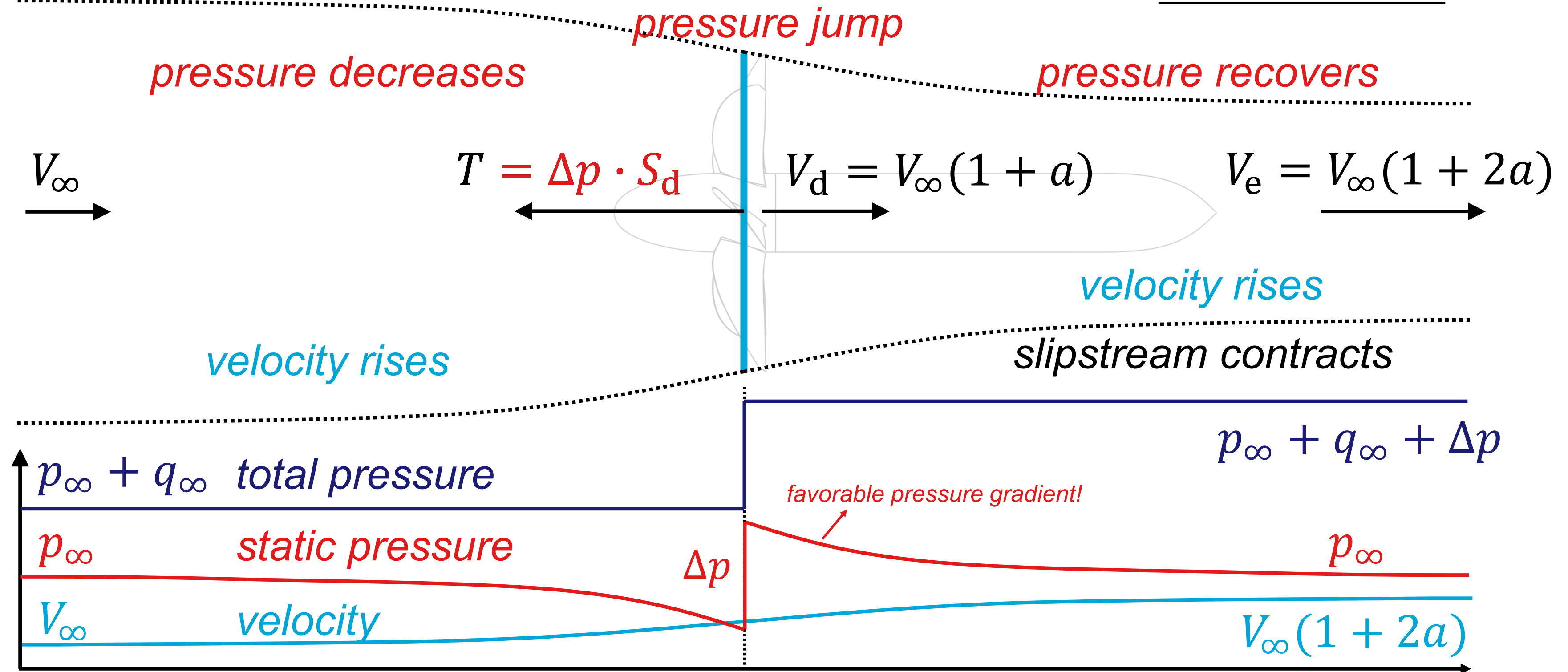


Credit: T.C.A. Stokkermans



What happens to the flowfield?

Everything opposite to the wind turbine case !





Propeller performance

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Key propeller performance indicators

- Forces/moments

- Thrust force
- Torque
- Shaft power

$$\begin{aligned} T \\ Q \\ P (= Q\Omega) \end{aligned}$$

- Operating conditions

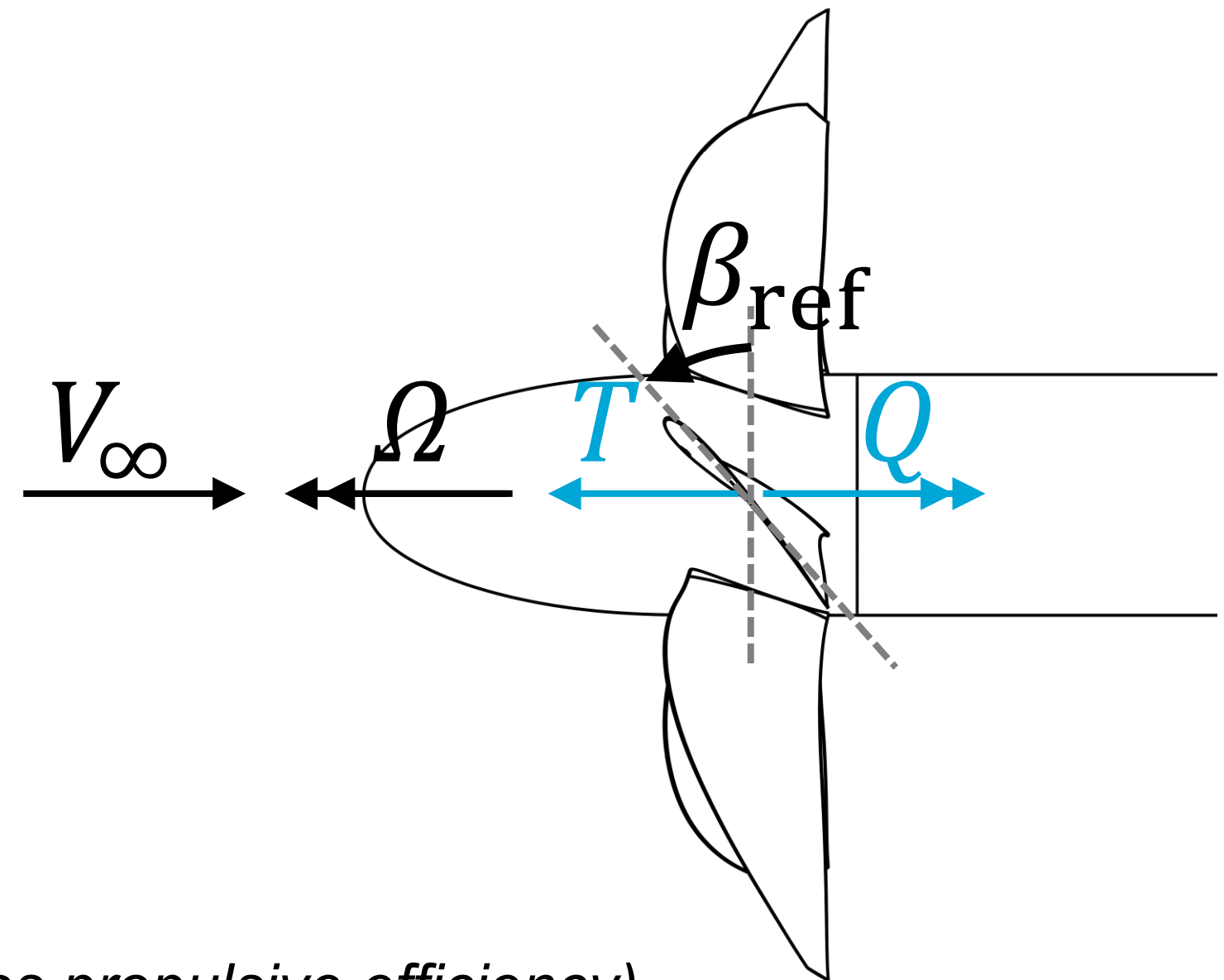
- Freestream velocity
- Rotational speed
- Blade pitch setting

$$\begin{aligned} V_\infty \\ \Omega (= 2\pi n) \\ \beta_{\text{ref}} \end{aligned}$$

- Efficiency

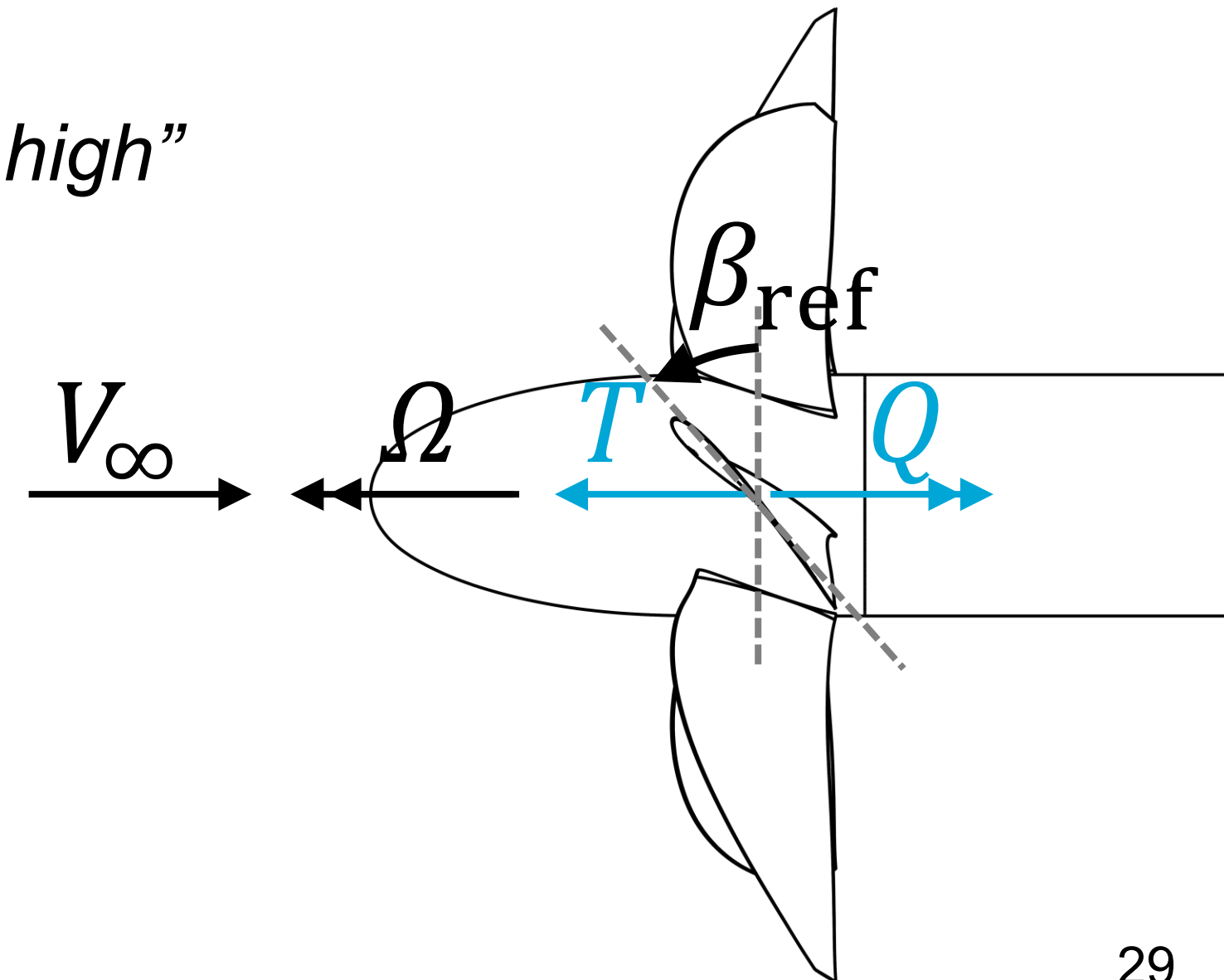
- Propeller efficiency

$$\eta = \frac{TV_\infty}{P} \quad (\text{not the same as } \underline{\text{propulsive}} \text{ efficiency})$$



How to describe propeller performance

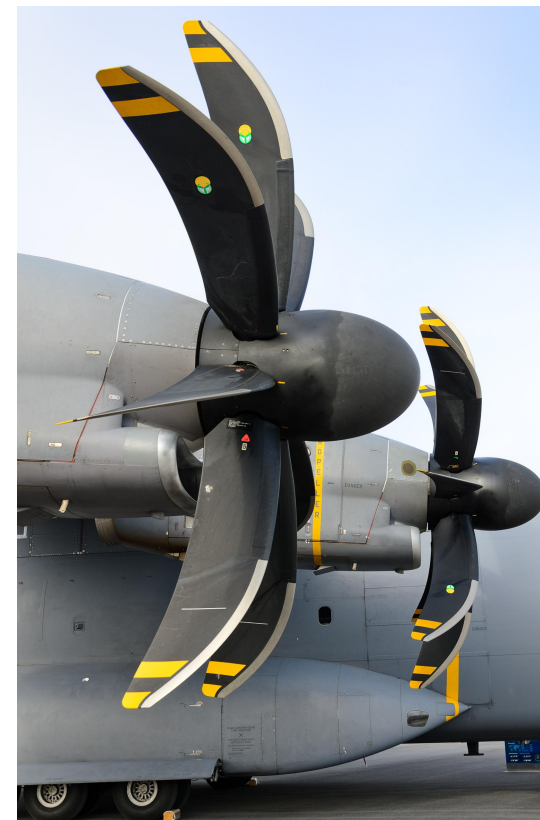
- Frequent questions/statements
 - “*What was the RPM?*”
 - “*The advance ratio was 0.3, so the thrust was high*”



Scaling

- Thrust T and shaft power P are function of
 - Operational conditions $V_\infty, n, \beta_{\text{ref}}$
 - Atmospheric conditions $\rho_\infty, a_\infty, \mu_\infty$
 - Propeller geometry D, c, N_B
- We need to **scale** the performance parameters to **compare between different designs, scales, operational conditions**
 - Dimensional analysis!
(Buckingham-Pi theorem)

T	Propeller thrust	kg m/s ²
P	Propeller shaft power	kg m ² /s ³
V_∞	Freestream velocity	m/s
n	Propeller rotational speed	rev/s
β_{ref}	Blade pitch setting	deg
ρ_∞	Air density	kg/m ³
a_∞	Speed of sound	m/s
μ_∞	Dynamic viscosity	kg/(m s)
D	Propeller diameter	m
c	Propeller blade chord	m
N_B	Number of blades	-



VS



Scaling of propeller performance

1. Choose parameters with **independent** dimension functions
2. Normalize parameters with **dependent** dimension functions with combinations of parameters with **independent** dimension functions → Pi-products

T	Propeller thrust	kg m/s ²
P	Propeller shaft power	kg m ² /s ³
V_∞	Freestream velocity	m/s
n	Propeller rotational speed	rev/s
β_{ref}	Blade pitch setting	deg
ϱ_∞	Air density	kg/m ³
a_∞	Speed of sound	m/s
μ_∞	Dynamic viscosity	kg/(m s)
D	Propeller diameter	m
c	Propeller blade chord	m
N_B	Number of blades	-

$$\Pi_1 = \frac{T}{\varrho_\infty V_\infty^2 D^2} = T_C \rightarrow \text{Thrust coefficient}$$

$$\Pi_2 = \frac{P}{\varrho_\infty V_\infty^3 D^2} = P_C \rightarrow \text{Power coefficient}$$

$$\Pi_3 = \frac{nD}{V_\infty} = \frac{1}{J} \rightarrow 1 / \text{Advance ratio}$$

$$\Pi_4 = \beta_{\text{ref}} \rightarrow \text{Pitch setting}$$

$$\Pi_5 = \frac{a_\infty}{V_\infty} = \frac{1}{M_\infty} \rightarrow 1 / \text{Mach number}$$

$$\Pi_6 = \frac{\mu_\infty}{\varrho_\infty V_\infty D} = \frac{1}{Re_D} \rightarrow 1 / \text{Reynolds number}$$

$$\Pi_7 = \frac{c}{D} = \sigma \frac{\pi}{2N_B} \rightarrow \text{Blade solidity}$$

$$\Pi_8 = N_B \rightarrow \text{Blade count}$$

Performance indicators

Operational /freestream conditions

Geometry

Scaled propeller performance

J	advance ratio
β_{ref}	pitch setting
M_{∞}	freestream Mach number
Re_D	Reynolds number w.r.t. diameter
σ	solidity
N_B	blade count

- Thrust coefficient $T_C = \frac{T}{\rho_{\infty} V_{\infty}^2 D^2} = f(J, \beta_{\text{ref}}, M_{\infty}, Re_D, \sigma, N_B)$
- Power coefficient $P_C = \frac{P}{\rho_{\infty} V_{\infty}^3 D^2} = f(J, \beta_{\text{ref}}, M_{\infty}, Re_D, \sigma, N_B)$
- Propeller efficiency $\eta = \frac{T_C}{P_C} (= J \frac{C_T}{C_P}) = f(J, \beta_{\text{ref}}, M_{\infty}, Re_D, \sigma, N_B)$

THRUST COEFFICIENT *ALWAYS CHECK WHICH DEFINITION IS USED...*

(same for power coefficient)



Thrust coefficient(s)

$$F_p \sim \frac{1}{\rho} \frac{\partial p}{\partial x} \sim \frac{p}{\rho L} \quad F_I \sim u \frac{\partial u}{\partial x} \sim \frac{U^2}{L}$$

$$\frac{\text{thrust pressure force } (= \frac{T/S_{\text{ref}}}{\rho_{\infty} \sqrt{S_{\text{ref}}}})}{\text{freestream inertia force } (\propto V_{\infty}^2 / \sqrt{S_{\text{ref}}})}$$

- Multiple definitions are used...

$$T_C = \frac{T}{\rho_{\infty} V_{\infty}^2 D^2} \quad \longleftrightarrow \quad T_C = \frac{C_T}{J^2} \quad \longleftrightarrow \quad C_T = T_C J^2$$

$$C_T = \frac{T}{\rho_{\infty} n^2 D^4}$$

$$T'_C = \frac{T}{\frac{1}{2} \rho_{\infty} V_{\infty}^2 S_{\text{ref}}}$$

pressure jump at disk ($\propto T/D^2$)

freestream dynamic pressure ($\propto \rho_{\infty} V_{\infty}^2$)

pressure jump at disk ($\propto T/D^2$)

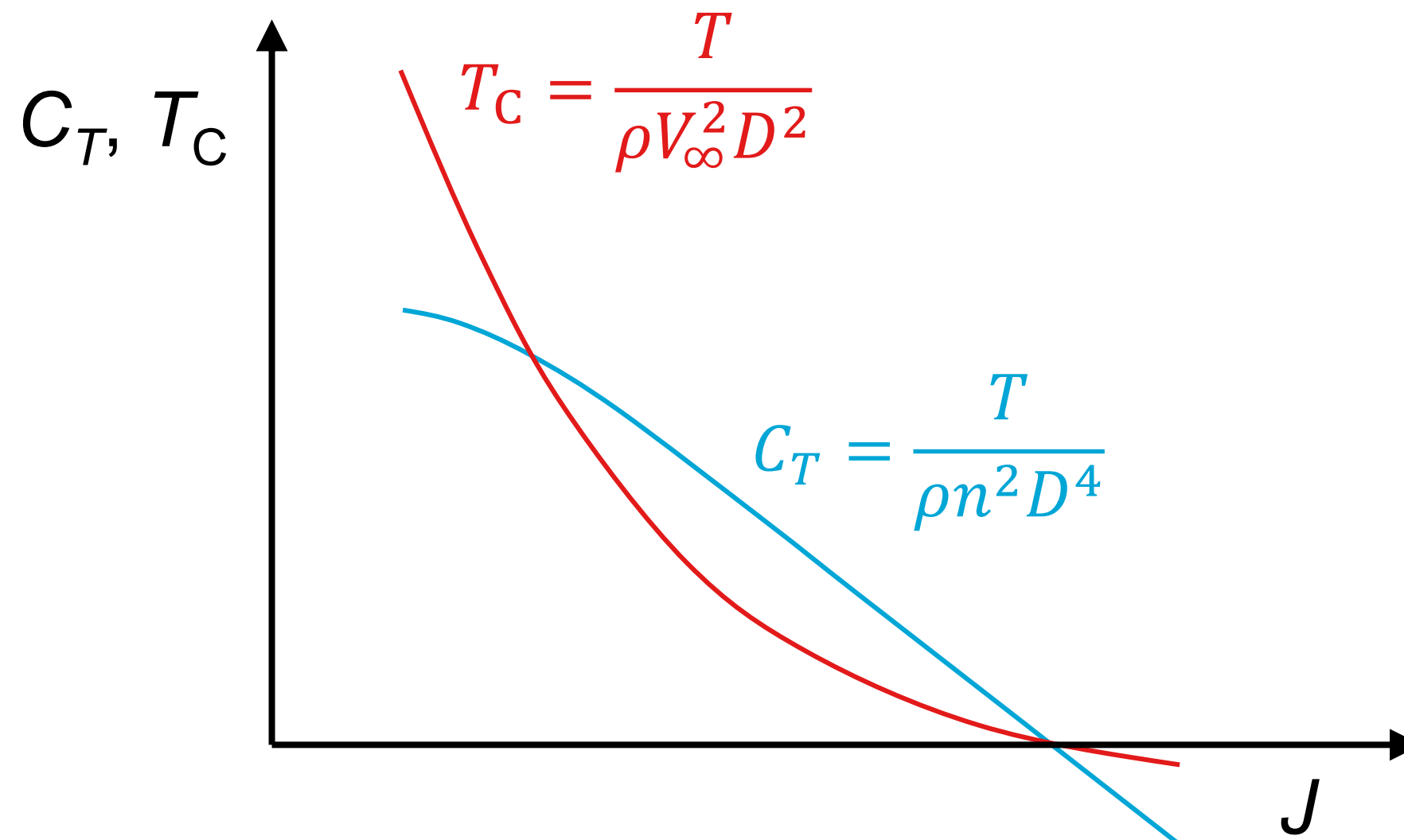
dynamic pressure relative to tip speed ($\propto \rho_{\infty} (nD)^2$)

$$\text{Also used: } T_C = \frac{T}{\frac{1}{2} \rho_{\infty} V_{\infty}^2 \frac{\pi}{4} D^2}$$

$\longleftrightarrow$ (Typically referred to as C_T in wind-turbine industry...)

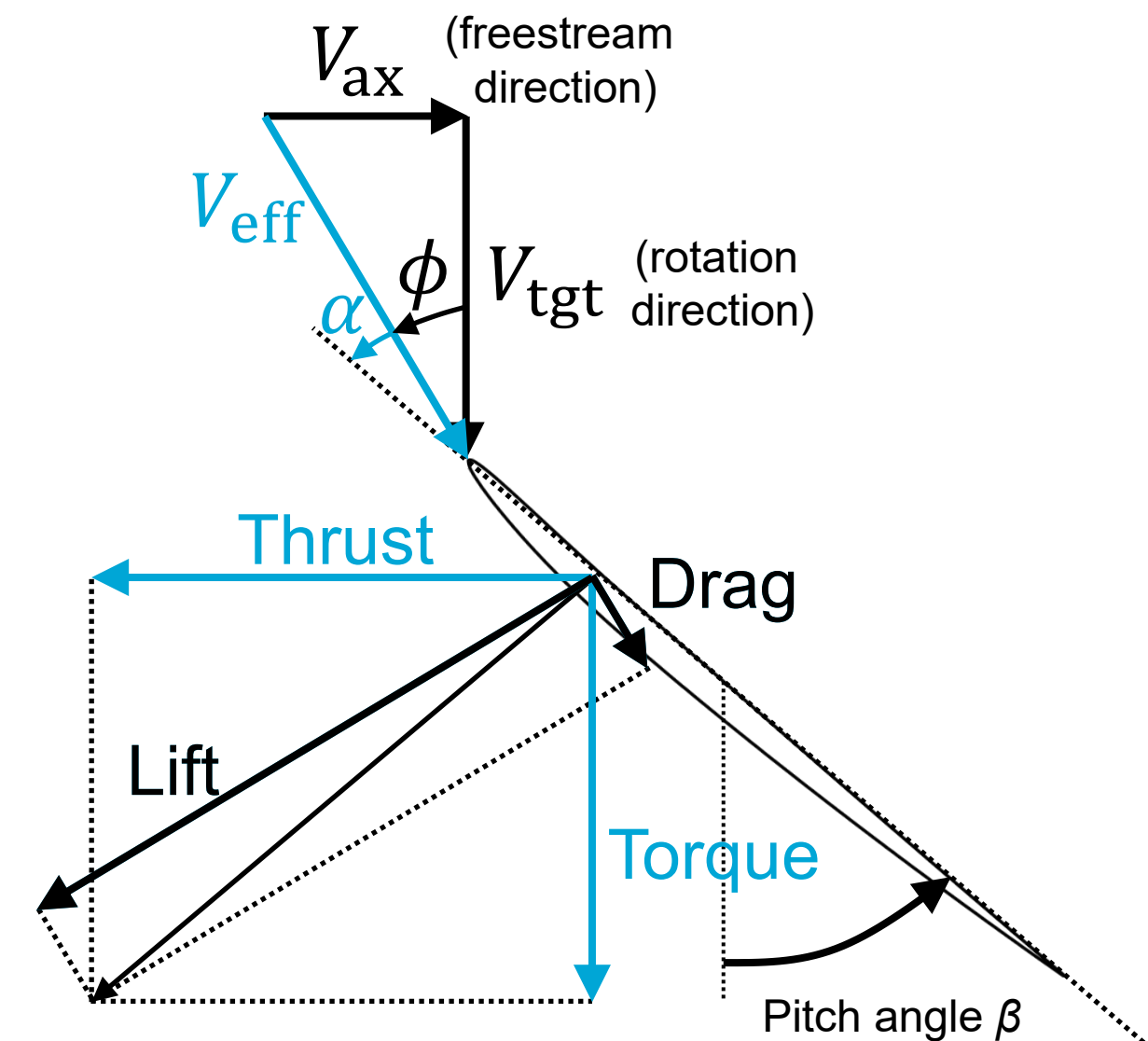
Thrust coefficients

- Thrust coefficients
 - C_T indicative of local blade loading $\rightarrow$ linear in advance ratio like C_L vs α
 - T_C indicative of disk loading, induced axial velocity in slipstream
 - ***Note different nomenclature vs wind-turbine world!***



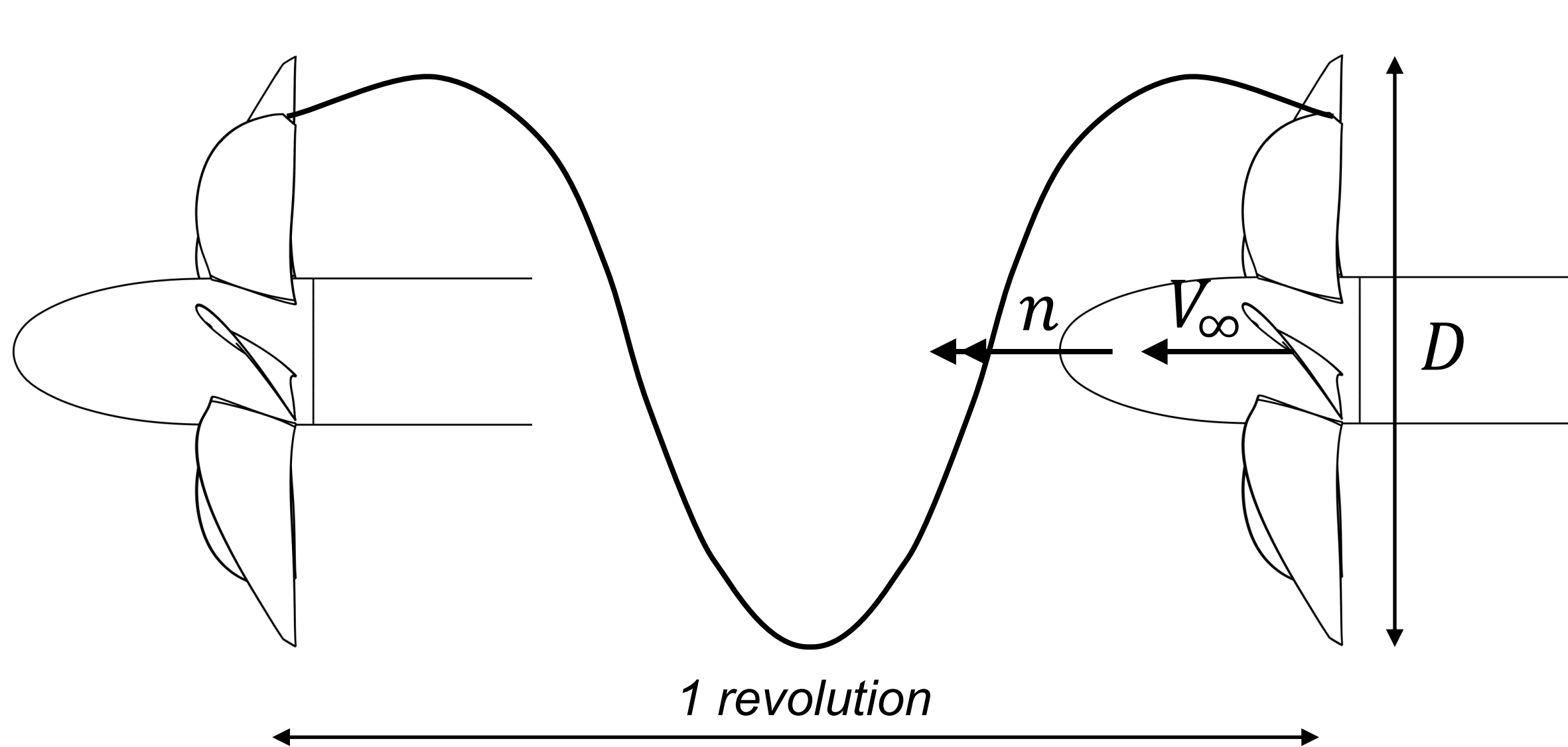
Similarity parameters for operating conditions

- Advance ratio $J = \frac{V_\infty}{nD}$
- Pitch angle β_{ref}
- Scaled operating conditions define inflow angles at the blade sections
- Local angles of attack ...
 - Increase with decreasing J
 - Increase with increasing β



Advance ratio

- Advance ratio $J = \frac{V_\infty}{nD} = \frac{V_\infty t_{1-\text{rev}}}{D} = \frac{\Delta x_{1-\text{rev}}}{D} = \frac{\text{distance traveled in 1 revolution}}{\text{diameter}}$



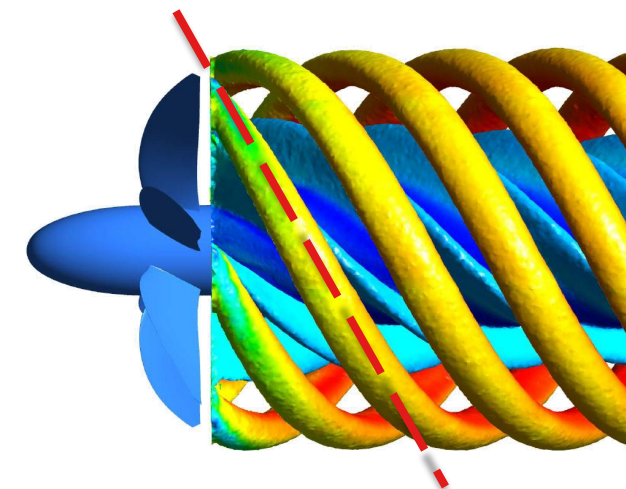
$$J = \frac{V_\infty}{nD} = \frac{V_\infty}{\frac{\Omega}{2\pi} 2R} = \frac{\pi V_\infty}{\Omega R} = \frac{\pi}{\lambda}$$

Tip-speed ratio

Advance ratio

- Advance ratio $J = \frac{V_\infty}{nD} = \frac{V_\infty t_{1-\text{rev}}}{D} = \frac{\Delta x_{1-\text{rev}}}{D} = \frac{\text{distance traveled in 1 revolution}}{\text{diameter}}$
 - Defines pitch of helix system in propeller slipstream!
 - Low $J \rightarrow$ high pitch: flow structures in slipstream condensed, convect slowly away from prop
 - High $J \rightarrow$ low pitch: flow structures in slipstream stretched, convect rapidly away from prop

Pitch angle $\sim 1/J$



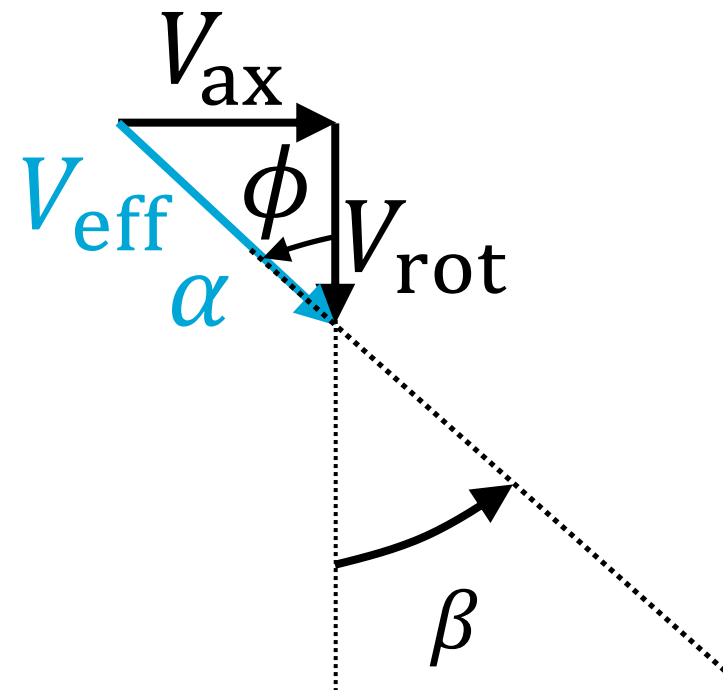
Credit: T.C.A.
Stokkermans

Thrust/power coefficients vs advance ratio

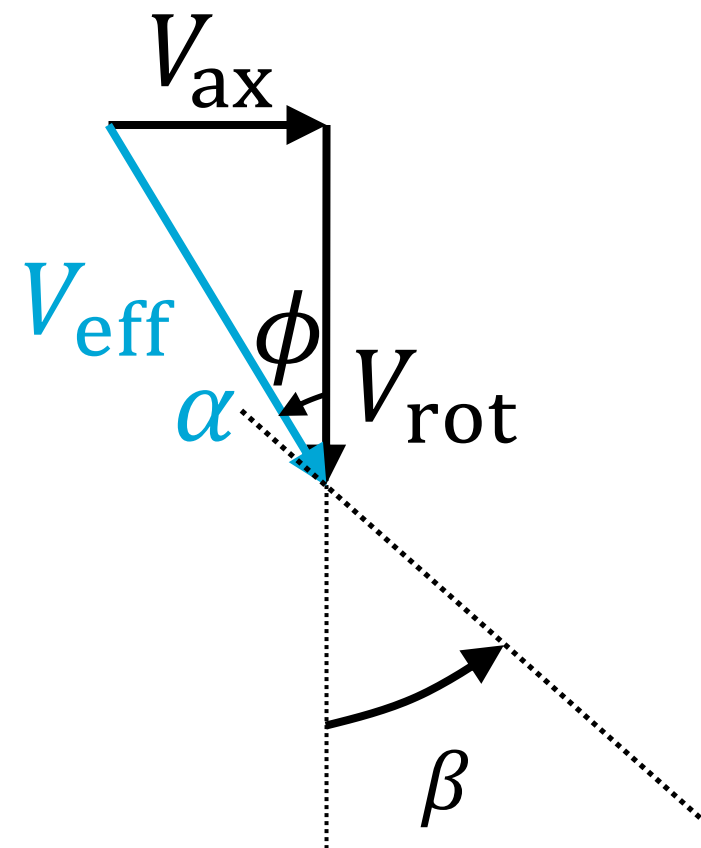
- Thrust/power coefficients increase with decreasing advance ratio

$$- J = \frac{V_\infty}{nD} \propto \frac{V_{ax}}{V_{rot}} \rightarrow \phi = \tan^{-1} \left(\frac{V_{ax}}{V_{rot}} \right) \propto J$$

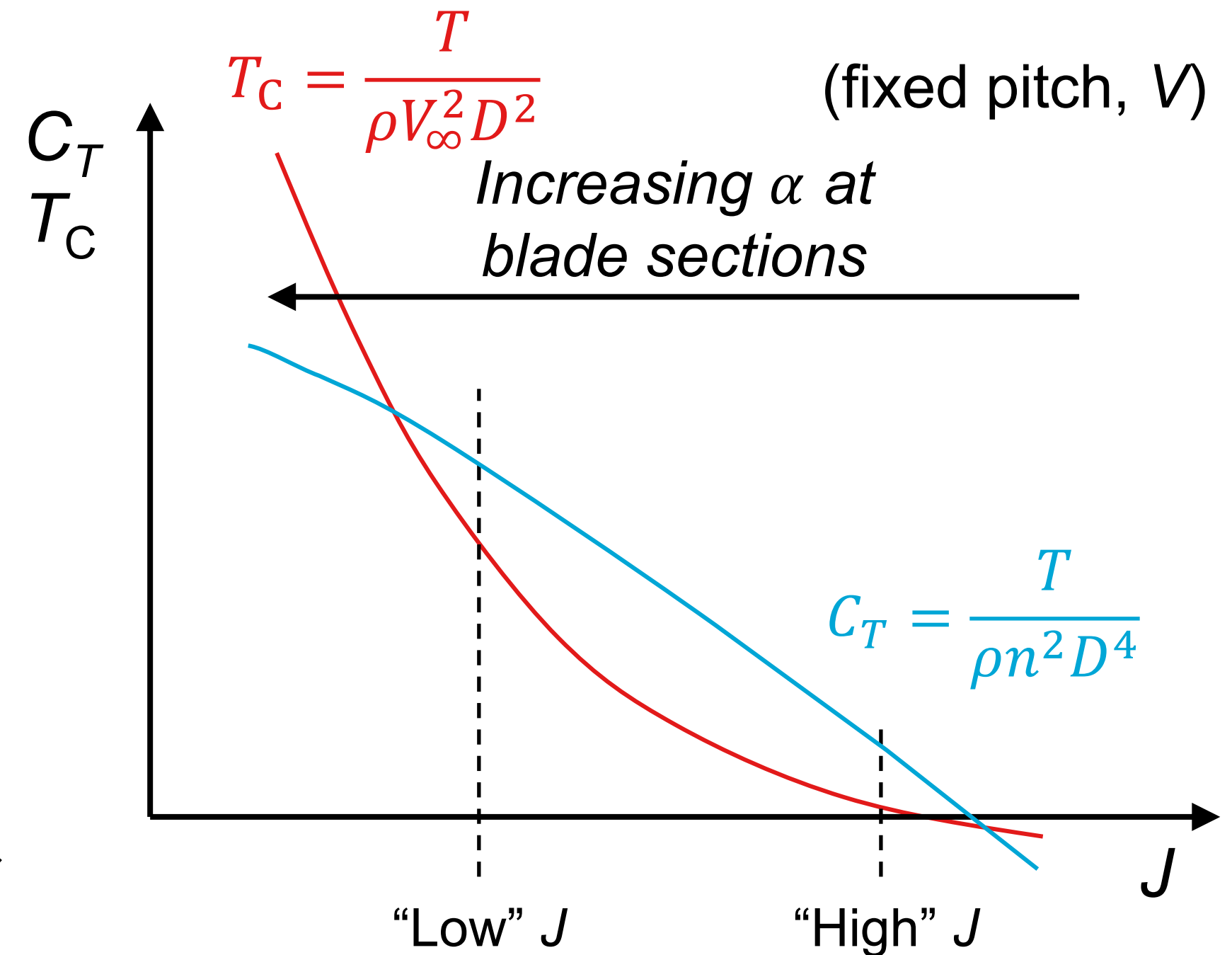
$$- \alpha = \beta - \phi \propto -J$$



High J: low α



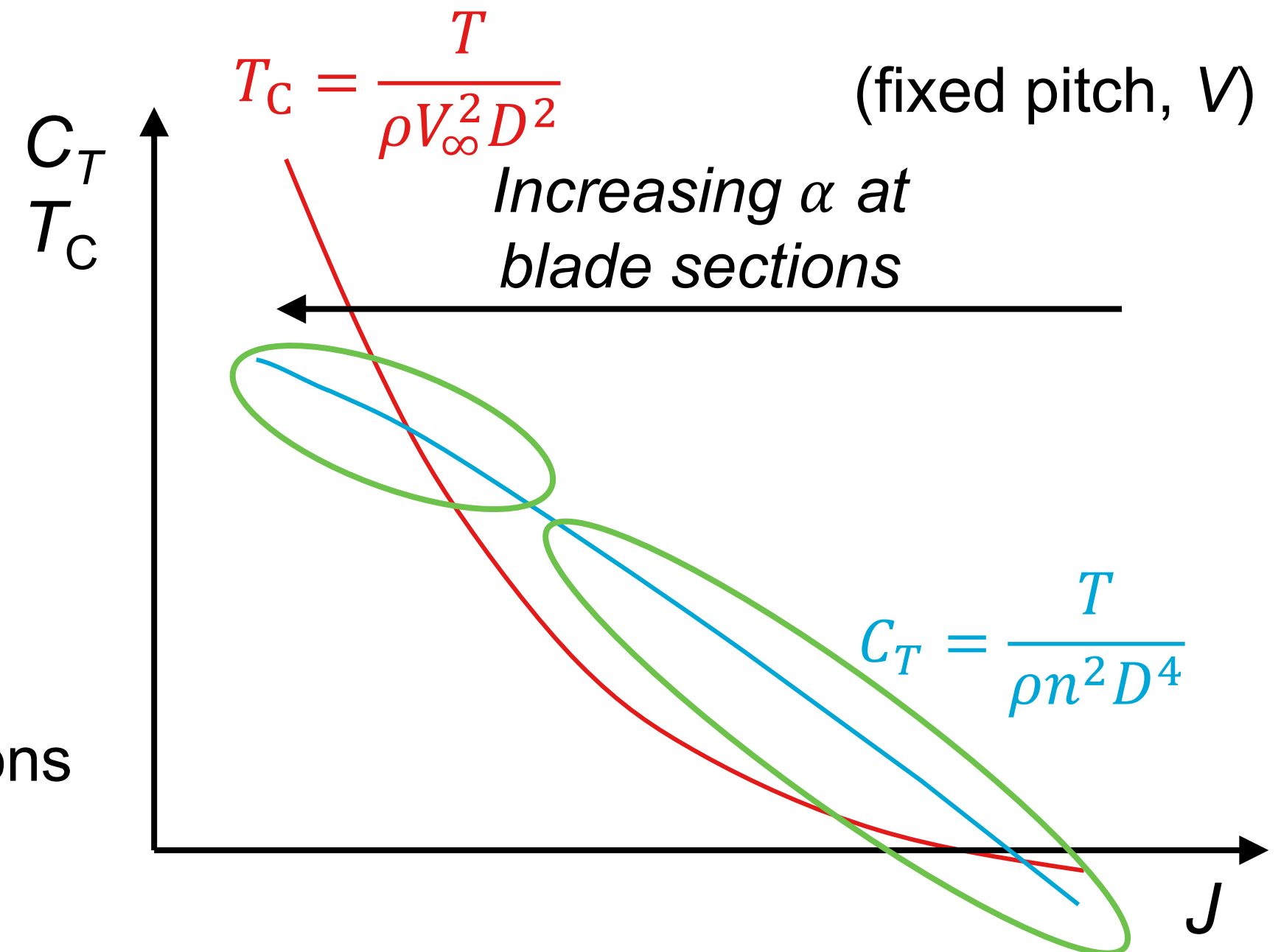
Low J: high α



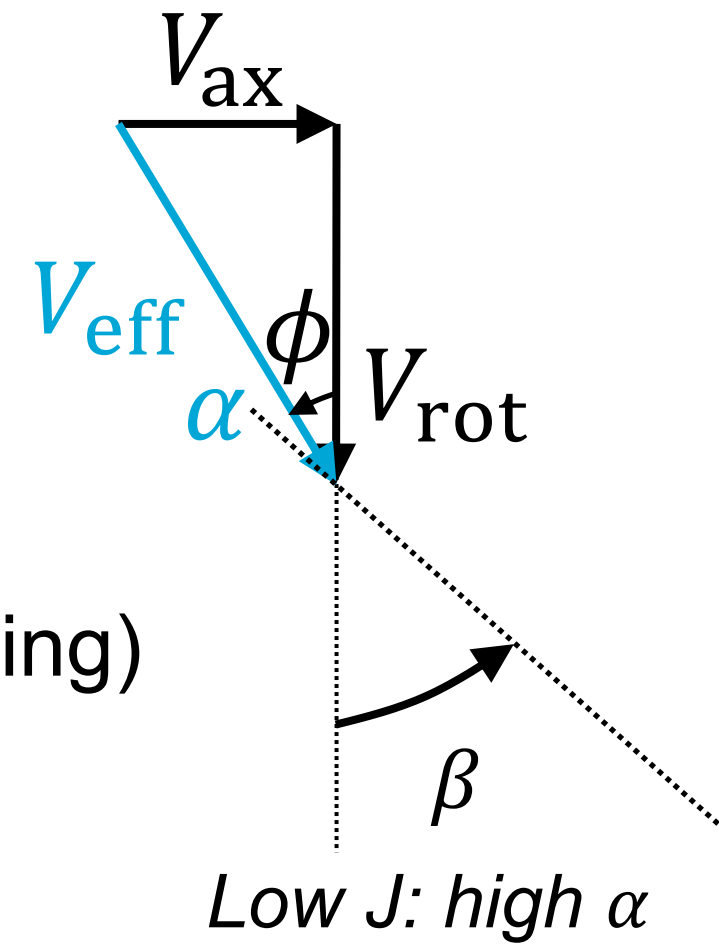
Thrust/power coefficients vs advance ratio

- Thrust/power coefficients increase with decreasing advance ratio
- Linear relation $C_T(J)$ at low loading
- Nonlinear relation $C_T(J)$ at high loading
 1. Separation at low J
 2. Nonlinear induction with J

1. Separation at low J
 - Low $J \rightarrow$ high $\alpha \rightarrow$ separation at blade sections

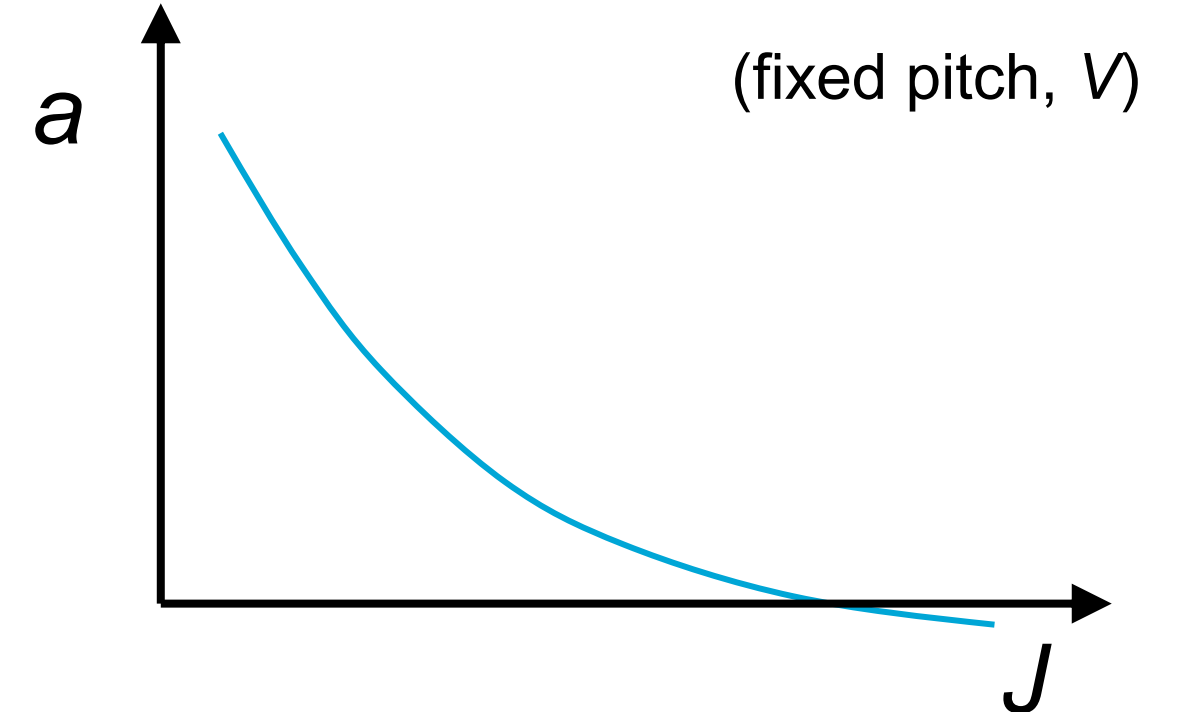
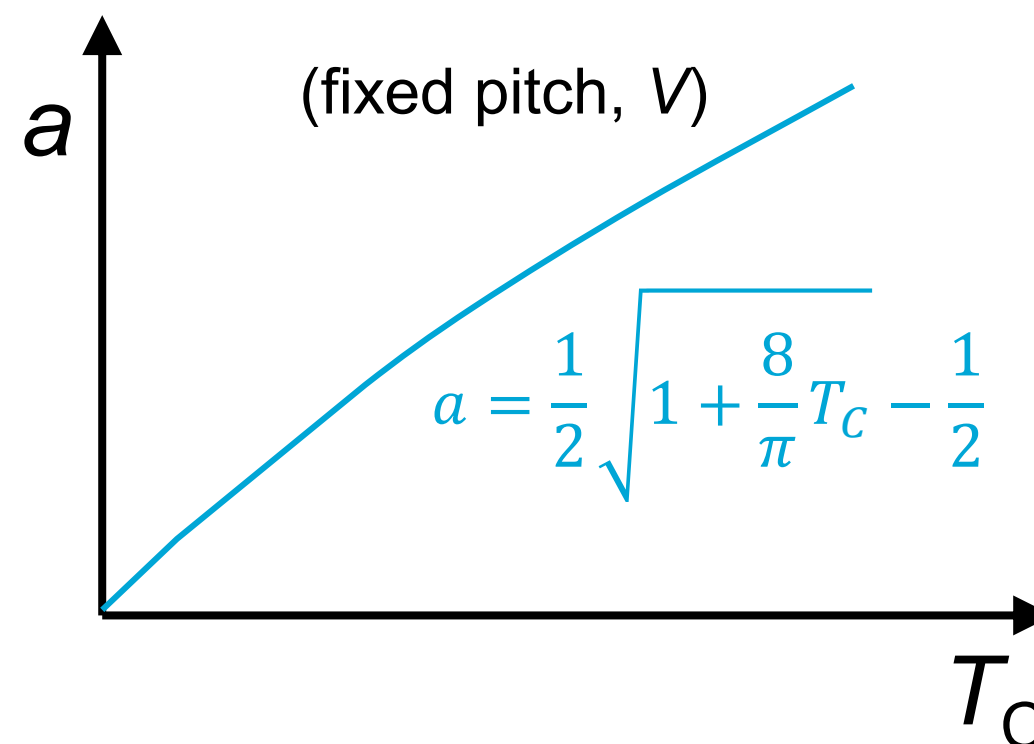
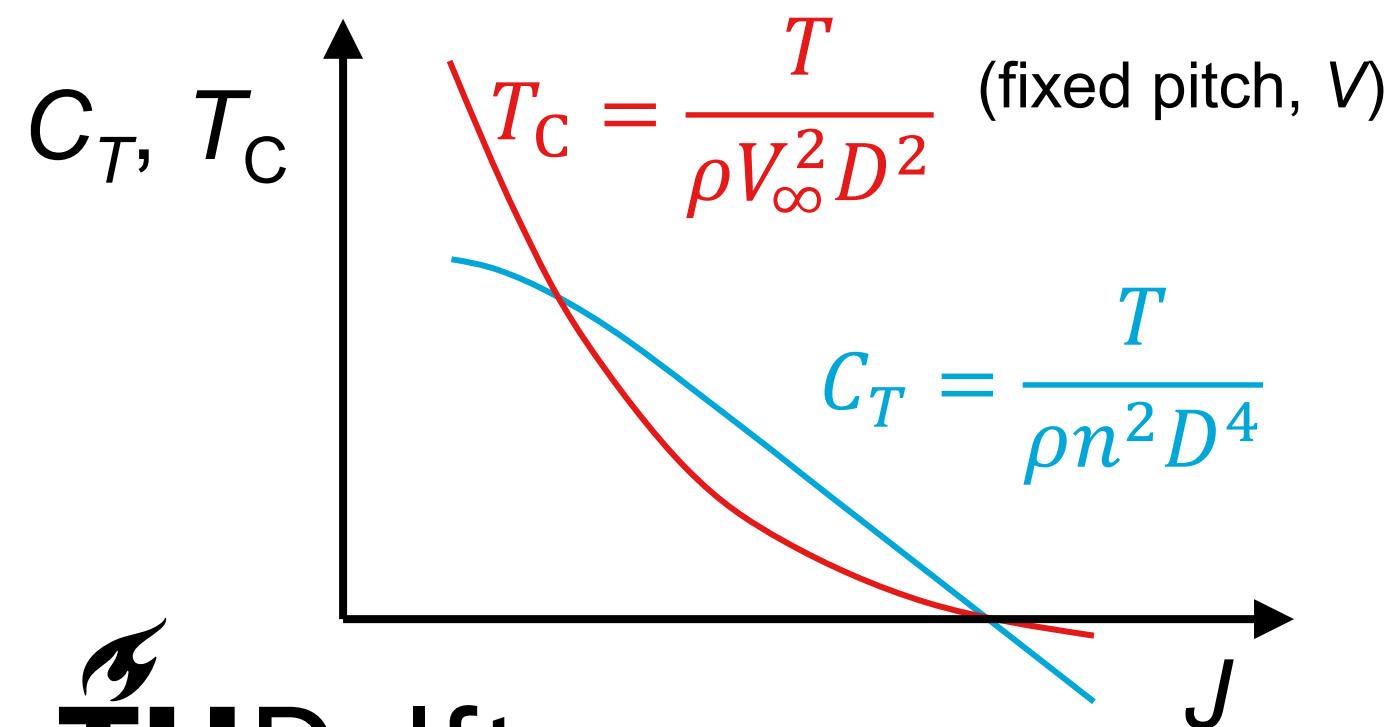


Thrust/power coeff. vs advance ratio



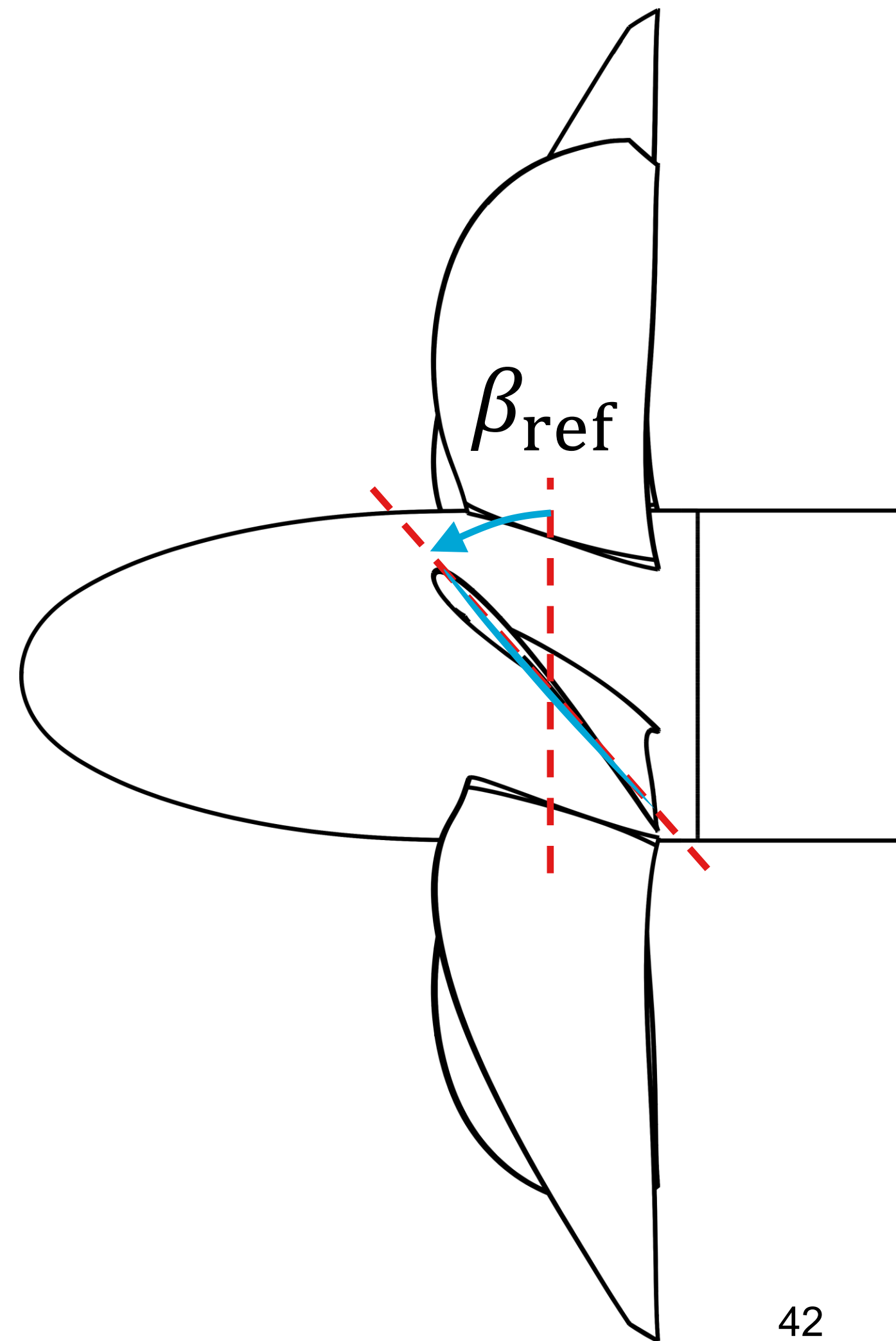
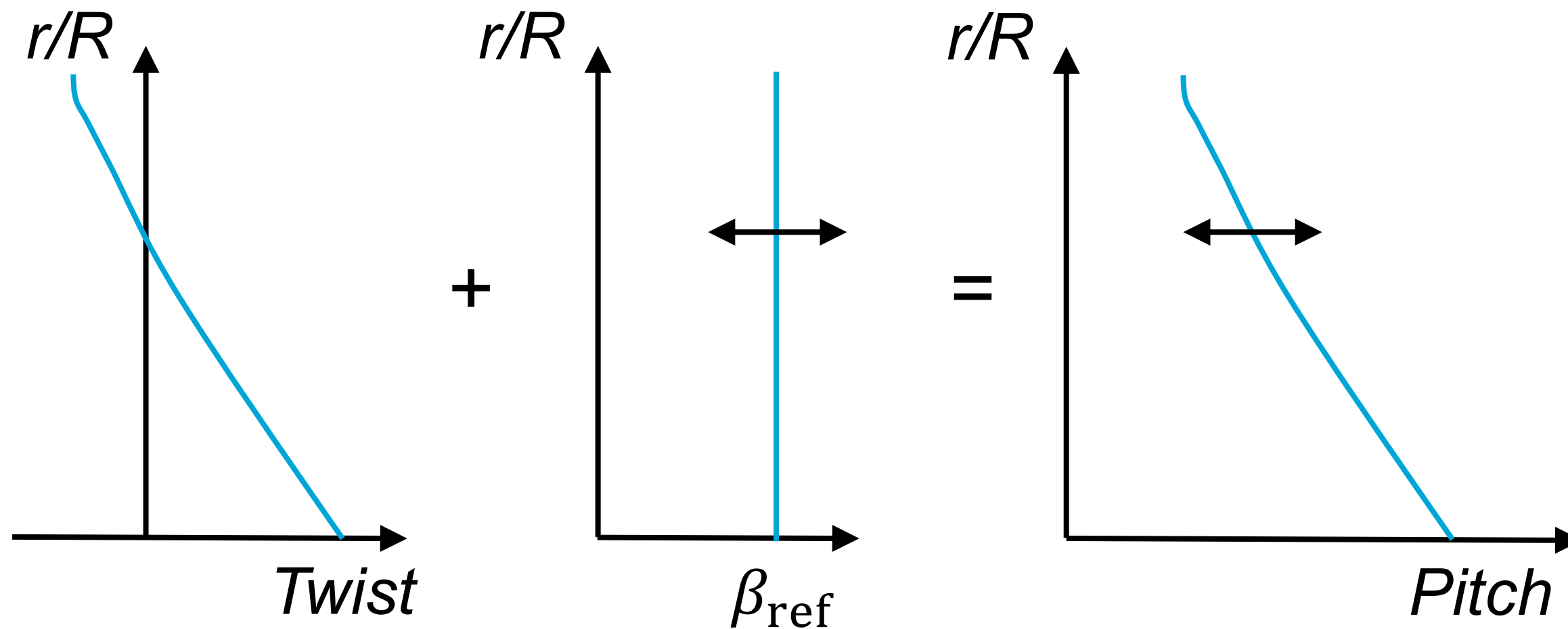
2. Nonlinear induction with J

- Axial inflow velocity defined by induction factor $a = v_a/V_\infty \propto T_C$ (disk loading)
 - Axial induction increases inflow angle $\rightarrow$ reduces α
 - Induction factor scales nonlinearly with T_C and J
- $\rightarrow$ for given change in J , less α increase at high thrust coefficient than at low thrust coefficient



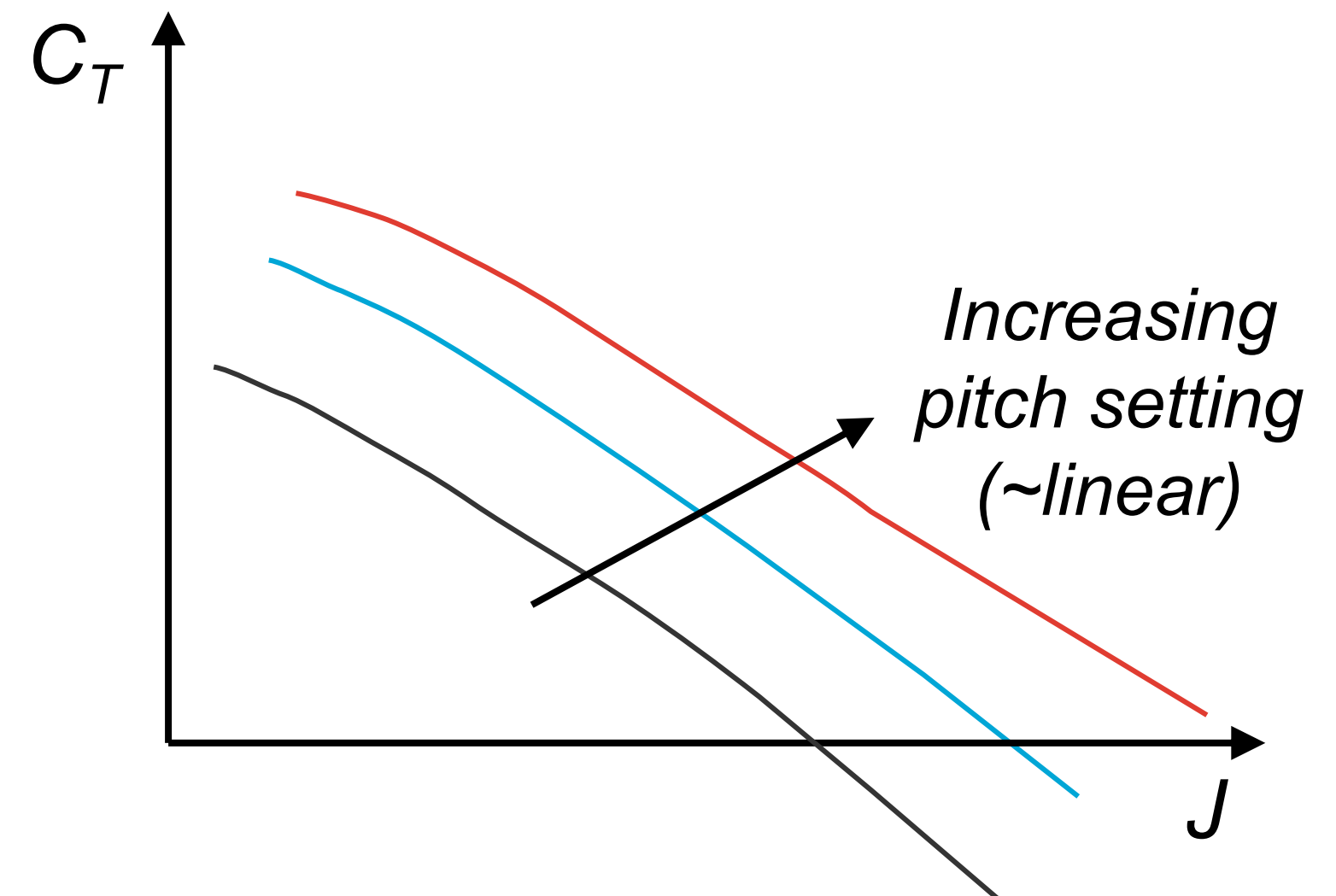
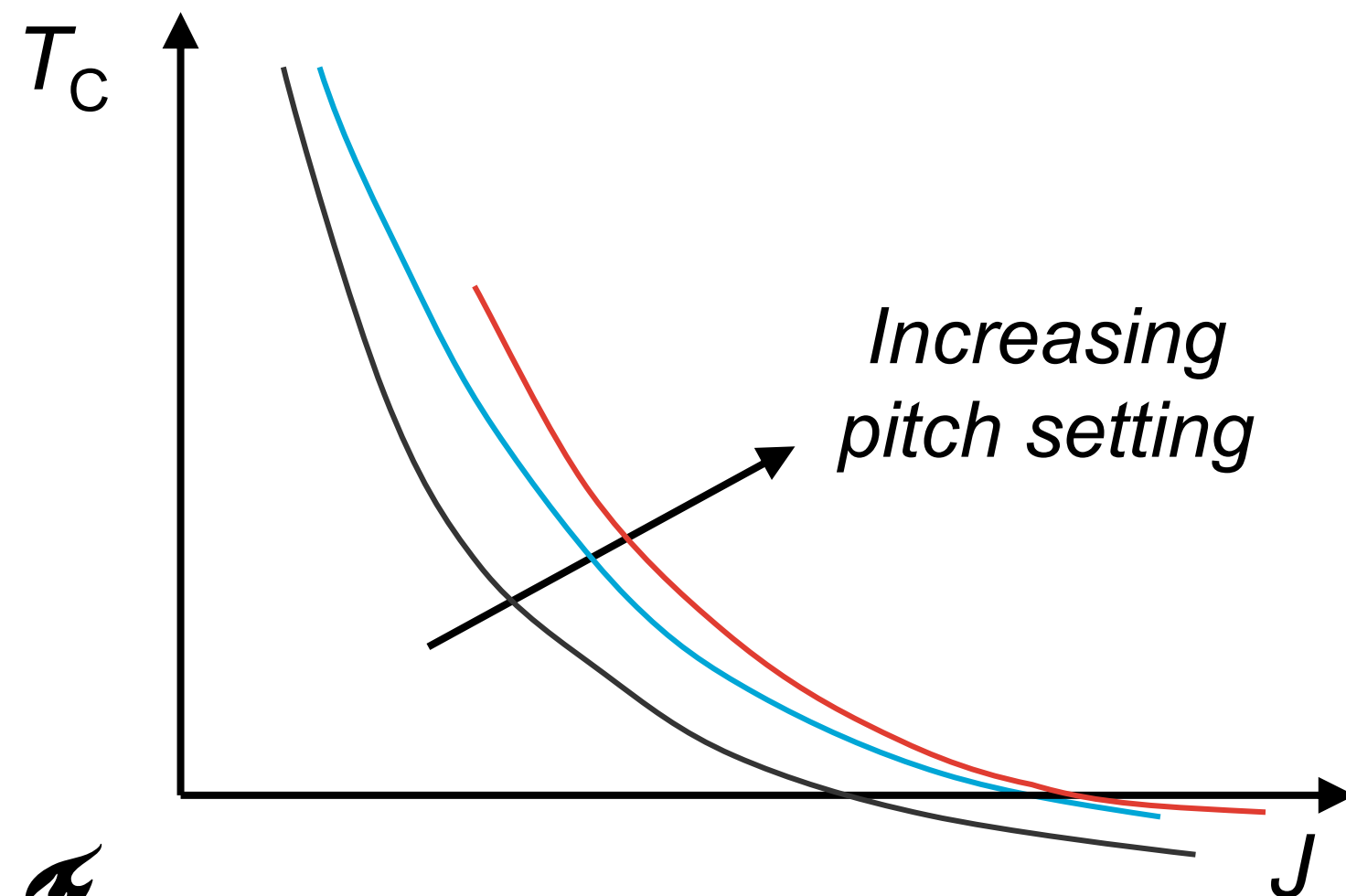
Pitch setting

- Reference pitch setting of entire blade
- “Translates” the blade twist distribution



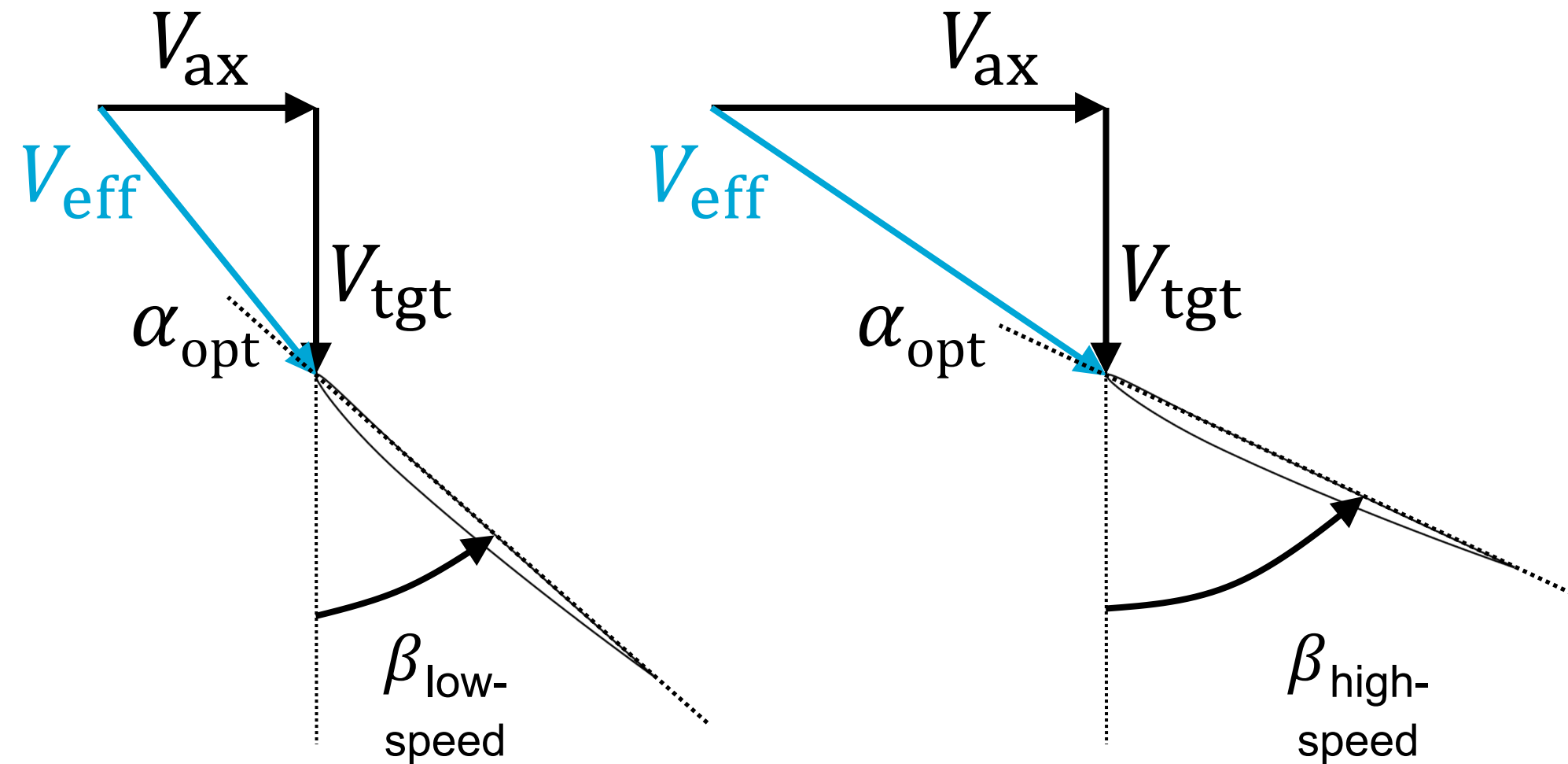
Thrust/power coefficients vs pitch setting

- Thrust/power coefficients increase with increasing pitch
 - Curves “shift to the right”
 - Advance ratio operating window changes when changing pitch



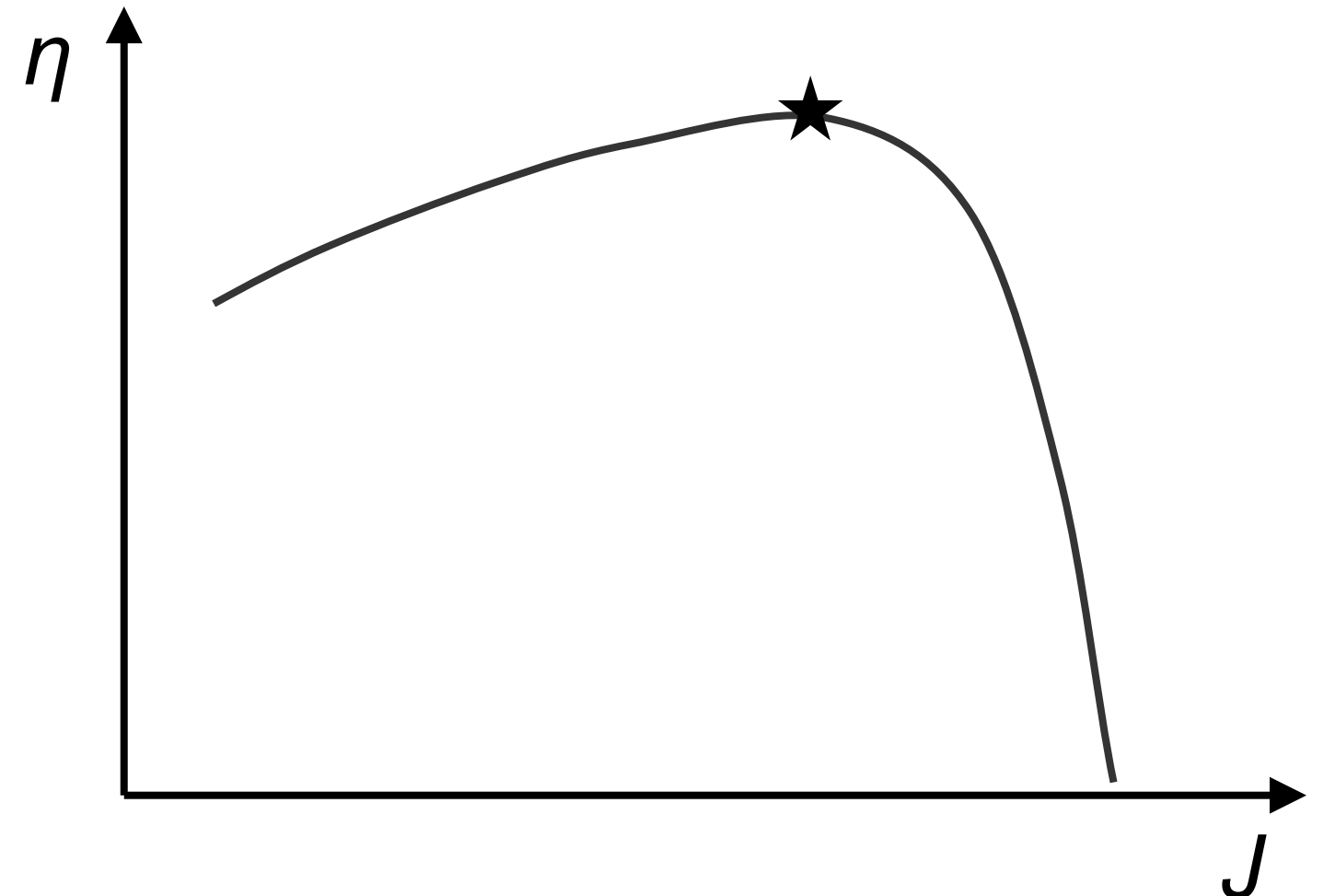
Fixed-pitch vs fixed-speed operation

- 2 possible ways of propeller operation
 1. Fixed pitch: constant pitch, variable rotational speed
 2. Fixed speed: constant rotational speed, variable pitch
 - Peak efficiency can be achieved for multiple mission segments
 - Engine speed can be optimized
 - Weight and maintenance penalty due to pitch-change mechanism



Propeller efficiency vs advance ratio

- $\eta = \frac{T_C}{P_C} \left(= J \frac{C_T}{C_P} \right) = f(J, \beta_{\text{ref}}, M_\infty, Re_D, \sigma, N_B)$
- Distinct shape with maximum at single advance ratio for given pitch
 - Modern full-scale props $\eta \approx 85 - 90\%$ in cruise!
 - Specific values depend on disk loading requirements and operational conditions

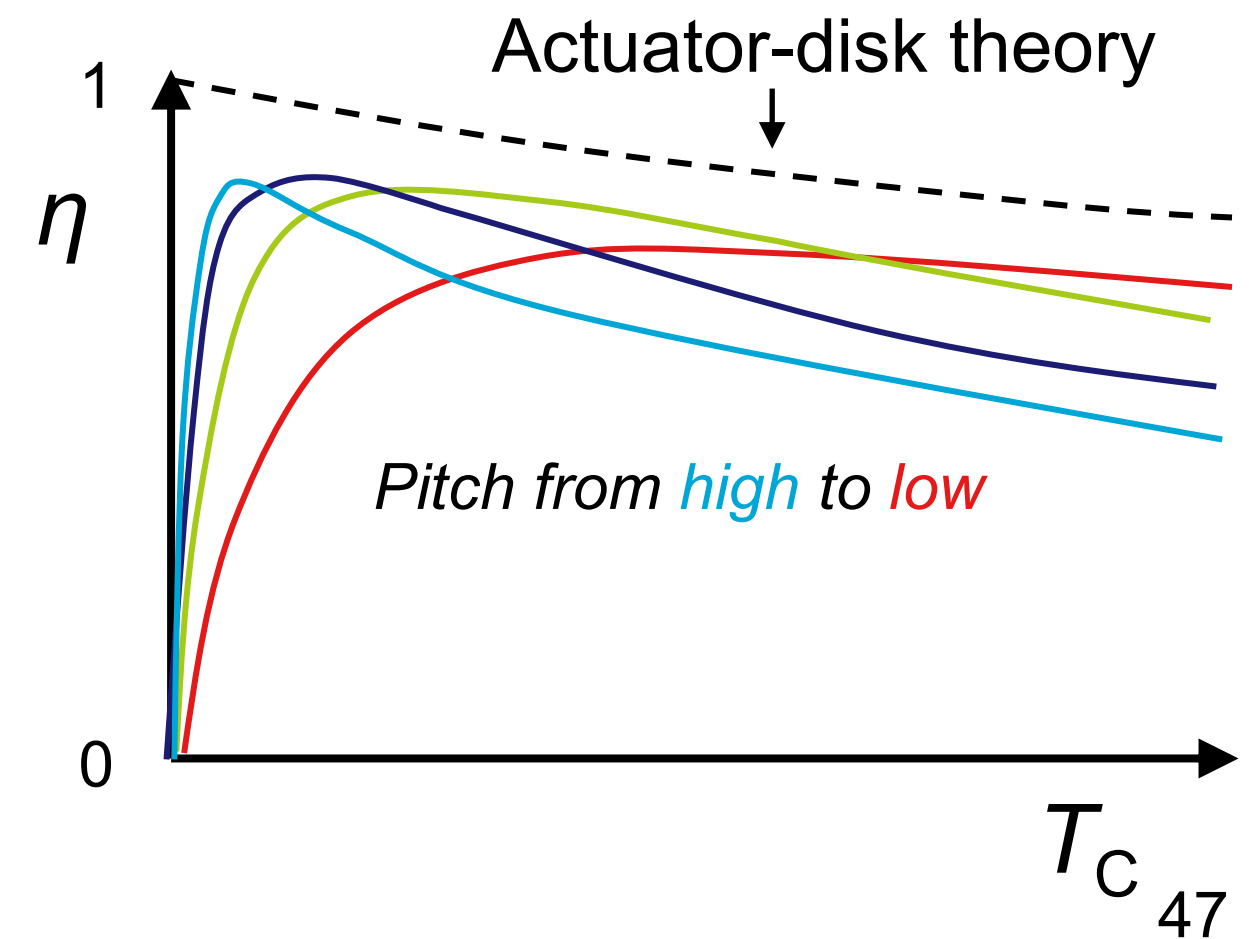
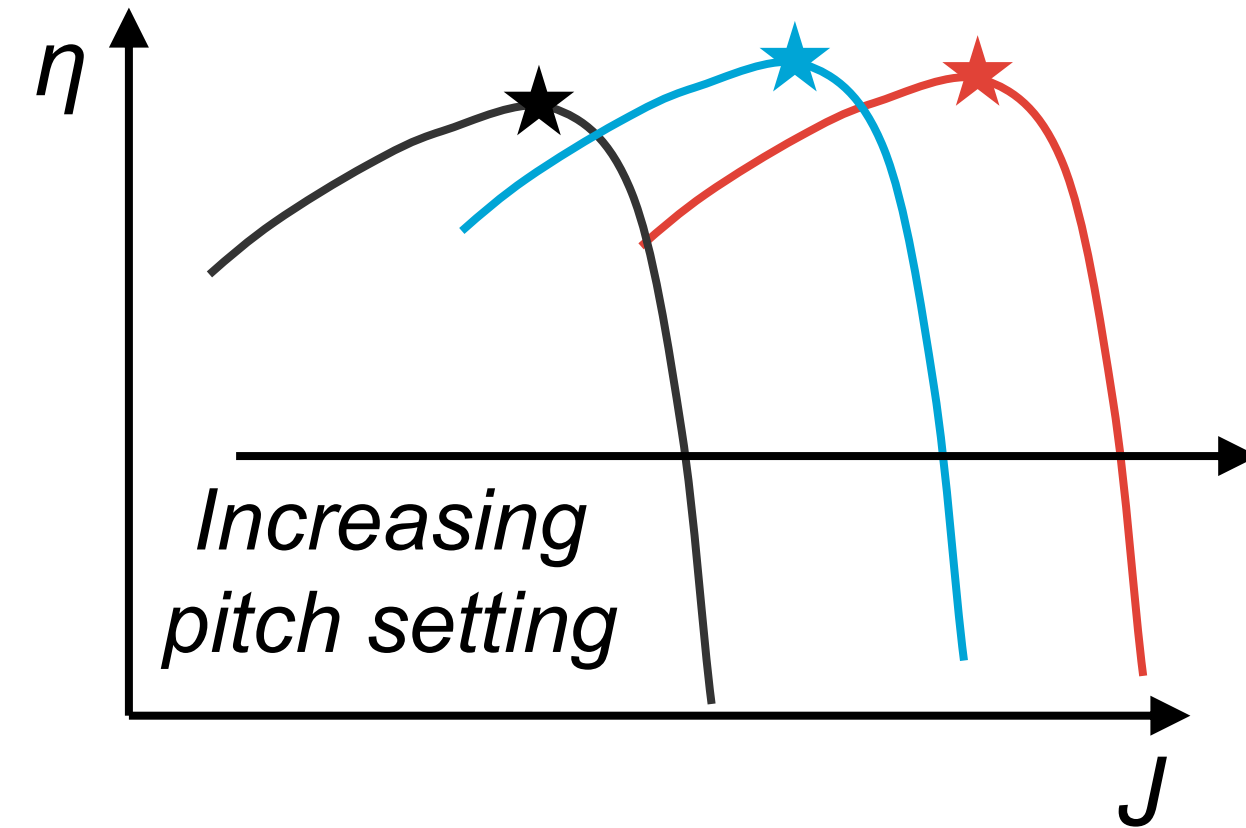


Propeller efficiency: loss mechanisms

- $\eta = \frac{T_C}{P_C} \left(= J \frac{C_T}{C_P} \right) = f(J, \beta_{\text{ref}}, M_\infty, Re_D, \sigma, N_B)$
 - Loss mechanisms
 1. Axial acceleration behind the propeller (= *propulsive* efficiency)
 2. Angular momentum due to torque (swirl)
 3. Blade profile losses (blade drag)
 - Dominant loss mechanism depends on operational conditions and loading
 - High loading (T_C), high freestream Ma-#* → axial acceleration, swirl
 - High loading (T_C), low freestream Ma-#* → axial acceleration
 - Low loading (T_C) → no dominant mechanism
- * assumption: tip Mach number constraining → higher advance ratio at higher freestream Ma-# (and not significant wave drag)

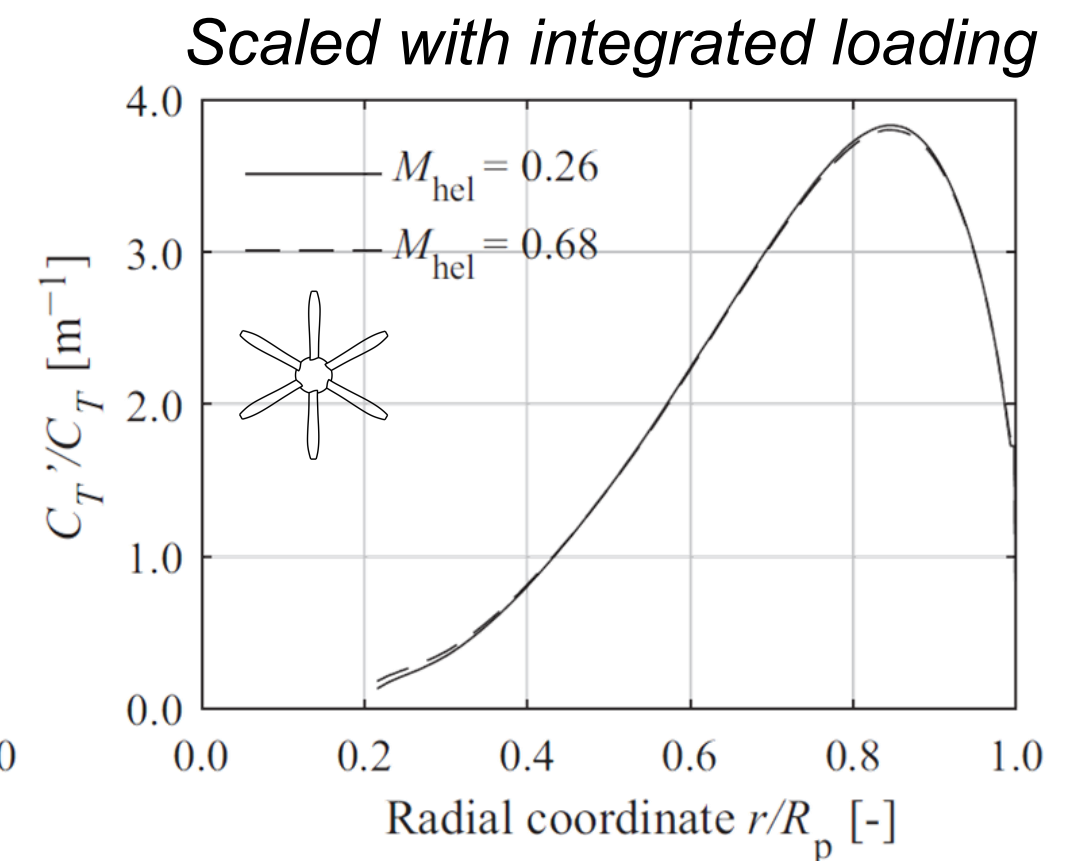
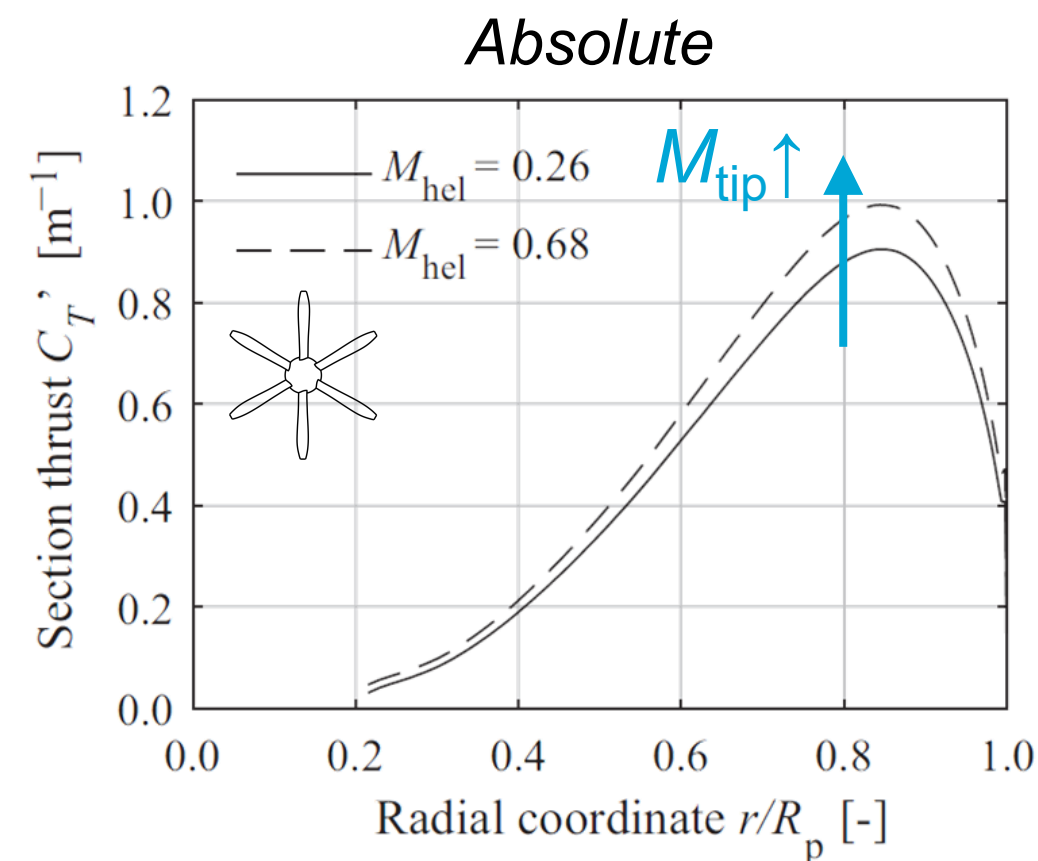
Propeller performance trends

- Optimum efficiency for each thrust setting at specific combination of advance ratio & pitch angle
 - Compromise between axial-induction losses, swirl losses, profile losses
- Optimum efficiency varies with thrust setting
 - Lower than theoretical efficiency (actuator-disk theory) due to swirl & profile losses
- Propeller design is compromise between performance at multiple thrust settings...
 - Variable pitch propeller required for maximum performance over mission



Impact of Mach number

- Performance increases with increasing Mach number until shock wave formation
 - Compressibility effects 'scale' the blade loading distribution ($C_p \sim \frac{1}{\sqrt{1-M_\infty^2}}$)
 - (Also: strong impact on acoustic performance)
 - Much more relevant than for wind turbines due to higher tip Mach number of propellers



Impact of Mach number

- Performance increases with increasing Mach number until shock wave formation
 - Compressibility effects ‘scale’ the blade loading distribution
 - (Also: strong impact on acoustic performance)
- **Scale-independent!** → freestream Mach number and advance ratio define effective Mach numbers experienced by the blade sections, not the diameter

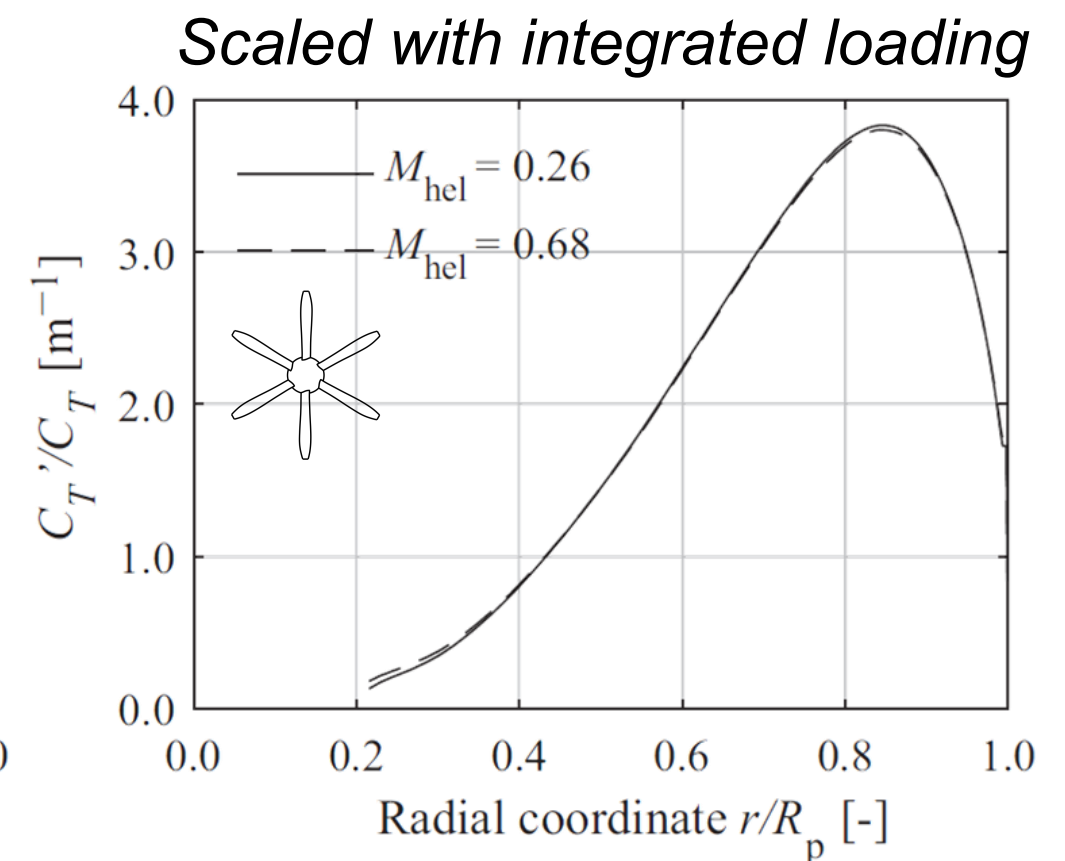
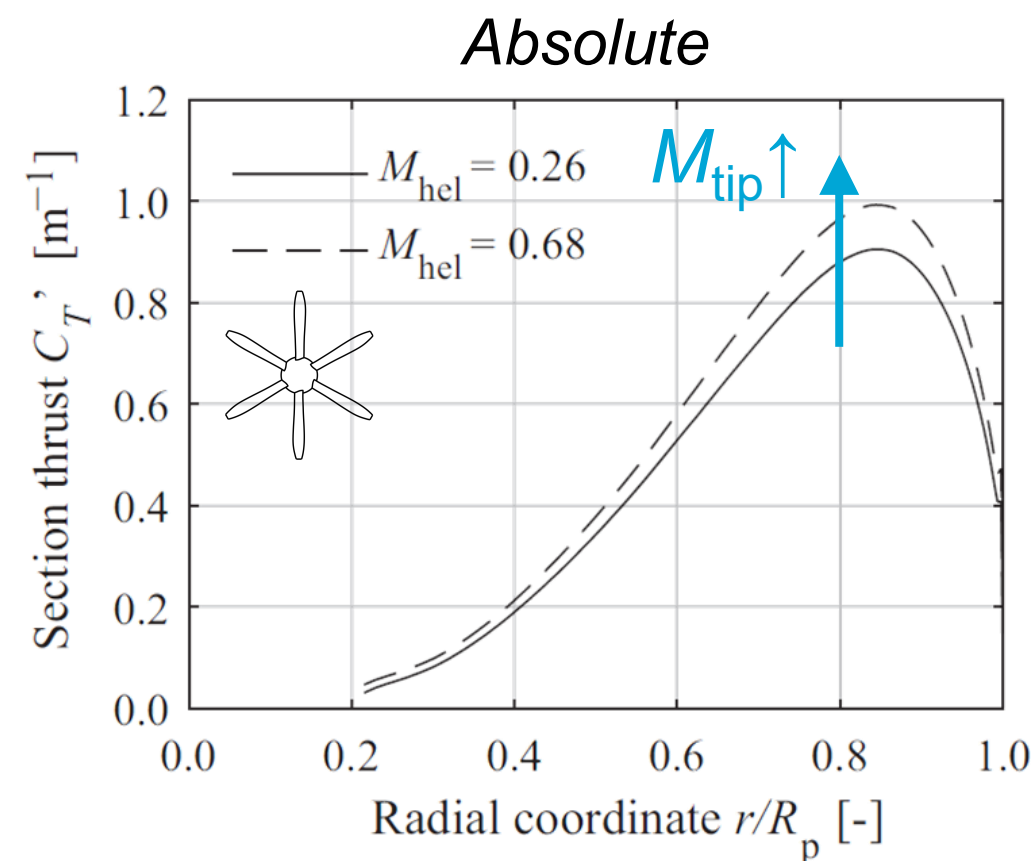
$$M_{\text{tip}} = \frac{\Omega R}{a_\infty} = \frac{\pi n D V_\infty}{a_\infty V_\infty} = M_\infty \frac{\pi}{J}$$

$$M_{\text{tip, hel}} = \sqrt{M_\infty^2 + M_{\text{tip}}^2} = M_\infty \sqrt{1 + \left(\frac{\pi}{J}\right)^2}$$

- Recent propellers for turboprops:

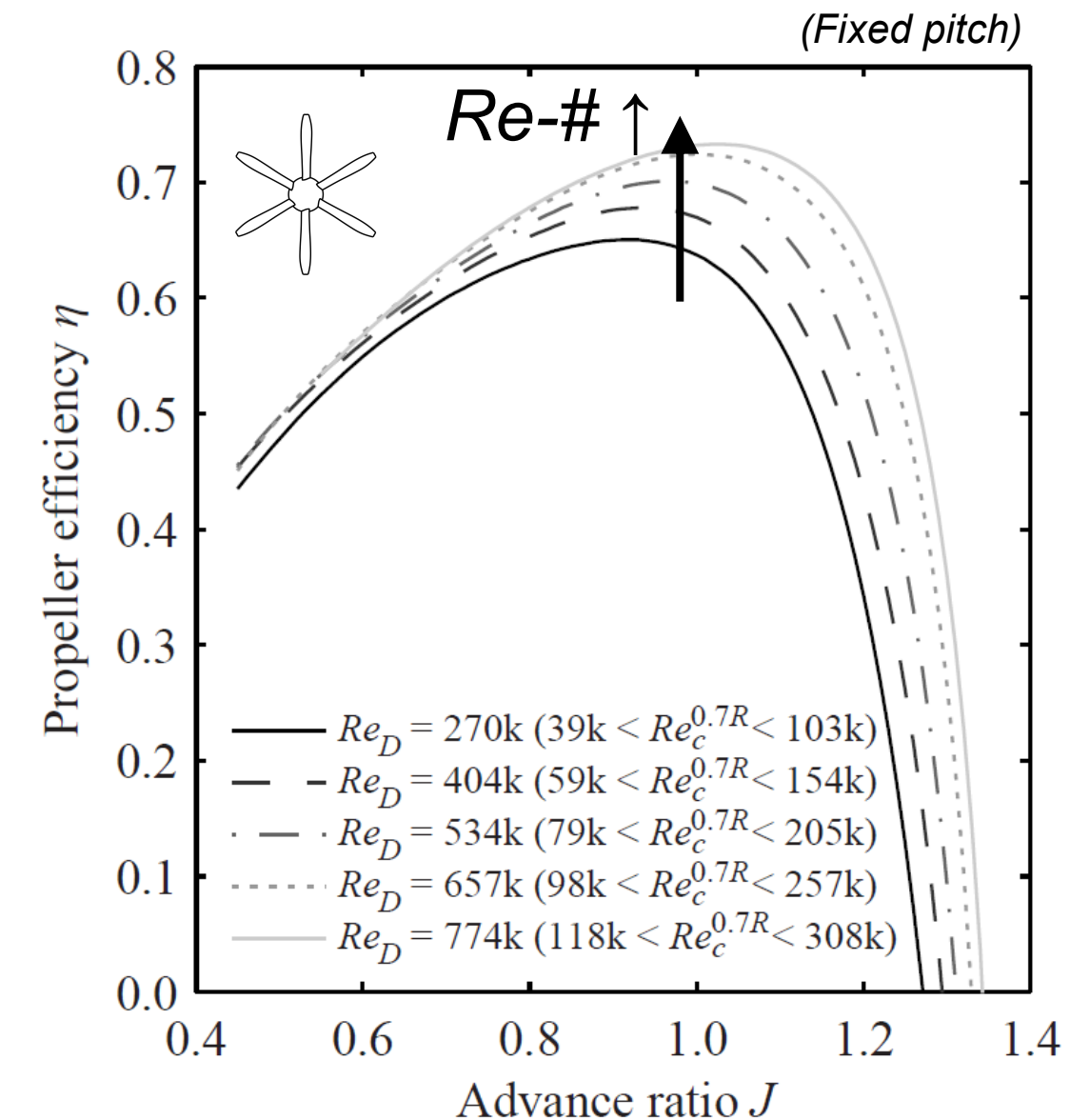
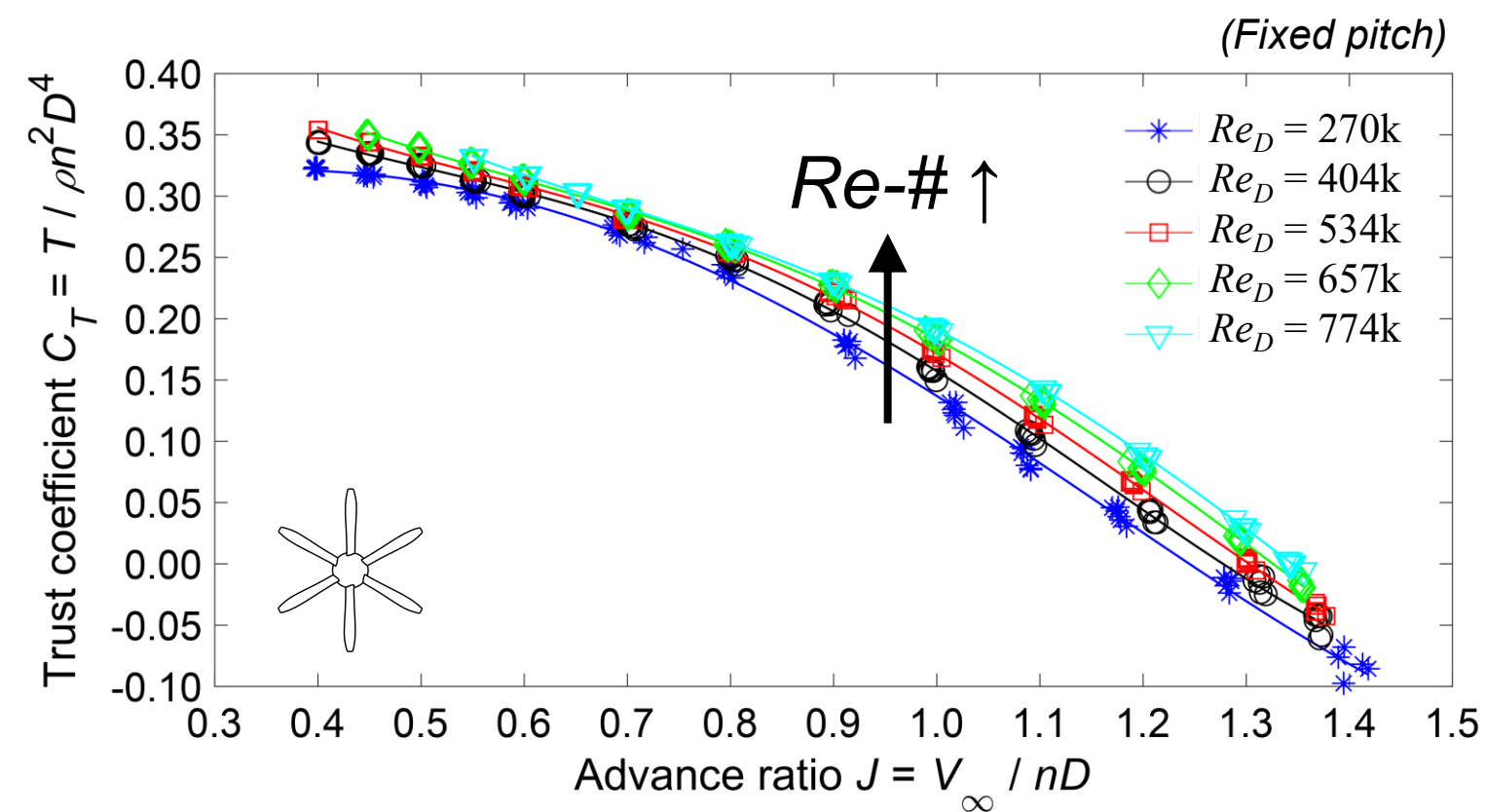
$$M_{\text{tip, hel}} \approx 0.8$$

- Achievable limit depends on freestream Mach number



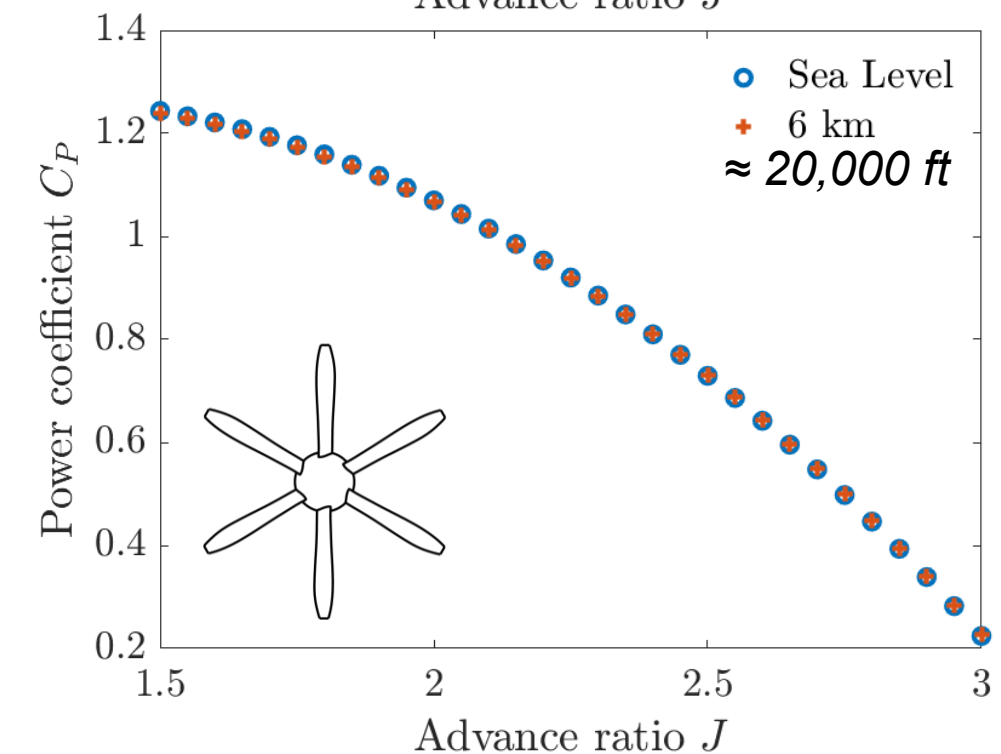
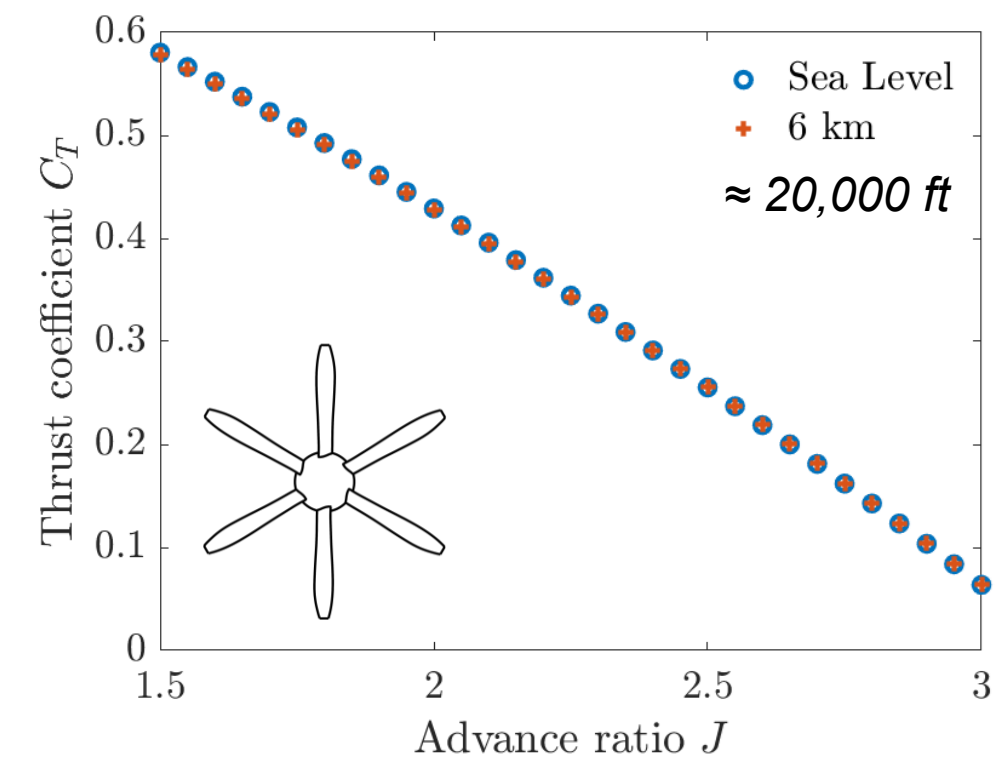
Impact of Reynolds number

- Performance increases with increasing Reynolds number
 - Increase in blade sectional lift, reduction in sectional friction drag
 - Keep in mind that usually Re-# and Ma-# are coupled...
- Scale-dependent effect! → Reynolds number $\sim$ blade chord
 - (effective velocity = scale-independent for J similarity)
 - Important for model-scale experiments!
 - Limited effect for $Re_c > 500,000$ → less relevant at full-scale



Impact of Reynolds number

- Example: impact of altitude on scaled propeller performance negligible for large propellers
 - TUD-XPROP geometry upscaled to $D = 4.064$ m
 - BEM simulations with RFOIL airfoil polars (modified version of XFOIL that accounts for rotational effects on boundary-layer behavior)
 - Cruise-like pitch setting
 - Comparison of scaled performance at sea-level vs $\sim 20,000$ ft
 - no differences observed



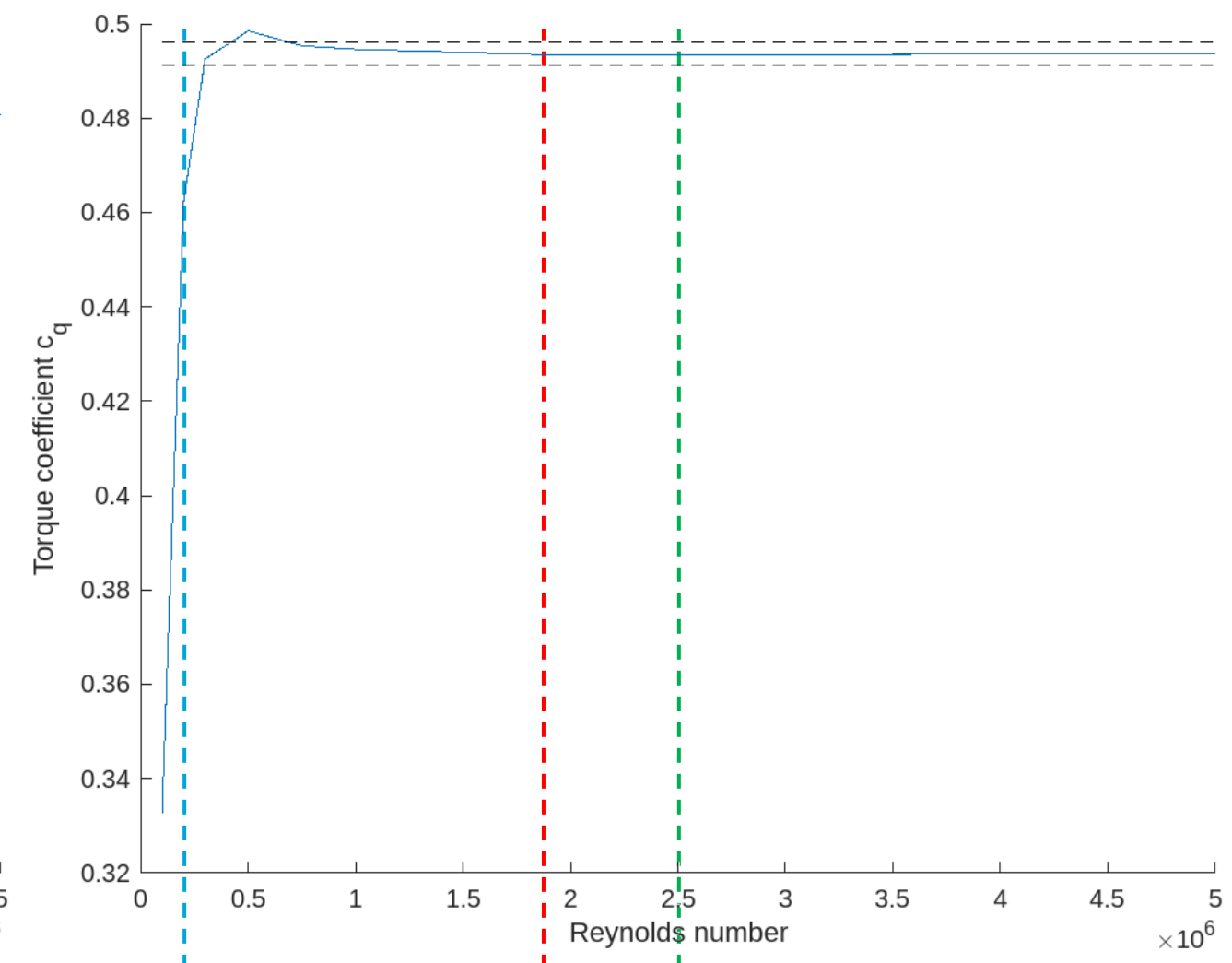
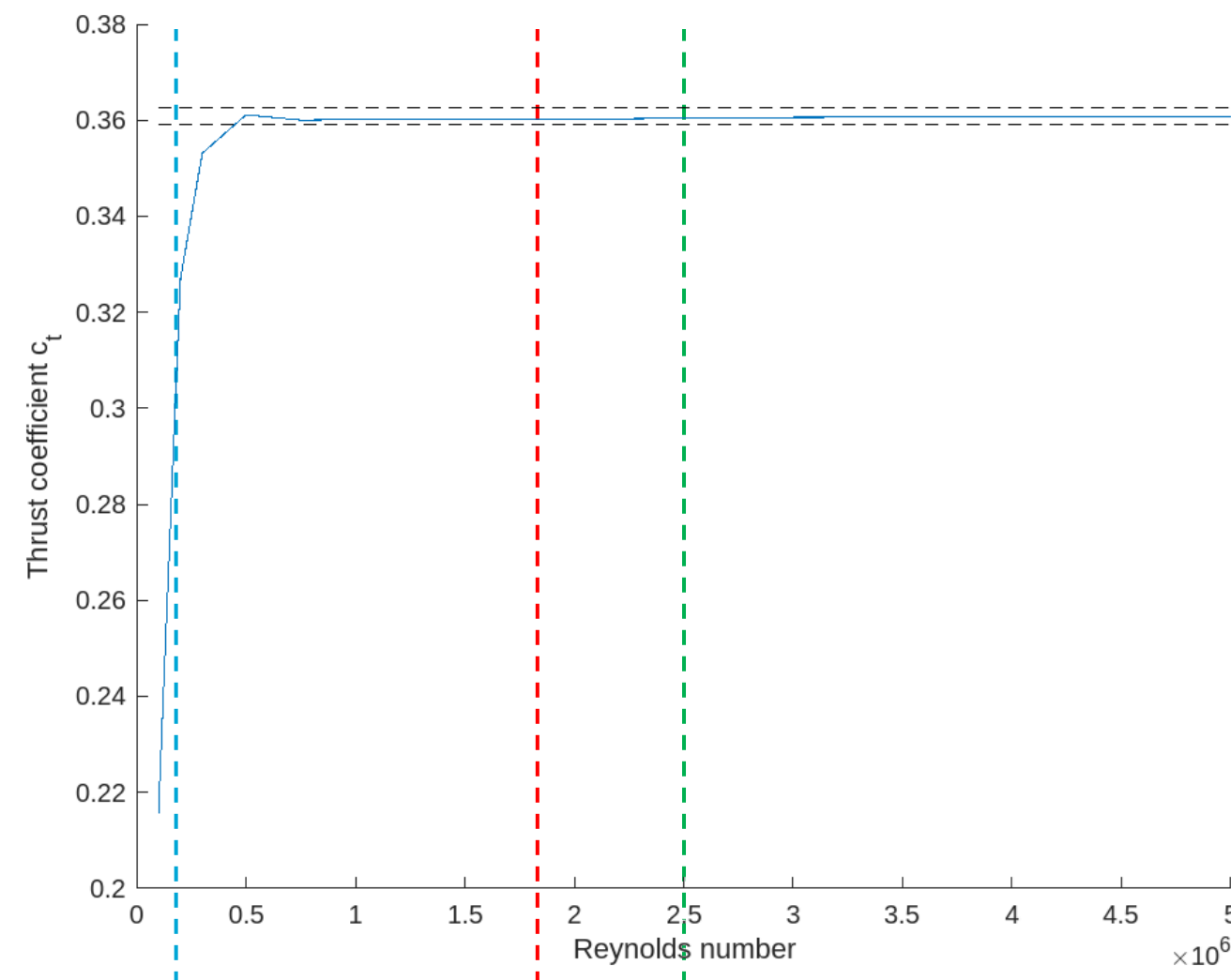
$$\beta_{0.7R} = 50^\circ, M_\infty = 0.4996$$

Impact of Reynolds number on propeller performance

- High-speed prop airfoil performance, representative for design near radial point of max loading
 - XFOIL simulations for Reynolds number range $0.1 - 10 \times 10^6$ (free transition), $M_{\text{inf}} = 0$
 - FS cruise @ FL250; all else at S/L ISA except “DNW” cruise at 2 bar
 - Thrust/torque coefficients computed from combination of c_l , c_d , flow angle ϕ
 - Thrust/torque coefficient converges well for $Re > 0.5-1.0 \times 10^6$ ($c_l/c_d \approx 100 \rightarrow$ lift dominated)

Cruise-like condition ($J = 3.0$)

Case	D [m]	V _{inf}	M _{inf}	Re70R
TUD	0.4	80	0.24	210k
DNW	0.6	203	0.60	1800k
FS	4.0	185	0.60	2500k

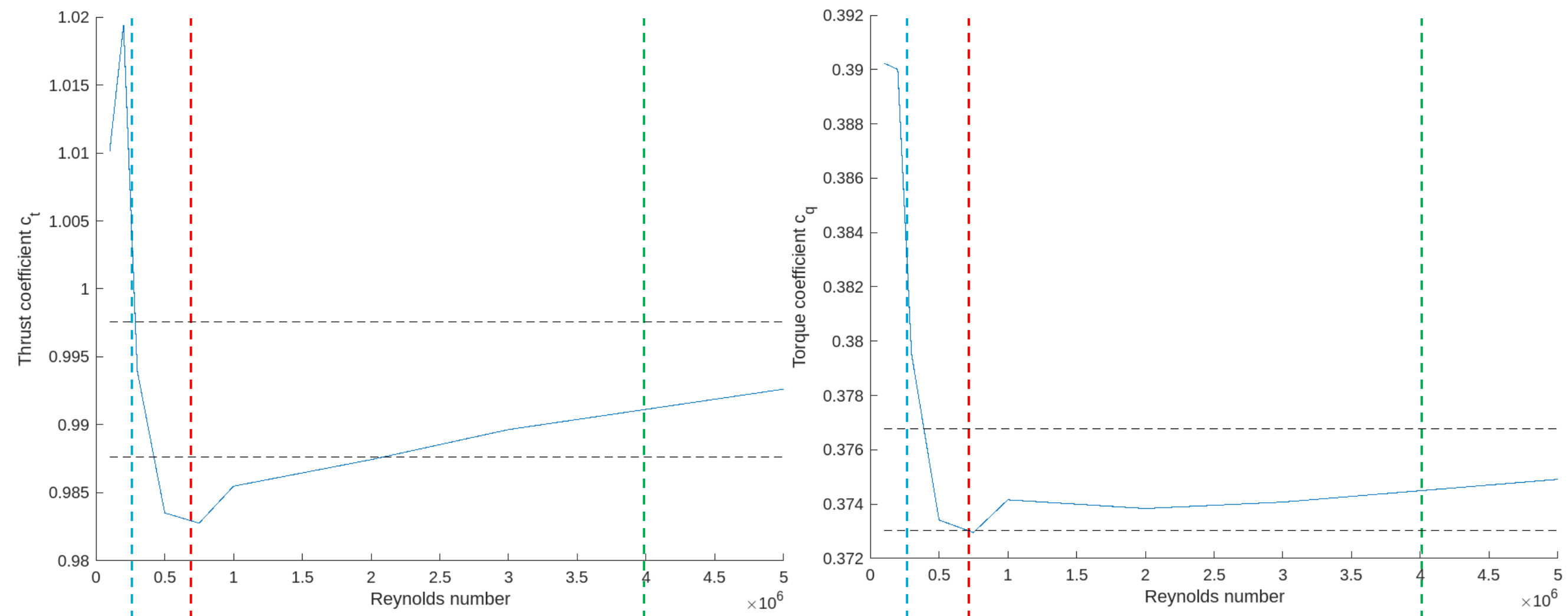


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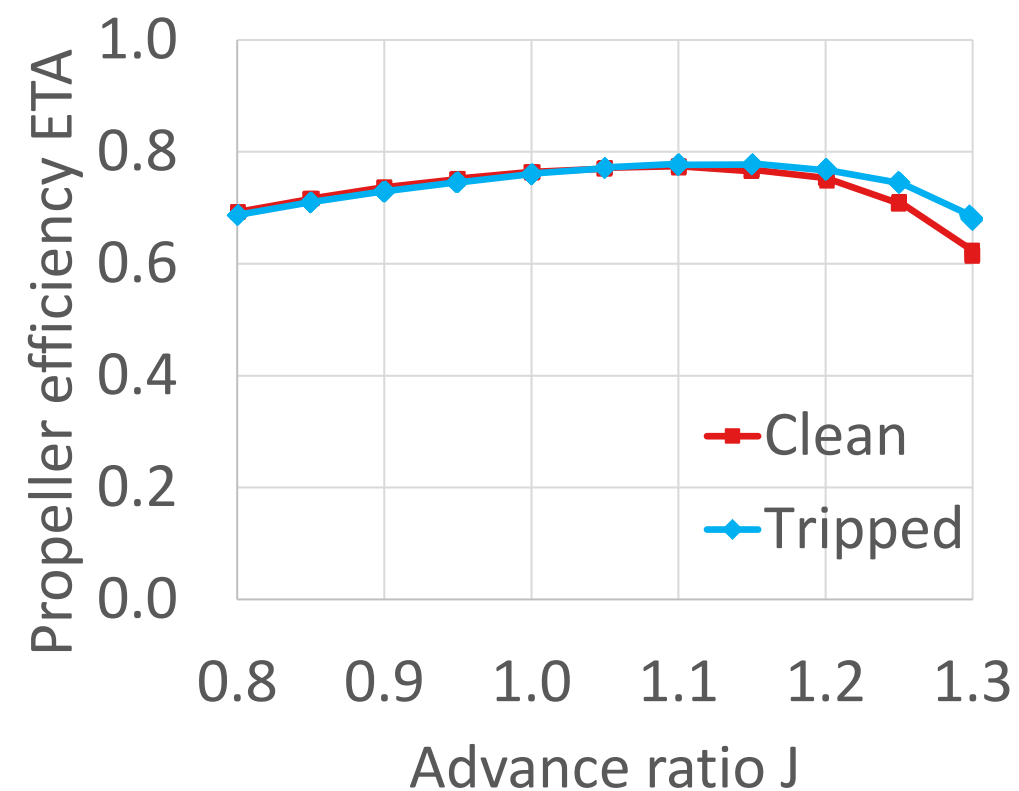
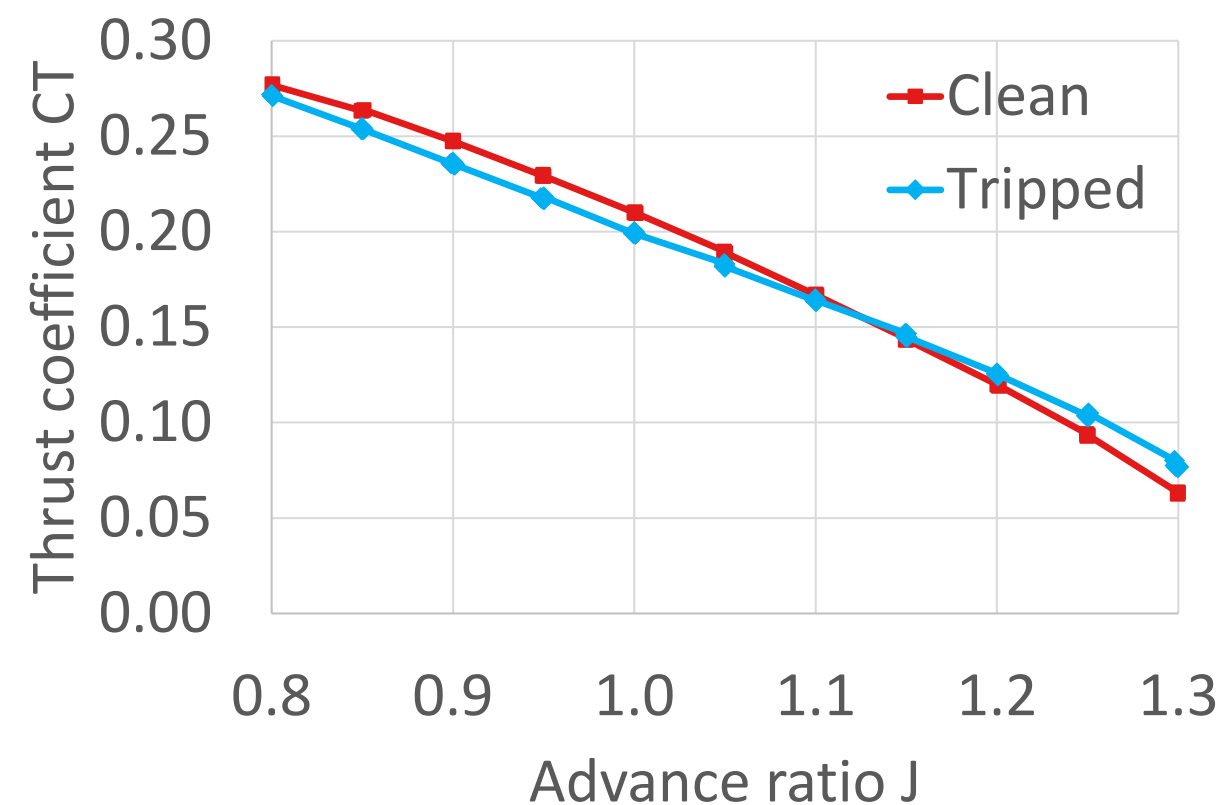
Climb-like condition ($J = 0.8$)

Case	D [m]	V _{inf}	M _{inf}	Re70R
TUD	0.4	40	0.12	240k
DNW	0.6	68	0.20	680k
FS	4.0	68	0.20	4000k



Impact of Reynolds number on propeller performance

- Low Reynolds number in model-scale tests also impacts laminar vs turbulent flow → tripping may be required for representative simulation
 - Tripping may avoid laminar separation at low to moderate loading conditions
 - Performance penalty at high loading due to increased boundary-layer growth
 - Ongoing research → tripping effectiveness to be traded with geometry modifications due to relatively high trip height w.r.t. blade thickness



Wrap-up

- Propellers are like rotating wings
 - Thrust and torque due to lift and drag produced at blade sections
 - Radial distributions of thrust and torque due to radially varying inflow conditions, tip effects
 - Self-induced velocities due to thrust and torque loading
- Propeller performance can be described by nondimensional parameters
 - Thrust coefficient, power coefficient, advance ratio, pitch angle, Mach number, Reynolds number, ...
 - Various coefficients available (C_T , T_C , T'_C) → choose based on use case (and always check definition)
 - Thrust/power coefficient increase with decreasing advance ratio, increasing pitch angle
 - Peak propeller efficiency at unique combination of advance ratio and pitch for given thrust
 - Compressibility effects scale blade-loading distribution until shock wave formation
 - Reynolds number effects not so relevant at full scale, but have strong impact on model-scale performance assessments

Questions?

